

GROUP ASSIGNMENT



CIMAGE PROFESSIONAL COLLEGE, PATNA

NUMERICAL ANALYSIS

SUBMITTED BY – Group 1, BCA (aku) Batch 1

SUBMITTED TO - KUNDAN SIR

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2.	KRISH	B-30	1	II
3.	SHABNAM	A-05	1	II
4.	VIVEK	A-02	1	II
5.	KRITI	A-40	1	II

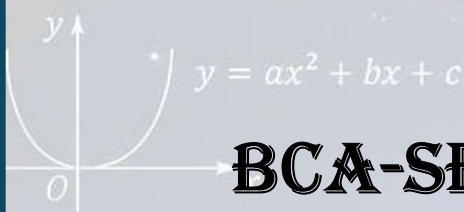
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

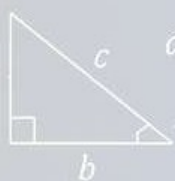
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Vishal Kumar Yadav



BCA-SEMESTER 2ND SYLLABUS



$a^2 + b^2 = c^2$ 1) **Solution of system of linear equations**

- a) **Direct Method**
 - i) **Gauss Elimination method**
 - ii) **Gauss Jordan Method**
 - iii) **LU Decomposition Method**
- b) **Indirect (Iterative) Method**
 - i) **Gauss Jacobi Method**
 - ii) **Gauss Seidel Method**

2) **Solution of Algebraic and Transcendental equation**

- a) **Newton Rapsons Method**
- b) **Iterative Method**
- c) **Regula Falsi Method**
- d) **Bisection (bolzano) Method**

3) **Interpolation Methods**

- a) **Lagrange Interpolation method**
- b) **Newton Forward Interpolation**
- c) **Newton Backward Interpolation**
- d) **Central Interpolation**

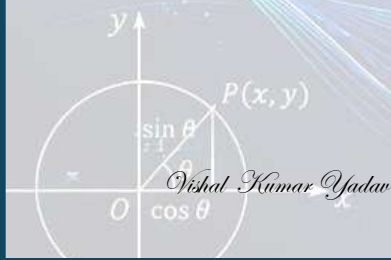
4) **Numerical Differentiation**

5) **Numerical Integration**

- a) **Trapezoidal Rule**
- b) **Simpson's 1/3 Rule**
- c) **Simpson's 3/8 Rule**

6) **Solution of ordinary differential equation**

- d) **Taylor Series Method**
- e) **Euler's Method**
- f) **Modified Euler's Method**
- g) **Runge – Kutta Method (2nd order)**
- h) **Runge – Kutta Method (4th order)**



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

GAUSS ELIMINATION METHOD

(THEORY)

Let, $a^2 + b^2 = c^2$

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Be the given system of linear equation.

Now,

We convert the given system of linear equation it's equivalence matrix form

$$AX=B$$

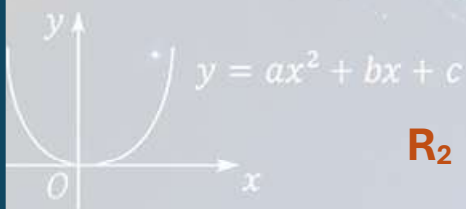
$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

or

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Now,

We convert the above matrix it's equivalent lower (upper) triangular matrix, by using row (column) transformation.



$$R_2 \longrightarrow R_2 - \frac{a_{21}}{a_{11}} R_1$$



$$a^2 + b^2 = c^2 \quad R_3 \longrightarrow R_3 - \frac{a_{31}}{a_{11}} R_1$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{pmatrix} = \begin{pmatrix} b_1 \\ b'_2 \\ b'_3 \end{pmatrix}$$

$$R_3 \longrightarrow R_3 - \frac{a'_{32}}{a'_{22}} R_2$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a''_{23} \\ 0 & 0 & a''_{33} \end{pmatrix} = \begin{pmatrix} b_1 \\ b'_2 \\ b''_3 \end{pmatrix}$$

Now,

we convert the above matrix it's equivalent linear equation form.

$$a_{11}x + a_{12}y + a_{13}z = b_1 \text{ ----- i}$$

$$a'_{22}y + a'_{23}z = b'_2 \text{ ----- ii}$$

$$a''_{33}z = b''_3 \text{ ----- iii}$$

From equation (iii) we get the value of z

Then, put the value of z in in equation (ii)

We get the value of y.

Then, put the value of y & z in equation (i)

We get the value of x.



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

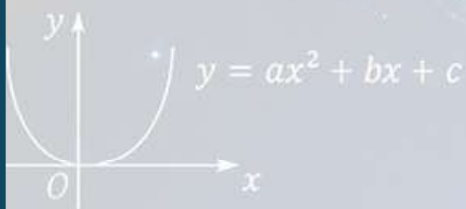
Page no - 4

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =

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$$y = ax^2 + bx + c$$

PRACTICAL - 1

Q.1) $2x + 3y - z = 5$

$4x + y + 2z = 6$

$-2x + 5y - z = -3$

Solution:-

We convert the given system of linear equation it's equivalence matrix form

$$\begin{pmatrix} 2 & 3 & -1 \\ 4 & 1 & 2 \\ -2 & 5 & -3 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ -3 \end{pmatrix}$$

$$R_2 \longrightarrow R_2 - \frac{4}{2}R_1$$

$$R_3 \longrightarrow R_3 + R_1$$

$$\begin{pmatrix} 2 & 3 & -1 \\ 0 & -5 & 4 \\ 0 & 8 & -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 2 \end{pmatrix}$$

$$R_3 \longrightarrow R_3 + \frac{8}{5}R_2$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

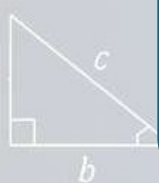
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 5

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

$$\begin{pmatrix} 2 & 3 & -1 \\ 0 & -5 & 4 \\ 0 & 0 & \frac{22}{5} \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ -\frac{22}{5} \end{pmatrix}$$

$$\frac{22}{5} z = -\frac{22}{5}$$

$$z = -1$$

$$-5y + 4z = -4$$

$$-5y + 4(-1) = -4$$

$$y = \frac{-4+4}{-5} = 0$$

$$2x + 0 + 1 = 5$$

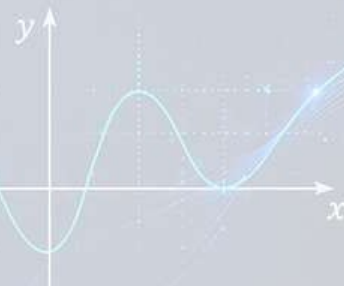
$$x = \frac{5-1}{2} = 2$$

ANSWER :

$$x = 2$$

$$y = 0$$

$$z = -1$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

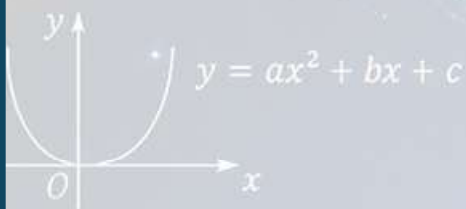
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 6

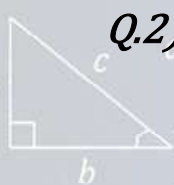
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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PRACTICAL - 2



$$Q.2) x + y + z = 6$$

$$2x + 3y + 7z = 20$$

$$x + 3y + 4z = 17$$

Solution:-

We convert the given system of linear equation its equivalence matrix form.

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 7 \\ 1 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 20 \\ 17 \end{pmatrix}$$

$$R_2 \longrightarrow R_2 - 2R_1$$

$$R_3 \longrightarrow R_3 - R_1$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \\ 17 \end{pmatrix}$$

$$R_3 \longrightarrow R_3 - 2R_2$$

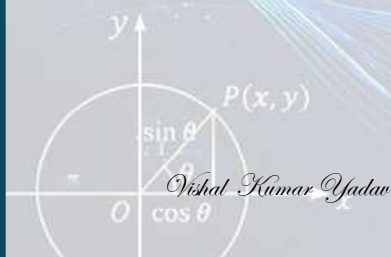


$$\sin^2 \theta + \cos^2 \theta = 1$$

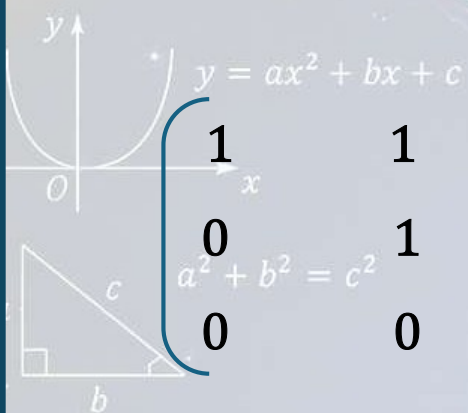
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 7

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & -7 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \\ -5 \end{pmatrix}$$

$$-7z = -5$$

$$z = \frac{5}{7}$$

$$y + 5z = 8$$

$$y + 5 \times \frac{5}{7} = 8$$

$$y = 8 - \frac{25}{7}$$

$$y = \frac{31}{7}$$

$$x + y + z = 6$$

$$x + \frac{31}{7} + \frac{5}{7} = 6$$

$$x = 6 - \frac{36}{7}$$

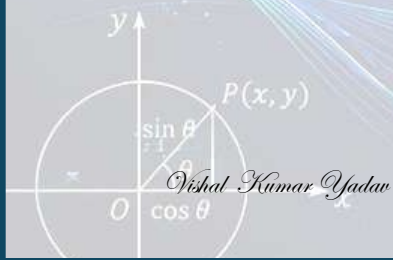
$$x = \frac{6}{7}$$

ANSWER :

$$x = \frac{6}{7}$$

$$y = \frac{31}{7}$$

$$z = \frac{5}{7}$$

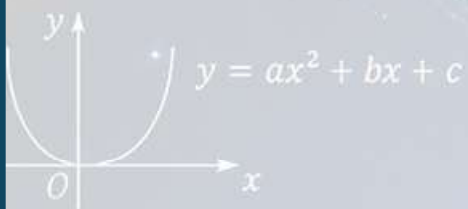


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 3

Q.3) $x - y + z = 7$
 $2x + y - 3z = -1$
 $-x + 2y + 2z = 7$

Solution:-

We convert the given system of linear equation it's equivalence matrix form.

$$\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ -1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \\ 7 \end{pmatrix}$$

$$R_2 \longrightarrow R_2 - 2R_1$$

$$R_3 \longrightarrow R_3 + R_1$$

$$\begin{pmatrix} 1 & -1 & 1 \\ 0 & 3 & -5 \\ 0 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 7 \\ -15 \\ 14 \end{pmatrix}$$

$$R_3 \longrightarrow R_3 - \frac{1}{3}R_2$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

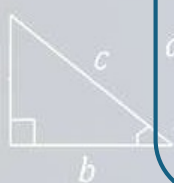
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

$$\begin{pmatrix} 1 & -1 & 1 \\ 0 & 3 & -5 \\ 0 & 0 & \frac{14}{3} \end{pmatrix} = \begin{pmatrix} 7 \\ -15 \\ 19 \end{pmatrix}$$

$$\frac{14}{3} z = 19$$

$$z = \frac{57}{14}$$

$$3y - 5 \times \frac{57}{14} = -15$$

$$Y = \frac{-210 + 285}{14 \times 3}$$

$$Y = \frac{75}{42} = \frac{25}{14}$$

$$x - y + z = 7$$

$$x - \frac{25}{14} + \frac{57}{14} = 7$$

$$x - \frac{25 - 57}{14} = 7$$

$$x = 7 + \frac{32}{14} = \frac{98 + 32}{14}$$

$$x = \frac{130}{14} = \frac{65}{7}$$

ANSWER :

$$x = \frac{65}{7}$$

$$y = \frac{25}{14}$$

$$z = \frac{57}{14}$$

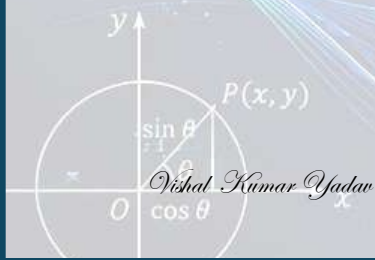


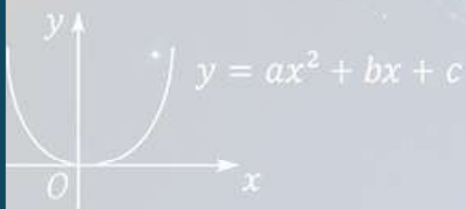
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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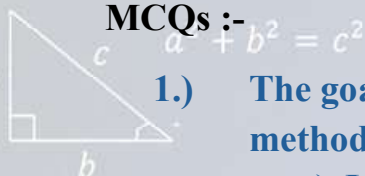
$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



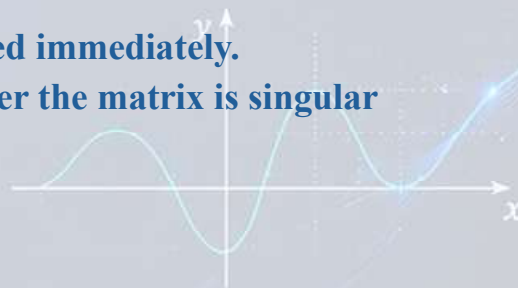


EXERCISE - 1

MCQs :-



- 1.) The goal of the forward elimination steps in the Gauss elimination method is to reduce the coefficient matrix to a (an):
 - a) Identity matrix
 - b) Diagonal matrix
 - c) Lower triangular matrix
 - d) Upper triangular matrix
- 2.) How is the transformation of the coefficient matrix to an upper triangular matrix primarily done?
 - a) Elementary row transformations
 - b) Elementary column transformations
 - c) Successive division
 - d) Matrix transposition
- 3.) What are the specific diagonal coefficients of the matrix obtained during the elimination process called?
 - a) Operatives
 - b) Pivots
 - c) Constants
 - d) Roots
- 4.) What happens when you encounter a division by zero during the forward elimination step?
 - a) The matrix is mathematically guaranteed to be singular.
 - b) The system has no solution.
 - c) The coefficient matrix must be inverted immediately.
 - d) It does not necessarily indicate whether the matrix is singular or non-singular.

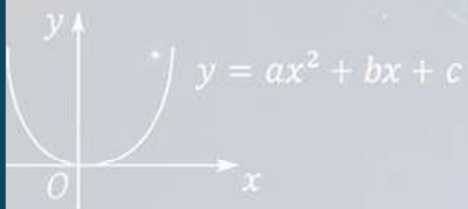


$$\sin^2 \theta + \cos^2 \theta = 1$$

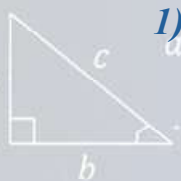
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXERCISE - 2



$$\begin{aligned} 1) \quad & \begin{cases} 5x - 2y + z = 4 \\ 7x + y - 5z = 8 \\ 3x + 7y + 4z = 10 \end{cases} \end{aligned}$$

$$\begin{aligned} 2) \quad & \begin{cases} 2x + y - z = -1 \\ x - 2y + 3z = 9 \\ 3x - y + 5z = 14 \end{cases} \end{aligned}$$

$$\begin{aligned} 3) \quad & \begin{cases} 2x + y + z = 10 \\ 3x + 2y + 3z = 18 \\ x + 4y + 9z = 16 \end{cases} \end{aligned}$$

$$\begin{aligned} 4) \quad & \begin{cases} 0.5x_1 + 1.1x_2 + 3.1x_3 = 6.0 \\ 2.0x_1 + 4.5x_2 + 0.4x_3 = 0.02 \\ 5.0x_1 + 1.0x_2 + 6.5x_3 = 1.0 \end{cases} \end{aligned}$$

$$\begin{aligned} 5) \quad & \begin{cases} 0.5x_1 + 1.1x_2 + 3.1x_3 = 6.0 \\ 0.5x_1 + 1.1x_2 + 3.1x_3 = 6.0 \\ 0.5x_1 + 1.1x_2 + 3.1x_3 = 6.0 \end{cases} \end{aligned}$$

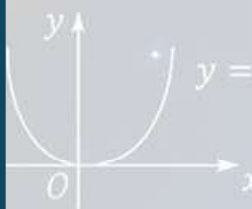


$$\sin^2 \theta + \cos^2 \theta = 1$$

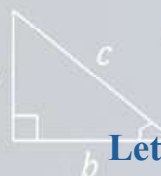
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

GAUSS JORDAN METHOD

(THEORY)

Let,

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Be the given system of linear equation.

Now,

We convert the given system of linear equation it's equivalence matrix form.

$$AX = B$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Now,

We convert the above matrix it's equivalent diagonal matrix by using row column transformation.

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$



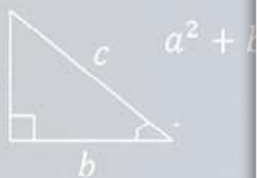
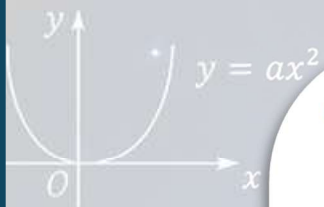
$P(x, y)$
 $\sin \theta$
 $\cos \theta$
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$R_2 \longrightarrow R_2 - \frac{a_{21}}{a_{11}} R_1$$

$$R_3 \longrightarrow R_3 - \frac{a_{31}}{a_{11}} R_1$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{pmatrix} = \begin{pmatrix} b_1 \\ b'_2 \\ b'_3 \end{pmatrix}$$

$$R_3 \longrightarrow R_3 - \frac{a_{32}'}{a_{22}'} R_2$$

$$R_1 \longrightarrow R_1 - \frac{a_{12}'}{a'_{22}} R_2$$

$$\begin{pmatrix} a'_{11} & 0 & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{pmatrix} = \begin{pmatrix} b'_1 \\ b'_2 \\ b''_3 \end{pmatrix}$$

$$R_1 \longrightarrow R_1 - \frac{a'_{13}}{a''_{33}} R_3$$

$$R_2 \longrightarrow R_2 - \frac{a'_{23}}{a''_{33}} R_3$$

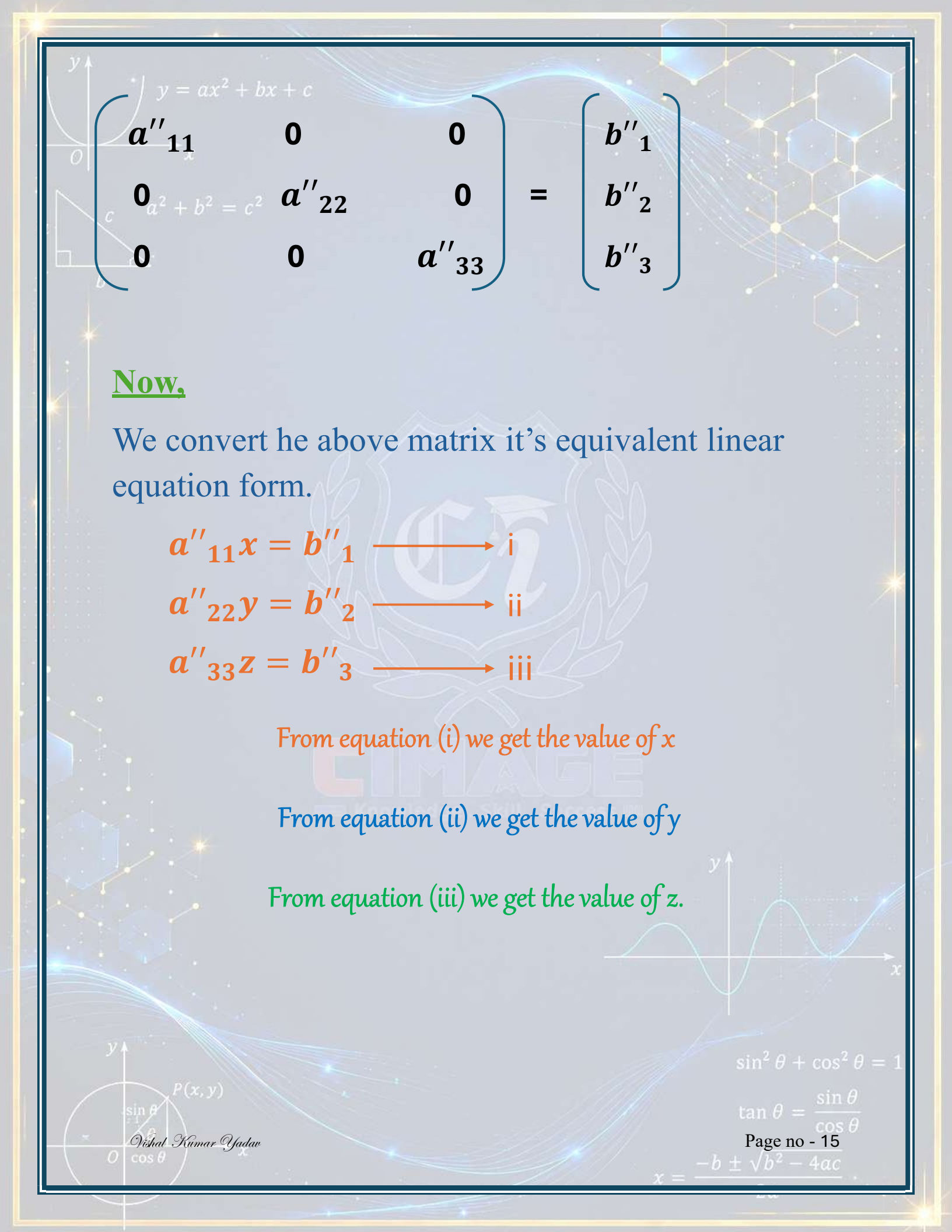


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 14

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$y = ax^2 + bx + c$
 $a^2 + b^2 = c^2$

$$\begin{pmatrix} a''_{11} & 0 & 0 \\ 0 & a''_{22} & 0 \\ 0 & 0 & a''_{33} \end{pmatrix} = \begin{pmatrix} b''_1 \\ b''_2 \\ b''_3 \end{pmatrix}$$

Now,

We convert the above matrix to its equivalent linear equation form.

$$a''_{11}x = b''_1 \longrightarrow \text{i}$$

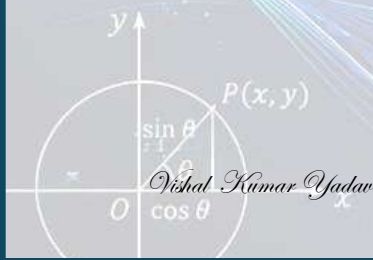
$$a''_{22}y = b''_2 \longrightarrow \text{ii}$$

$$a''_{33}z = b''_3 \longrightarrow \text{iii}$$

From equation (i) we get the value of x

From equation (ii) we get the value of y

From equation (iii) we get the value of z .

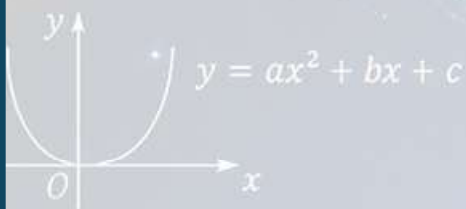


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 15

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Q.1) $x + 5y + 7z = 13$

$3x + 4y + 3z = 10$

$4x + 3y + 5z = 12$

Solution:-

We convert the given system of linear equation its equivalence matrix form.

$$\begin{pmatrix} 1 & 5 & 7 \\ 3 & 4 & 3 \\ 4 & 3 & 5 \end{pmatrix} = \begin{pmatrix} 13 \\ 10 \\ 12 \end{pmatrix}$$

$R_2 \longrightarrow R_2 - 3R_1$

$R_3 \longrightarrow R_3 - 4R_1$

$$\begin{pmatrix} 1 & 5 & 13 \\ 0 & -11 & -18 \\ 0 & -17 & -23 \end{pmatrix} = \begin{pmatrix} 13 \\ -29 \\ -40 \end{pmatrix}$$

$R_3 \longrightarrow R_3 - \frac{17}{11}R_2$

$R_1 \longrightarrow R_1 + \frac{5}{11}R_2$



$\sin^2 \theta + \cos^2 \theta = 1$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

Page no - 16

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$y = ax^2 + bx + c$$


$$\begin{pmatrix} 1 & 0 & -\frac{13}{11} \\ 0 & -11 & -18 \\ 0 & 0 & \frac{53}{11} \end{pmatrix} = \begin{pmatrix} -\frac{2}{11} \\ -29 \\ \frac{53}{11} \end{pmatrix}$$

$$R_1 \longrightarrow R_1 + \frac{13}{53} R_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -11 & 0 \\ 0 & 0 & \frac{53}{11} \end{pmatrix} = \begin{pmatrix} 1 \\ -11 \\ \frac{53}{11} \end{pmatrix}$$

$$x = 1$$

$$-11y = -11$$

$$y = \frac{-11}{-11} = 1$$

$$\frac{5}{11}z = \frac{5}{11}$$

$$z = 1$$

ANSWER :

$$X = 1$$

$$Y = 1$$

$$Z = 1$$

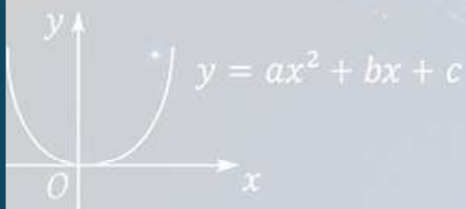
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 17

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





PRACTICAL - 2

Q.2) $x + y + z = 6$

$2x + 3y + 7z = 20$

$x + 3y + 4z = 13$

Solution:-

We convert the given system of linear equation it's equivalence matrix form.

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 7 \\ 1 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 20 \\ 13 \end{pmatrix}$$

$R_2 \longrightarrow R_2 - 2R_1$

$R_3 \longrightarrow R_3 - R_1$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 7 \\ 0 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \\ 7 \end{pmatrix}$$

$R_1 \longrightarrow R_1 - R_2$

$R_3 \longrightarrow R_3 - 2R_2$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 18

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$



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$y = ax^2 + bx + c$
 $a^2 + b^2 = c^2$

$$\begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & 5 \\ 0 & 0 & -7 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \\ -9 \end{pmatrix}$$

$R_1 \longrightarrow R_1 - \frac{4}{7}R_3$

$$R_2 \longrightarrow R_2 + \frac{5}{7}R_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -7 \end{pmatrix} = \begin{pmatrix} \frac{22}{7} \\ \frac{11}{7} \\ -9 \end{pmatrix}$$

$$x = \frac{22}{7}$$

$$y = \frac{11}{7}$$

$$-7z = -9$$

$$z = \frac{9}{7}$$

ANSWER :

$$x = \frac{22}{7}$$

$$y = \frac{11}{7}$$

$$z = \frac{9}{7}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

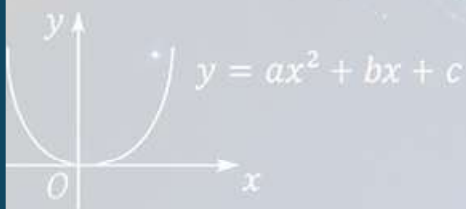
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 19

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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PRACTICAL - 3

Q.3) $x + 2y + 3z = 14$

$2x + 3y + z = 13$

$3x + y + 2z = 13$

Solution:-

We convert the given system of linear equation it's equivalence matrix form.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 14 \\ 13 \\ 13 \end{pmatrix}$$

$R_2 \longrightarrow R_2 - 2R_1$

$R_3 \longrightarrow R_3 - 3R_1$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -5 \\ 0 & -5 & -7 \end{pmatrix} = \begin{pmatrix} 14 \\ -15 \\ -29 \end{pmatrix}$$

$R_1 \longrightarrow R_1 + 2R_2$

$R_3 \longrightarrow R_3 - 5R_2$



$\sin^2 \theta + \cos^2 \theta = 1$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

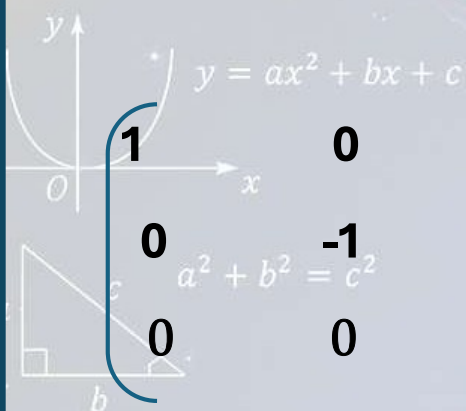
Page no - 20

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x =$



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$$\begin{pmatrix} 1 & 0 & -7 \\ 0 & -1 & -5 \\ 0 & 0 & 18 \end{pmatrix} = \begin{pmatrix} -16 \\ -15 \\ 46 \end{pmatrix}$$

$$R_1 \longrightarrow R_1 + \frac{7}{18} R_3$$

$$R_2 \longrightarrow R_2 + \frac{5}{18} R_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 18 \end{pmatrix} = \begin{pmatrix} \frac{17}{9} \\ -\frac{20}{9} \\ 46 \end{pmatrix}$$

$$18z = 46$$

$$z = \frac{23}{9}$$

$$-y = -\frac{20}{9}$$

$$x = \frac{17}{9}$$

ANSWER :

$$x = \frac{17}{9}$$

$$y = \frac{20}{9}$$

$$z = \frac{23}{9}$$

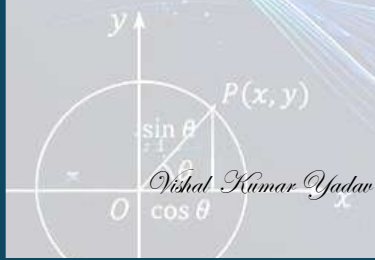


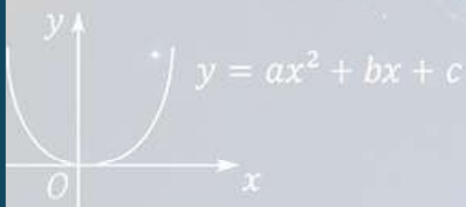
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

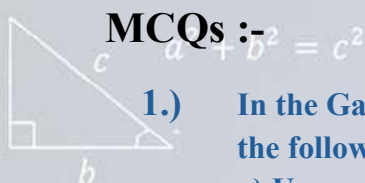
Page no - 21

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



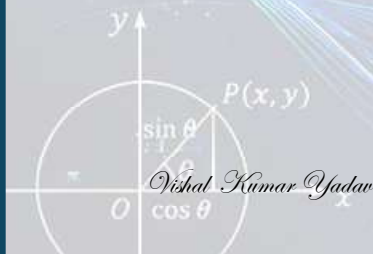


EXERCISE - 1



MCQs :- $a^2 + b^2 = c^2$

- 1.) In the Gauss-Jordan method, the augmented matrix is transformed into which of the following forms?
 - a) Upper triangular matrix
 - b) Identity matrix
 - c) Lower triangular matrix
 - d) Row Echelon form
- 2.) When solving a system of equations ($AX = B$) using the Gauss-Jordan method, the final solution is obtained from:
 - a) Back substitution
 - b) Successive approximations
 - c) Directly from the last column of the augmented matrix
 - d) Cramer's Rule
- 3.) Compared to standard Gaussian Elimination, Gauss-Jordan elimination avoids which step?
 - a) Forward elimination
 - b) Partial pivoting
 - c) Back-substitution
 - d) Row scaling
- 4.) Besides solving linear systems, what is the Gauss-Jordan method widely used to calculate?
 - a) Matrix determinants
 - b) Matrix inverses
 - c) Eigenvalues
 - d) Matrix rank only
- 5.) What is the final target structure for the coefficient matrix in the Gauss-Jordan method?
 - a) Upper triangular form
 - b) Lower triangular form
 - c) Reduced row-echelon form
 - d) Hessenberg form



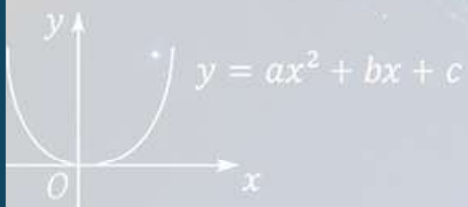
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 22

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

EXERCISE - 2



1) $x + 2y = 5$
 $a^2 + b^2 = c^2$
 $2x + y = 4$

2) $2x + y - z = -1$

$x - 2y + 3z = 9$

$x - y + 5z = 14$

3) $x + y + z = 3$

$2x + 3y + 5z = 8$

$4x + 0y + 5z = 10$

4) $2x + y - z = 8$

$-3x - y + 2z = -11$

$-2x + y + 2z = -3$

5) $x - y + z = 2$

$2x + y + 3z = 9$

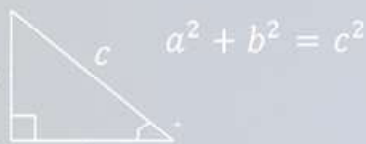
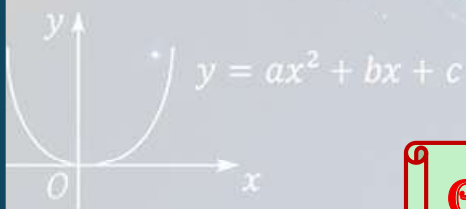


$\sin^2 \theta + \cos^2 \theta = 1$

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

Page no - 23

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$



GAUSS JACOBI METHOD

(THEORY)

Let,

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Be the given diagonal dominant linear equation.

Its initial approximation (initial solution) is not given.

$$x_0 = 0, \quad y_0 = 0, \quad z_0 = 0$$

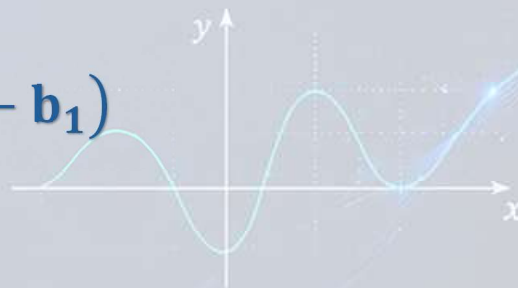
From eq_n (i)

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$x = \frac{-1}{a_{11}} (a_{12}y + a_{13}z - b_1)$$

From eq_n (ii)

$$a_{21}x + a_{22}y + a_{23}z = b_2$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

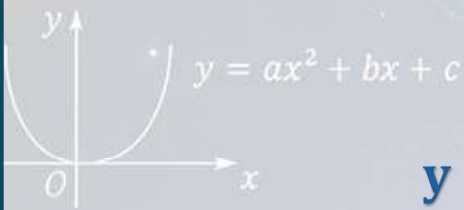
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 24

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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$$y = \frac{-1}{a_{22}} (a_{21}x + a_{23}z - b_2)$$



From eq_n (iii)

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

$$z = \frac{-1}{a_{33}} (a_{31}x + a_{32}y - b_3)$$

i.e,

$$x^{k+1} = \frac{-1}{a_{11}} (a_{12}y^k + a_{13}z^k - b_1)$$

$$y^{k+1} = \frac{-1}{a_{22}} (a_{21}x^k + a_{23}z^k - b_2)$$

$$z^{k+1} = \frac{-1}{a_{33}} (a_{31}x^k + a_{32}y^k - b_3)$$

K+1, Denotes number of iterations.

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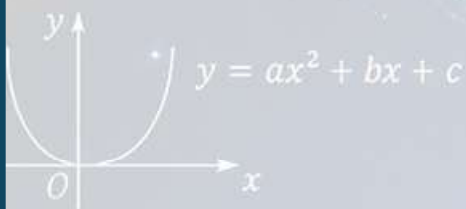


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 25

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Q.1) solve the system of linear eqⁿ by using gauss jacobi method.

$$12x + y + z = 14$$

$$x + 13y + 2z = 17$$

$$2x + y + 14z = 18$$

Solution:-

From eq_n (i)

$$12x + y + z = 14$$

$$x = \frac{-1}{12} (y^k + z^k - 14)$$

From eq_n (ii)

$$x + 13y + 2z = 17$$

$$y = \frac{-1}{13} (x^k + 2z^k - 17)$$

From eq_n (iii)

$$2x + y + 14z = 18$$

$$z = \frac{-1}{14} (2x^k + y^k - 18)$$

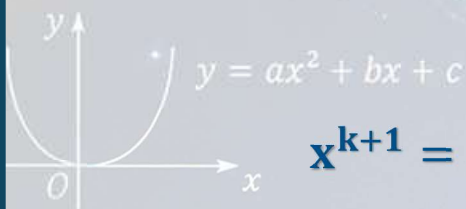


$$\sin^2 \theta + \cos^2 \theta = 1$$

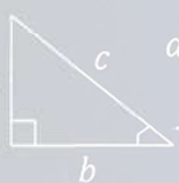
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 26

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$x^{k+1} = \frac{-1}{12} (y^k + z^k - 14)$$



$$a^2 + b^2 \quad y^{k+1} = \frac{-1}{13} (x^k + 2z^k - 17)$$

$$z^{k+1} = \frac{-1}{14} (2x^k + y^k - 18)$$

Now,

1st iteration , put k=0

$$x^1 = \frac{-1}{12} (y^0 + z^0 - 14)$$

$$= \frac{-1}{12} (0 + 0 - 14)$$

$$= 1.166$$

$$y^1 = \frac{-1}{13} (x^0 + 2z^0 - 17)$$

$$= \frac{-1}{13} (0 + 2 \times 0 - 14)$$

$$= 1.307$$

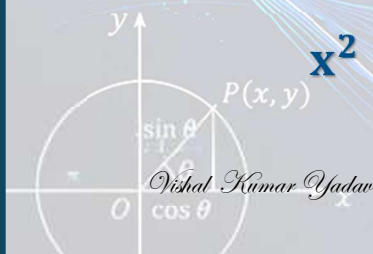
$$z^1 = \frac{-1}{14} (2x^0 + y^0 - 18)$$

$$= \frac{-1}{14} (0 + 0 - 18)$$

$$= 1.285$$

2nd iteration , put k=1

$$x^2 = \frac{-1}{12} (y^1 + z^1 - 14)$$

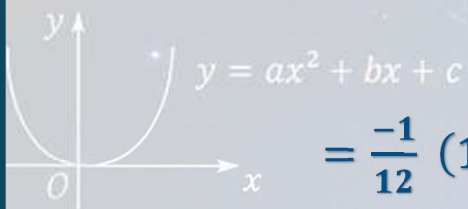


$$\sin^2 \theta + \cos^2 \theta = 1$$

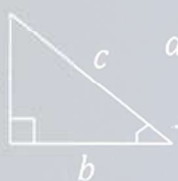
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 27

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$= \frac{-1}{12} (1.307 + 1.285 - 14)$$



$$a^2 + b^2 = c^2 = 0.950$$

$$y^2 = \frac{-1}{13} (x^1 + 2z^1 - 17)$$

$$= \frac{-1}{13} (1.166 + 2 \times 1.285 - 17)$$

$$= 1.020$$

$$z^2 = \frac{-1}{14} (2x^1 + y^1 - 18)$$

$$= \frac{-1}{14} (2 \times 1.166 + 1.307 - 18)$$

$$= 1.025$$

Answer

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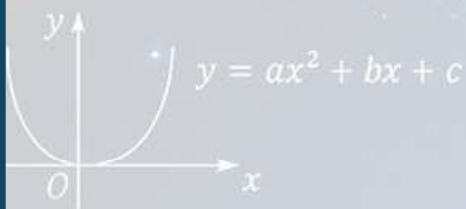


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 28

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 2

Q.2) solve the system of linear eqⁿ by using gauss jacobi method

$$5x + y + z = 10$$

$$x + 6y + z = 20$$

$$x + y + 7z = 30$$

Solution:-

From eq_n (i)

$$5x + y + z = 10$$

$$x = \frac{-1}{5} (y + z - 10)$$

From eqn (ii)

$$x + 6y + z = 20$$

$$y = \frac{-1}{6} (x + z - 20)$$

From eqn (iii)

$$x + y + 7z = 30$$

$$z = \frac{-1}{7} (x + y - 30)$$

i.e,

$$x^{k+1} = \frac{-1}{5} (y^k + z^k - 10)$$

$$y^{k+1} = \frac{-1}{6} (x^k + z^k - 20)$$

$$z^{k+1} = \frac{-1}{7} (x^k + y^k - 30)$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

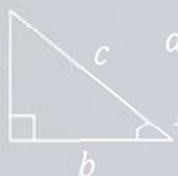




$$y = ax^2 + bx + c$$

Now,

1st iteration, put $k=0$



$$a^2 + b^2 = c^2$$

$$x^1 = \frac{-1}{5} (y^0 + z^0 - 10)$$

$$= \frac{-1}{5} (0 + 0 - 10)$$

$$= 2$$

$$y^1 = \frac{-1}{6} (x^0 + z^0 - 20)$$

$$= \frac{-1}{6} (0 + 0 - 20)$$

$$= 3.3333$$

$$z^1 = \frac{-1}{7} (x^0 + y^0 - 30)$$

$$= \frac{-1}{7} (0 + 0 - 30)$$

$$= 4.2857$$

Answer

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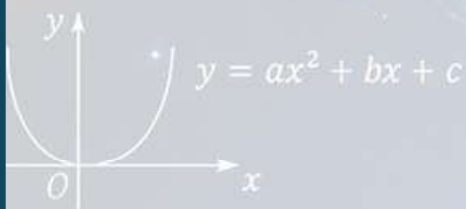
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 30

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





PRACTICAL - 3

Q.3) solve the system of linear eqⁿ by using gauss jacobi method

$$8x + y - z = 8$$

$$x + 7y + z = 15$$

$$-x + y + 6z = 20$$

Solution:-

From eq_n (i)

$$8x + y - z = 8$$

$$x = \frac{-1}{8} (y - z - 8)$$

From eq_n (ii)

$$x + 7y + z = 15$$

$$y = \frac{-1}{7} (x + z - 15)$$

From eq_n (iii)

$$-x + y + 6z = 20$$

$$z = \frac{-1}{6} (-x + y - 20)$$

i.e,

$$x^{k+1} = \frac{-1}{8} (y^k - z^k - 8)$$

$$y^{k+1} = \frac{-1}{7} (x^k + z^k - 15)$$

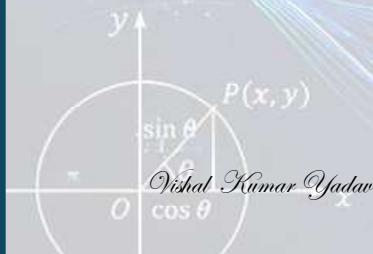
$$z^{k+1} = \frac{-1}{6} (-x^k + y^k - 20)$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 31

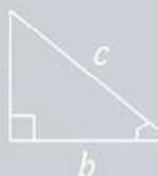
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





$$y = ax^2 + bx + c$$

Now,
1st iteration, put $k=0$



$$a^2 + b^2 = c^2$$

$$\begin{aligned} x^1 &= \frac{-1}{8} (y^0 - z^0 - 8) \\ &= \frac{-1}{8} (0 - 0 - 8) \\ &= 1 \end{aligned}$$

$$\begin{aligned} y^1 &= \frac{-1}{7} (x^0 + z^0 - 15) \\ &= \frac{-1}{6} (0 + 0 - 15) \\ &= 2.1428 \end{aligned}$$

$$\begin{aligned} z^1 &= \frac{-1}{6} (-x^0 + y^0 - 20) \\ &= \frac{-1}{6} (-0 + 0 - 20) \\ &= 3.3333 \end{aligned}$$

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Answer

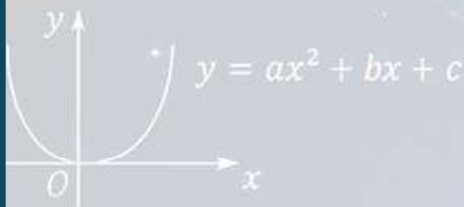


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

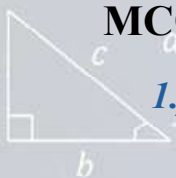
Page no - 32

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

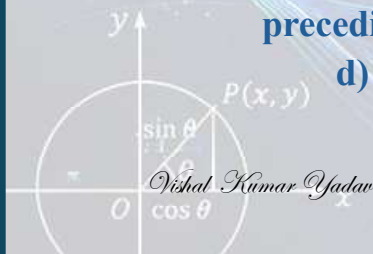


EXERCISE - 1

MCQs :- $a^2 + b^2 = c^2$



- 1.) The Gauss-Jacobi method is classified as which type of numerical technique?
 - a) Direct method
 - b) Iterative (indirect) method
 - c) Graphical method
 - d) Elimination method
- 2.) What is the necessary condition for a system of linear equations to converge using the Gauss-Jacobi method?
 - a) The coefficient matrix must be an identity matrix
 - b) The coefficient matrix must be strictly diagonally dominant
 - c) The determinant of the coefficient matrix must be zero
 - d) The matrix must be an upper triangular matrix.
- 3.) What is the main structural difference between the Gauss-Jacobi method and the Gauss-Seidel method?
 - a) Gauss-Jacobi solves nonlinear equations, while Gauss-Seidel solves linear equations
 - b) Gauss-Jacobi uses previous iteration values for all variables, while Gauss-Seidel uses the most recently updated values
 - c) Gauss-Jacobi requires an initial guess, while Gauss-Seidel does not
 - d) Gauss-Seidel can be solved in parallel, while Gauss-Jacobi cannot
- 4.) In the Gauss-Jacobi method, how are the new values for variables in a single iteration step calculated?
 - a) By back substitution after eliminating variables
 - b) Sequentially using the most up-to-date available values
 - c) Simultaneously using values only from the immediately preceding iteration
 - d) By taking the average of the previous two iterations



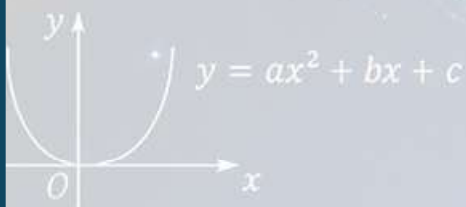
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 33

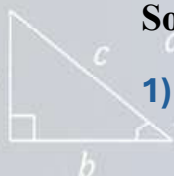
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =



EXERCISE - 2

Solve the system of linear equation by Gauss jacobi method



$$\begin{aligned} 1) \quad & 5x_1 - 2x_2 + 3x_3 = -1 \\ & -3x_1 + 9x_2 + x_3 = 2 \\ & 2x_1 - x_2 - 7x_3 = 3 \end{aligned}$$

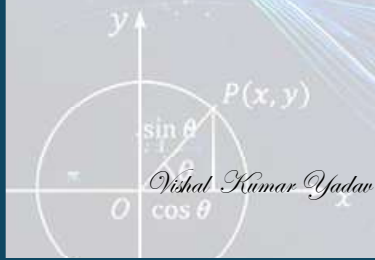
$$\begin{aligned} 2) \quad & 3x + y + z = 10 \\ & x + 4y + 2z = 12 \\ & x + 2y + 5z = 20 \end{aligned}$$

$$\begin{aligned} 3) \quad & 4x + y + z = 9 \\ & x + 3y + z = 10 \\ & x + y + 5z = 20 \end{aligned}$$

$$\begin{aligned} 4) \quad & 6x + y - z = 5 \\ & x + 7y + 2z = -4 \\ & -2x + y + 8z = 27 \end{aligned}$$

$$\begin{aligned} 5) \quad & 5x - y + z = 10 \\ & -x + 6y + 2z = 12 \\ & x + y + 7z = 30 \end{aligned}$$

$$\begin{aligned} 6) \quad & 3x + y + z = 11 \\ & x + 5y + z = 14 \\ & x + y + 7z = 23 \end{aligned}$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 34

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

GAUSS SEIDEL METHOD

(THEORY)

Let,

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Be the given diagonally dominant linear equation.

It initial approximation (initial solution) is given

$$x_0 = a, \quad y_0 = b, \quad z_0 = c$$

It initial approximation (initial solution) is not given then we take.

$$x_0 = 0, \quad y_0 = 0, \quad z_0 = 0$$

From eq_n (i)

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$x = \frac{-1}{a_{11}} (a_{12}y + a_{13}z - b_1)$$

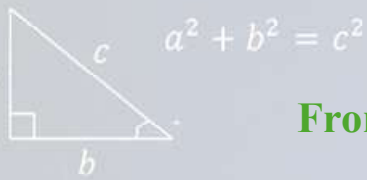
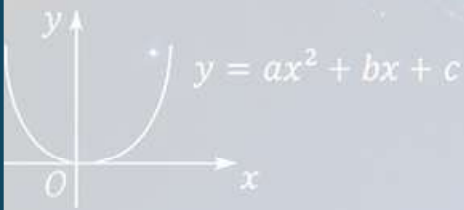
From eq_n (ii)

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$y = \frac{-1}{a_{22}} (a_{21}x + a_{23}z - b_2)$$

From eqn (iii)

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

$$z = \frac{-1}{a_{33}} (a_{31}x + a_{32}y - b_3)$$

i.e,

$$x^{k+1} = \frac{-1}{a_{11}} (a_{12}y^k + a_{13}z^k - b_1)$$

$$y^{k+1} = \frac{-1}{a_{22}} (a_{21}x^{k+1} + a_{23}z^k - b_2)$$

$$z^{k+1} = \frac{-1}{a_{33}} (a_{31}x^{k+1} + a_{32}y^{k+1} - b_3)$$

K+1, Denotes number of iterations

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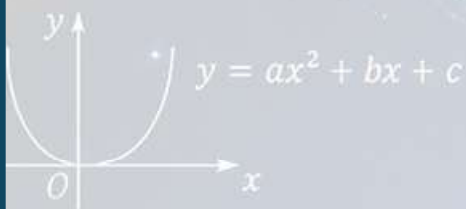
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 36

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Q.1) solve the system of linear eqⁿ by using gauss seidel method.

$$10x + y - z = 11$$

$$2x + 10y + z = 13$$

$$2x + 3y + 10z = 14$$

Solution:-

From eq_n (i)

$$10x + y - z = 11$$

$$x = \frac{-1}{10} (y - z - 11)$$

From eq_n (ii)

$$2x + 10y + z = 13$$

$$y = \frac{-1}{10} (2x + z - 13)$$

From eq_n (iii)

$$2x + 3y + 10z = 14$$

$$z = \frac{-1}{10} (2x + 3y - 14)$$

If initial approximation (initial solution) is not given then we take.

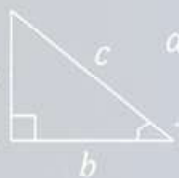
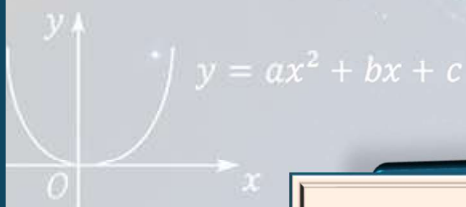
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 37

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





$$x_0 = 0, \quad y_0 = 0, \quad z_0 = 0$$

$$x^{k+1} = \frac{-1}{10} (y^k - z^k - 11)$$

$$y^{k+1} = \frac{-1}{10} (2x^{k+1} + z^k - 13)$$

$$z^{k+1} = \frac{-1}{10} (2x^{k+1} + 3y^{k+1} - 14)$$

1st iteration , put $k=0$

$$x^1 = \frac{-1}{10} (y^0 - z^0 - 8)$$

$$= \frac{-1}{10} (0 - 0 - 11)$$

$$= 1.1$$

$$y^1 = \frac{-1}{10} (2x_1 + z^0 - 13)$$

$$= \frac{-1}{10} (2 \times 1.1 + 0 - 13)$$

$$= 1.08$$

$$z^1 = \frac{-1}{10} (2x^1 + 3y^1 - 14)$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 38

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

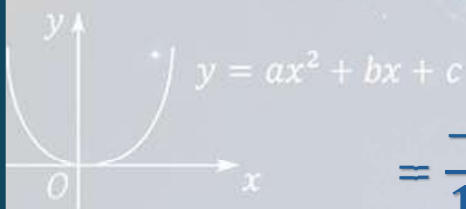
$$x =$$

$P(x, y)$

$\sin \theta$

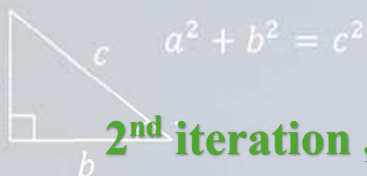
$\cos \theta$

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$$= \frac{-1}{10} (2 \times 1.1 + 3 \times 1.08 - 14)$$

$$= 0.856$$



2nd iteration , put k=1

$$x^2 = \frac{-1}{10} (y^1 - z^1 - 11)$$

$$= \frac{-1}{10} (1.08 - 0.856 - 11)$$

$$= 1.0776$$

$$y^2 = \frac{-1}{10} (2x_2 + z^1 - 13)$$

$$= \frac{-1}{10} (2 \times 1.0776 + 0.856 - 13)$$

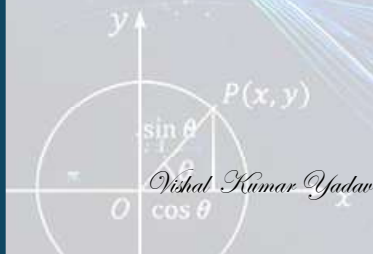
$$= 0.9988$$

$$z^2 = \frac{-1}{10} (2x^2 + 3y^2 - 14)$$

$$= \frac{-1}{10} (2 \times 1.0776 + 3 \times 0.9088 - 14)$$

$$= 0.8848$$

Answer



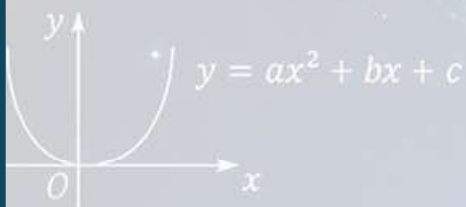
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 2

Q.2) solve the system of linear eqⁿ by using gauss seidel method.

$$4x - y + z = 7$$

$$-x + 4y - 2z = -21$$

$$x + y + 4z = 15$$

Solution:-

From eq_n (i)

$$4x - y + z = 7$$

$$x = \frac{-1}{4} (-y + z - 7)$$

From eq_n (ii)

$$-x + 4y - 2z = -21$$

$$y = \frac{-1}{4} (-x - 2z + 21)$$

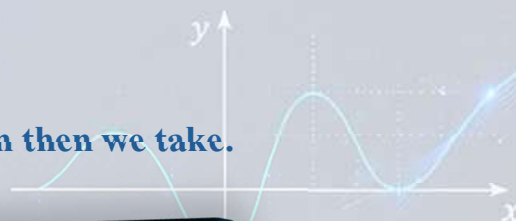
From eq_n (iii)

$$x + y + 4z = 15$$

$$z = \frac{-1}{4} (x + y - 15)$$

It initial approximation (initial solution) is not given then we take.

$$x_0 = 0, \quad y_0 = 0, \quad z_0 = 0$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 40

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

$$x^{k+1} = \frac{-1}{4} (-y^k + z^k - 7)$$


$$y^{k+1} = \frac{-1}{4} (-x^{k+1} - 2z^k + 21)$$

$$z^{k+1} = \frac{-1}{4} (x^{k+1} + y^{k+1} - 15)$$

1st iteration , put k=0

$$x^1 = \frac{-1}{4} (-y^0 - z^0 - 7)$$

$$= \frac{-1}{4} (-0 - 0 - 7)$$

$$= 1.75$$

$$y^1 = \frac{-1}{4} (-x_1 - 2z^0 + 21)$$

$$= \frac{-1}{4} (-1.75 - 2 \times 0 + 21)$$

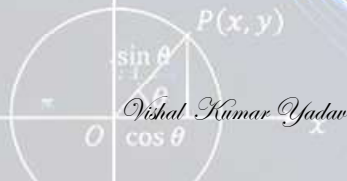
$$= -4.8125$$

$$z^1 = \frac{-1}{4} (x^1 + y^1 - 15)$$

$$= \frac{-1}{4} (1.75 - 4.8125 - 15)$$

$$= 4.5156$$

2nd iteration , put k=1



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

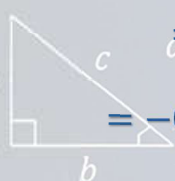
Page no - 41

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

$$x^2 = \frac{-1}{4x} (-y^1 + z^1 - 7)$$



$$= \frac{-1}{4} (4.8125 + 4.5156 - 7)$$

$$= -0.5820$$

$$y^2 = \frac{-1}{4} (-x_2 - 2z^1 + 21)$$

$$= \frac{-1}{4} (-1.75 - 2 \times 4.5156 + 21)$$

$$= -3.1377$$

$$z^2 = \frac{-1}{4} (x^2 + y^2 - 15)$$

$$= \frac{-1}{4} (-0.5820 - 3.1377 - 15)$$

$$= 4.6799$$

Answer

CIMAGE

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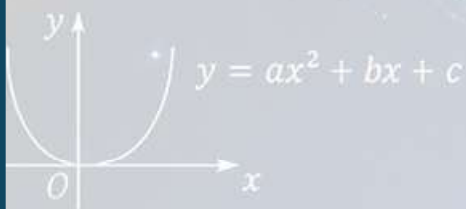
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 3

Q.3) solve the system of linear eqⁿ by using gauss seidel method.

$$5x + y + z = 10$$

$$x + 6y + 2z = 20$$

$$x + y + 7z = 30$$

Solution:-

From eq_n (i)

$$5x + y + z = 10$$

$$x = \frac{1}{5} (-y - z + 10)$$

From eq_n (ii)

$$x + 6y + 2z = 20$$

$$y = \frac{1}{6} (-x - z + 20)$$

From eq_n (iii)

$$x + y + 7z = 30$$

$$z = \frac{1}{7} (-x - y + 30)$$

It initial approximation (initial solution) is not given then we take.

$$\sin^2 \theta + \cos^2 \theta = 1$$

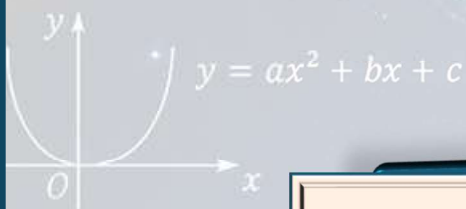
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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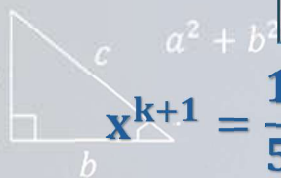
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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$$x_0 = 0, \quad y_0 = 0, \quad z_0 = 0$$



$$x^{k+1} = \frac{1}{5} (-y^k - z^k + 10)$$

$$y^{k+1} = \frac{1}{6} (-x^{k+1} - z^k + 20)$$

$$z^{k+1} = \frac{1}{7} (-x^{k+1} - y^{k+1} + 30)$$

1st iteration , put k=0

$$x^1 = \frac{1}{5} (-y^0 - z^0 + 10)$$

$$= \frac{1}{5} (-0 - 0 + 10)$$

$$= 2$$

$$y^1 = \frac{1}{6} (-x^1 - z^0 + 20)$$

$$= \frac{1}{6} (-2 - 0 + 20)$$

$$= 3$$

$$z^1 = \frac{1}{7} (-x^1 - y^1 + 30)$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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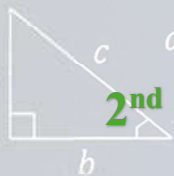
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

$$= \frac{1}{7} (-2 - 3 + 30)$$

$$= 3.5714$$



2nd iteration , put k=0

$$x^2 = \frac{1}{5} (-y^0 - z^0 + 10)$$

$$= \frac{1}{5} (-0 - 0 + 10)$$

$$= 2$$

$$y^2 = \frac{1}{6} (-x^1 - z^0 + 20)$$

$$= \frac{1}{6} (-2 - 0 + 20)$$

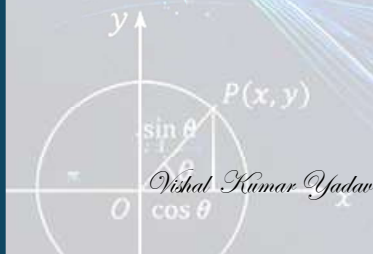
$$= 3$$

$$z^2 = \frac{1}{7} (-x^1 - y^1 + 30)$$

$$= \frac{1}{7} (-2 - 3 + 30)$$

$$= 3.5714$$

Answer

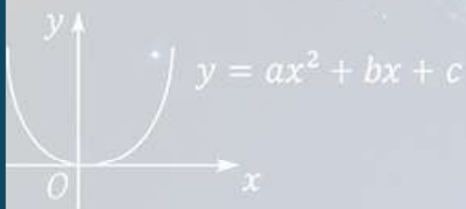


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

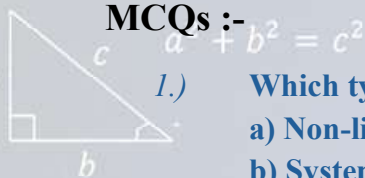
Page no - 45

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

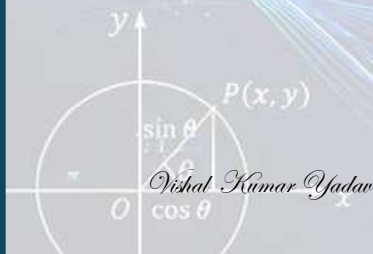


EXERCISE - 1

MCQs :-



- 1.) Which type of system is primarily solved using the Gauss-Seidel method?
 - a) Non-linear algebraic equations
 - b) Systems of linear equations
 - c) Ordinary differential equations
 - d) Partial differential equations.
- 2.) What is a sufficient and necessary condition for the convergence of the Gauss-Seidel method?
 - a) The coefficient matrix must be an identity matrix.
 - b) The coefficient matrix must be strictly diagonally dominant.
 - c) The system must only contain positive values.
 - d) The determinant of the matrix must be zero.
- 3.) In the Gauss-Seidel method, when are the newly calculated values of the variables utilized?
 - a) Only at the end of the entire iteration
 - b) Immediately in the calculation of the subsequent variables in the same iteration
 - c) Only after every other variable fails to change
 - d) Never, it only uses values from the previous iteration.
- 4.) How does the convergence speed of the Gauss-Seidel method generally compare to the Jacobi method?
 - a) It takes roughly the same number of iterations
 - b) It takes more iterations to converge
 - c) It converges faster (requires fewer iterations)
 - d) It never converges while Jacobi does
- 5.) Which classification best describes the Gauss-Seidel method?
 - a) Direct method
 - b) Iterative method
 - c) Elimination method
 - d) Analytical method.

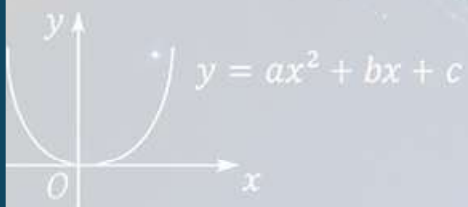


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXERCISE- 2

Solve the system of linear equation by Gauss seidel method.



1.) $27x + 6y - z = 85$
 $6x + 15y + 2z = 72$
 $x + y + 54z = 110$

2.) $2x - y = -1$
 $-x + 4y + 2z = 3$
 $2y + 6z = 5$

3.) $3x + y + z = 10$
 $x + 4y + 2z = 12$
 $x + 2y + 5z = 20$

4.) $4x + y + z = 9$
 $x + 3y + z = 10$
 $x + y + 5z = 20$

5.) $3x + y + z = 11$
 $x + 5y + z = 14$
 $x + y + 7z = 23$

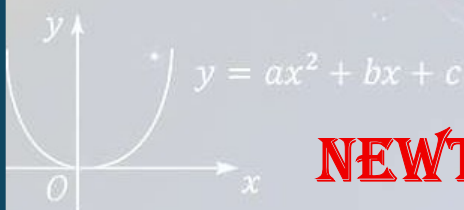


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

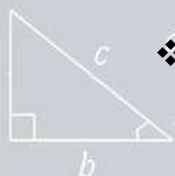
Page no - 47

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



NEWTON - RAPHSON METHOD

THEORY



❖ Explain Newton-Raphson method for finding root of the equation.

Let $f(x) = 0$ be the non-linear equation. If initial approximation is x_0 .

If initial approximation is not given, we tabulated $f(x)$, find the two values of x , $x = a$ and $x = b$ such that

$$f(a) \cdot f(b) < 0$$

then we take only any one value b/w $[a, b]$ as its initial approximation.

Let x_1 is the 1st iteration that is

$$x_1 = x_0 + h(i)$$

Now, we expand $f(x)$ by using Taylor series expansion in the neighbour route of x_0 .

With the help of Taylor series expansion we get

$$f(x) = 0$$

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!}f''(x_0) + \dots = 0$$

Neglecting the term containing more than one degree in h ,

$$f(x_0) + hf'(x_0) = 0$$

$$h = -\frac{f(x_0)}{f'(x_0)}$$

Put this value in equation (1)

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Now, x_2 is second iteration



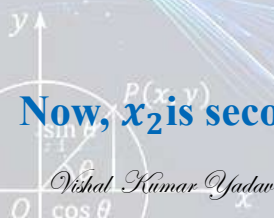
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

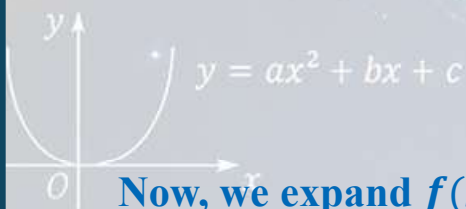
Page no - 48

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$

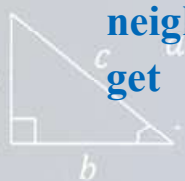


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$$x_2 = x_1 + h(ii)$$

Now, we expand $f(x)$ by using Taylor Series expansion in the neighbour route of x_1 , with the help of Taylor Series expansion we get



$$f(x) = f(x_1) + hf'(x_1) + \frac{h^2}{2!}f''(x_1) + \dots$$

$$f(x_1) + hf'(x_1) + \frac{h^2}{2!}f''(x_1) + \dots = 0$$

Neglecting the term containing more than one degree,

$$f(x_1) + hf'(x_1) = 0$$

$$h = -\frac{f(x_1)}{f'(x_1)}$$

Put the value in equation

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Continue the process

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$n + 1$ denotes no of iteration.

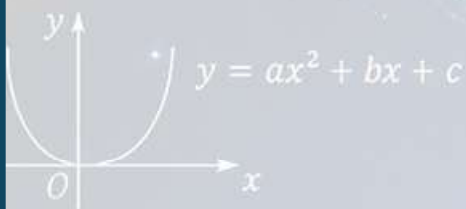


$$\sin^2 \theta + \cos^2 \theta = 1$$

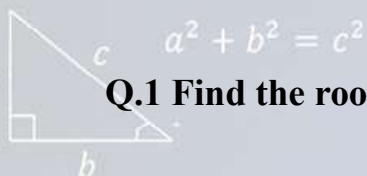
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL -1



Q.1 Find the root of the equation

$$f(x) = x^3 - x - 2$$

be a non-linear eq. its initial approximation is not given be tabulated $f(x)$

Solution:-

Let $x = 1$ and $x = 2$

X	0	1	2
F(x)	-2	-2	4

Such that,

its root lies b/w $]1,2[$

Now,

we take

$$x_0 = 1.5$$

we use initial approximation.

Given that

$$f(x) = x^3 - x - 2$$

$$f'(x) = 3x^2 - 1$$

$$x_0 = 1.5$$

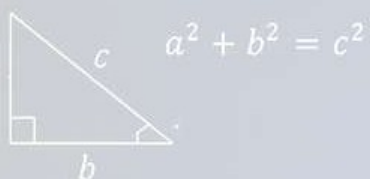
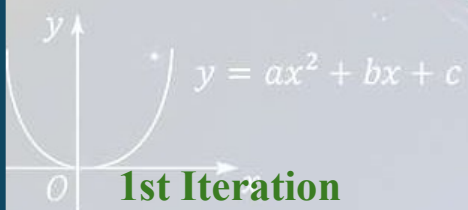


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 50

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$= 1.5 - \frac{f(1.5)}{f'(1.5)}$$

$$= 1.5 + \frac{0.125}{5.75}$$

$$= 1.5217$$

2nd Iteration

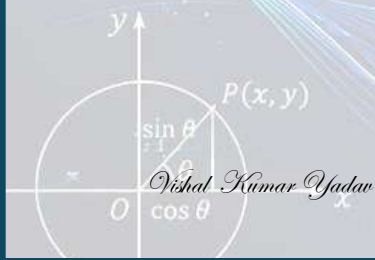
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.5217 - \frac{f(1.5217)}{f'(1.5217)}$$

$$= 1.5217 - \frac{0.0019}{5.9467}$$

$$= 1.5210$$

CIMAGE *Answer*
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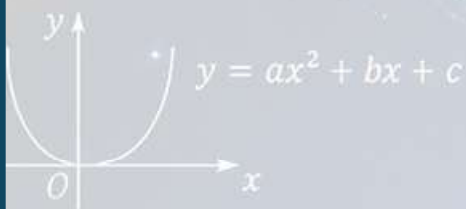


$$\sin^2 \theta + \cos^2 \theta = 1$$

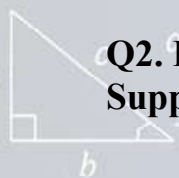
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 51

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 2



Q2. Find $\sqrt{12}$ to five places of decimal by Newton Raphson method
Suppose,

$$x = \sqrt{12}$$

Solution:-

Both side square

$$x^2 = 12$$

$$x^2 - 12 = 0$$

$$f(x) = x^2 - 12 = 0$$

$$f'(x) = 2x$$

Newton-Raphson Formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n + \frac{12 - x_n^2}{2x_n}$$

$$x_{n+1} = \frac{x_n + \frac{12}{x_n}}{2}$$

Find the nearest perfect square

$$3^2 = 9, 4^2 = 16$$

$$9 < 12 < 16$$

12 lies between 9 and 16.

$$\sqrt{12}$$

lies between 3 and 4.

$$x_0 = \frac{3 + 4}{2}$$



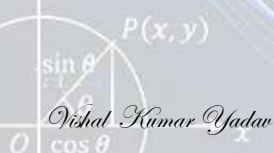
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

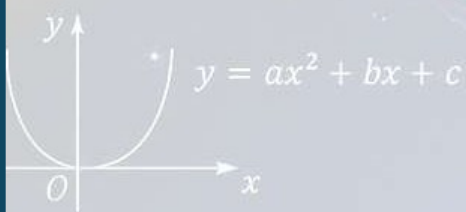
Page no - 52

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



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1st Iteration

$$= \frac{7}{2} = 3.5$$

$$\boxed{x_0 = 3.5}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 3.5 - \frac{(3.5)^2 - 12}{2(3.5)}$$

$$= 3.5 - 0.035714$$

$$x_1 = 3.464286$$

2st Iteration

$$= 3.464286 - \frac{(3.464286)^2 - 12}{2(3.464286)}$$

$$= 3.464286 - \frac{0.001276}{6.928572}$$

$$= 3.464286 - 0.000184$$

$$\boxed{x_2 = 3.464102}$$

3rd Iteration

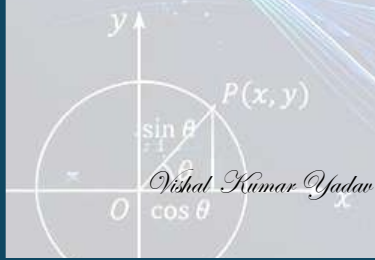
$$= 3.464102 - \frac{(3.464102)^2 - 12}{2(3.464102)}$$

$$= 3.464102 - \frac{0.000003856}{6.928204}$$

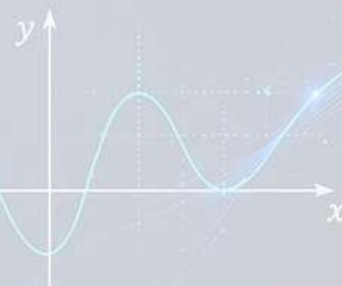
$$= 3.464102 - 0.000000556$$

$$\boxed{x_3 = 3.464101444}$$

Answer



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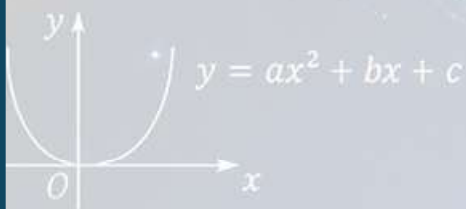


$$\sin^2 \theta + \cos^2 \theta = 1$$

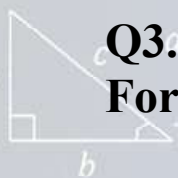
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 3



Q3. Find $\sqrt{16}$ two places of decimal by Newton-Raphson Formula

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Solution:-

Let

$$x = \sqrt{16}$$

$$x^2 = 16$$

$$x^2 - 16 = 0$$

$$f(x) = x^2 - 16$$

$$f'(x) = 2x$$

So,

$$x_0 = 4$$

1st iteration

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 4 - \frac{(4)^2 - 16}{2(4)}$$

$$= 4 - \frac{16 - 16}{8}$$

$$= 4 - \frac{0}{8}$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

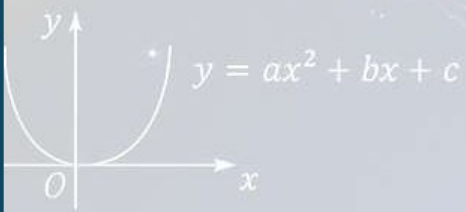
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 54

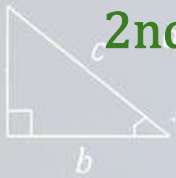
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$





$$x_1 = 4$$



2nd iteration

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 4 - \frac{(4)^2 - 16}{2(4)} \\ &= 4 - \frac{16 - 16}{8} \\ &= 4 - \frac{0}{8} \\ x_2 &= 4 \end{aligned}$$

$$\sqrt{16} = 4.0000$$

Answer

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$$\sin^2 \theta + \cos^2 \theta = 1$$

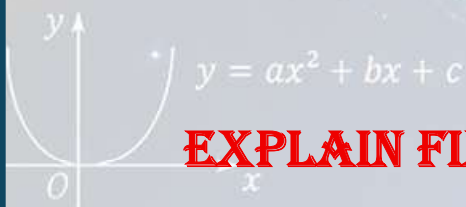
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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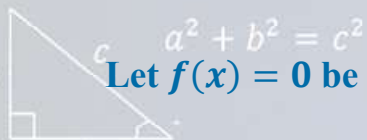
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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EXPLAIN FIXED POINT OR ITERATION METHOD (THEORY)



Let $f(x) = 0$ be the given non-linear equation.

If initial approximation is not given, we tabulate $f(x)$ and find two values of x , $x = a$ and $x = b$ such that

$$f(a) \cdot f(b) < 0$$

Its root lies between open interval $[a, b]$.

We can take any one value which lies between open interval $[a, b]$ and make x_0 as initial root.

We expand

$$f(x) = 0$$

in the form of

$$x = \phi(x)$$

such that

$$\phi'(x_0) < 0$$

and

$$\phi'(x_0) \in (-1, 1)$$

First iteration

$$x_1 = \phi(x_0)$$

Second iteration

$$x_2 = \phi(x_1)$$

Third iteration

$$x_3 = \phi(x_2)$$

and n^{th} iteration

$$x_{n+1} = \phi(x_n)$$



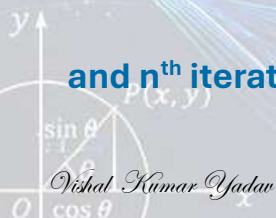
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

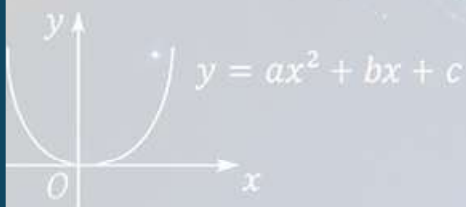
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



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EXERCISE - 1



Q1. Iterative Method is used to find:

- (A) Area
- (B) Derivative
- (C) Root of equation
- (D) Matrix

Q2. The basic form of Iterative Method is:

- (A) $f(x) = 0$
- (B) $x = \phi(x)$
- (C) $y = mx + c$
- (D) $ax + b = 0$

Q3. The convergence condition is:

- (A) $|\phi'(x)| > 1$
- (B) $|\phi'(x)| = 1$
- (C) $|\phi'(x)| < 1$
- (D) $|\phi'(x)| = 0$

Q4. First iteration is:

- (A) $x_0 = \phi(x_1)$
- (B) $x_1 = \phi(x_0)$
- (C) $x_2 = \phi(x_1)$
- (D) $x_n = \phi(x)$

Q5. Second iteration is:

- (A) $x_2 = \phi(x_1)$
- (B) $x_2 = \phi(x_0)$
- (C) $x_1 = \phi(x_2)$
- (D) None



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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 57

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

EXERCISE - 2

1. Find the root of the equation $x = \cos x$ using Iteration method. Take initial approximation $x_0 = 0.5$.

Solve $x = e^{-x}$ using Fixed Point Iteration method.

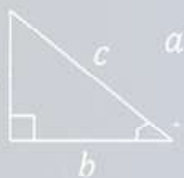
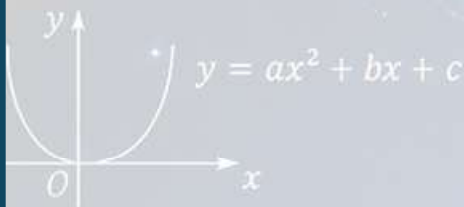
2. Find the root of $x = \sqrt{1+x}$ using Iteration method.

3. Solve $x = \frac{1}{1+x}$ using Iteration method.

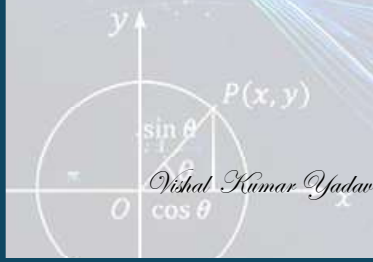
4. Find the root of $x = \frac{x^2+4}{5}$ using Iteration method.

5. Solve $x = \sqrt{2x+3}$ using Fixed Point Iteration method.

6. Find the root of $x = \ln(3x+1)$ using Iteration method.



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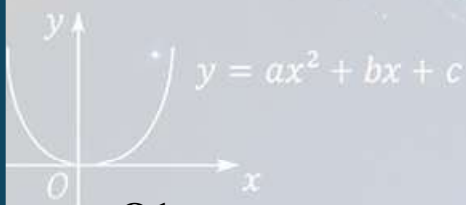


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

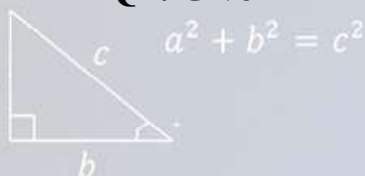
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Q1. Given



$$x = \frac{1}{1+x} + \frac{1}{3}$$

Iteration Formula:

$$x_{n+1} = \frac{1}{1+x_n} + \frac{1}{3}$$

Take

$$x_0 = 1$$

Solution:-

Iterations

$$x_1 = \frac{1}{1+1} + \frac{1}{3} = 0.5000 + 0.3333 = 0.8333$$

2ⁿ Iteration

$$x_2 = \frac{1}{1+0.8333} + \frac{1}{3} = 0.5455 + 0.3333 = 0.8788$$

3rd Iteration

$$x_3 = \frac{1}{1+0.8788} + \frac{1}{3} = 0.5323 + 0.3333 = 0.8656$$

4th Iteration

$$x_4 = \frac{1}{1+0.8656} + \frac{1}{3} = 0.5360 + 0.3333 = 0.8693$$

5th Iteration

$$x_5 = \frac{1}{1+0.8693} + \frac{1}{3} = 0.5350 + 0.3333 = 0.8683$$

6th Iteration

$$x_6 = \frac{1}{1+0.8683} + \frac{1}{3} = 0.5352 + 0.3333 = 0.8685$$

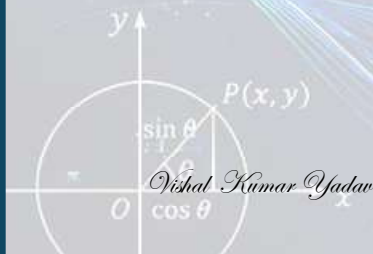
Answer

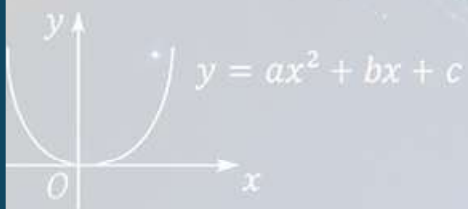
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

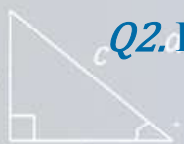
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





PRACTICAL - 2



Q2. Find the root of

$$x = \cos x$$

Take

$$x_0 = 0.5$$

Iteration Formula:

$$x_{n+1} = \cos x_n$$

Iterations

$$\begin{aligned} x_1 &= \cos(0.5) \\ &= 0.8776 \end{aligned}$$

2nd Iteration

$$\begin{aligned} x_2 &= \cos(0.8776) \\ &= 0.6390 \end{aligned}$$

3rd Iteration

$$\begin{aligned} x_3 &= \cos(0.6390) \\ &= 0.8027 \end{aligned}$$

4th Iteration

$$\begin{aligned} x_4 &= \cos(0.8027) \\ &= 0.6948 \end{aligned}$$

5th Iteration

$$\begin{aligned} x_5 &= \cos(0.6948) \\ &= 0.7682 \end{aligned}$$

6th Iteration

$$\begin{aligned} x_6 &= \cos(0.7682) \\ &= 0.7192 \end{aligned}$$

$$x = 0.739$$

Answer

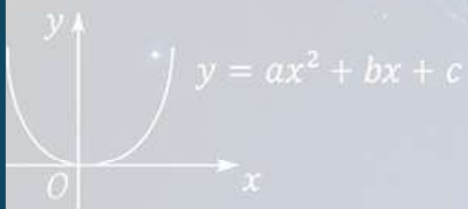
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

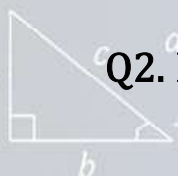
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





PRACTICAL - 3



Q2. Find the root of

$$x = \frac{x + 2}{4}$$

Take

$$x_0 = 1$$

Solution:-

Iteration Formula:

$$x_{n+1} = \frac{x_n + 2}{4}$$

1st Iterations

$$x_1 = \frac{1 + 2}{4} = 0.7500$$

2nd Iteration

$$x_2 = \frac{0.7500 + 2}{4} = 0.6875$$

3rd Iteration

$$x_3 = \frac{0.6875 + 2}{4} = 0.6719$$

4th Iteration

$$x_4 = \frac{0.6719 + 2}{4} = 0.6680$$

Answer

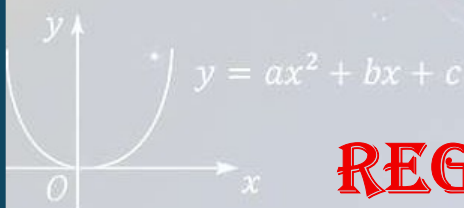


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

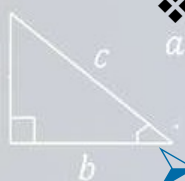
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



REGULA FALSI METHOD

❖ Explain Regula Falsi Method for finding root of the equation.



➤ Let $f(x)=0$ be a non linear eqn.

If initial approximation is given $[x_0, x_1]$

If initial approximation is not given be tabulated $f(x)$ and find two value of

$$x = x_0 \text{ \& \; } x = x_1,$$

Such that,

$$f(x_0) \cdot f(x_1) < 0$$

$[x_0, x_1]$ use as initial approximation

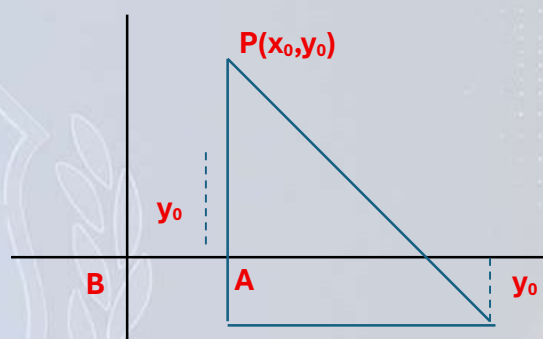
This method depends on the principle of very small distance the graph of smooth curve

In $\triangle PNB \sim \triangle PCD$

$$AB = \frac{CD \cdot PA}{PC}$$

$$CD = AE$$

$$= \frac{PA \cdot AE}{PA + AC}$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

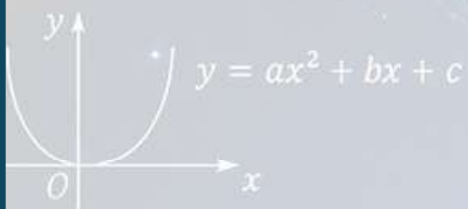
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



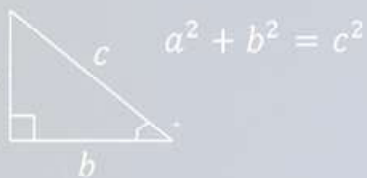
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$$= \frac{PA(OE - OA)}{PA + ED}$$

$$AC = ED$$

$$AB = \frac{|y_0|(x_1 - x_0)}{|y_0| + |y_1|}$$



B is the approximation root of the equation.

$$OB = OA + AB$$

$$x_2 = x_0 + \frac{|y_0|(x_1 - x_0)}{|y_0| + |y_1|}$$

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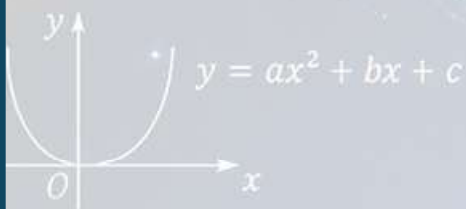


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL -1

Q1. Find the root of the equation

$$x^3 - x - 3 = 0$$

by using Regula-Falsi method.

Solution:-

Let

$$f(x) = x^3 - x - 3$$

if initial approximation is not given we tabulated

X	1	2
F(x)	-3	3

$$f(1) \cdot f(2) < 0$$

its root lies b/w]1,2[
]1,2[we can use as initial approximation.

By → Regula-Falsi

1st Iteration

$$\begin{aligned} x_2 &= x_0 + \frac{|y_0| (x_1 - x_0)}{|y_0| + |y_1|} \\ &= 1 + \frac{|3| (2 - 1)}{|-3| + |3|} \\ x_2 &= 1.5 \end{aligned}$$

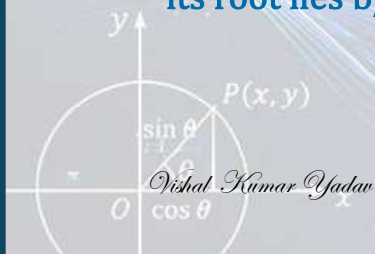
$$f(1.5) = (1.5)^3 - 1.5 - 3$$

$$= -1.125$$

$$f(1.5) \cdot f(2) < 0$$

its root lies b/w]1.5,2[

Answer

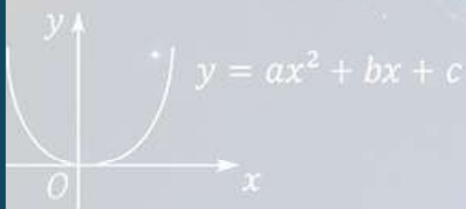


$$\sin^2 \theta + \cos^2 \theta = 1$$

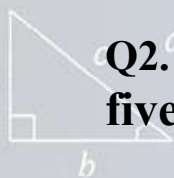
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 64

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL -2



Q2. Compute the real root of $x \log_{10} x - 1.2 = 0$ correct to five decimal places by Regula Falsi

Solution:-

Let

$$f(x) = x \log_{10} x - 1.2 = 0$$

if initial approximation are not given we tabulated

X	2	3
F(x)	-0.5980	0.2313

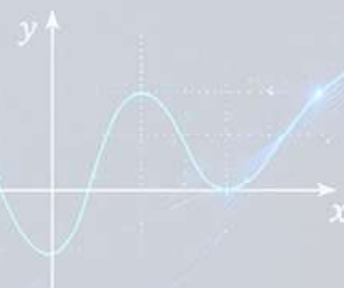
Since,

$$f(2) \cdot f(3) < 0$$

its root lies b/w]2,3[

By Regula false method

$$\begin{aligned}
 x &= x_0 + \frac{|y_0| (x_1 - x_0)}{|y_0| + |y_1|} \\
 &= 2 + \frac{|0.5980| (3 - 2)}{|0.5980| + |0.2313|} \\
 &= 2 + 0.7211 \\
 x_1 &= 2.7211
 \end{aligned}$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =





$$y = ax^2 + bx + c$$

Now,



$$a^2 + b^2 = c^2$$

$$f(2.7211) = 2.7211 \log_{10}(2.7211) - 1.2$$

$$= -0.0172$$

Since

$$f(2.7211) \cdot f(3) < 0$$

2nd

$$x_3 = 2.7211 + \frac{|0.0172| (3 - 2.7211)}{|0.0172| + |0.2313|}$$

$$= 2.7211 + 0.0193$$

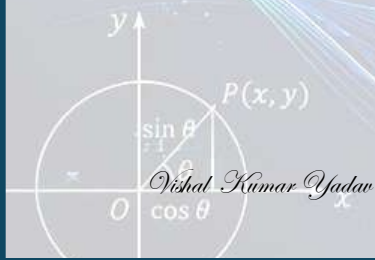
Now

$$x_3 = 2.7404$$

$$f(2.7404) = 2.7404 \log_{10}(2.7404) - 1.2$$

$$= -0.0002$$

CIMAGE *Answer*
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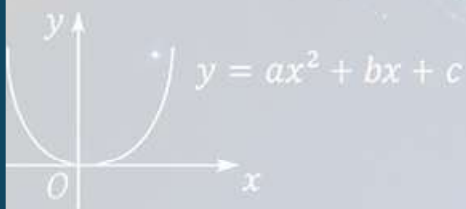


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 66

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 3

Q3. Find the root of $(x)=x^3-x-1$ by Regula falsi method correct to 3 iteration.

Solution:-

X	1	2
$F(x)$	-1	5

Root lies between

$$\begin{aligned} &(1, 2) \\ x_0 &= 1, x_1 = 2 \\ y_0 &= f(1) = -1 \\ y_1 &= f(2) = 5 \end{aligned}$$

Formula

$$x = x_0 + \frac{|y_0| (x_1 - x_0)}{|y_0| + |y_1|}$$

Iteration 1

$$x_2 = 1 + \frac{1(2 - 1)}{1 + 5}$$

$$x_2 = 1 + \frac{1}{6}$$

$$x_2 = 1.1667$$

$$f(1.1667) = (1.1667)^3 - 1.1667 - 1$$

$$= 1.5879 - 1.1667 - 1$$

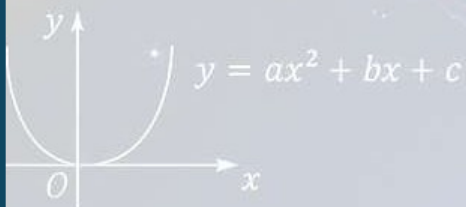
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





$$= -0.5788$$

It's root lies between



$$a^2 + b^2 = c^2$$

$$[1.1667, 2]$$

Iteration 2

$$x_3 = 1.1667 + \frac{0.5788(2 - 1.1667)}{0.5788 + 5}$$

$$= 1.1667 + \frac{0.4823}{5.5788}$$

$$= 1.2532$$

$$f(1.2532) = (1.2532)^3 - 1.2532 - 1$$

$$= 1.9682 - 1.2532 - 1$$

$$= -0.2850$$

It's root lies between

$$[1.2532, 2]$$

Iteration 3

$$x_4 = 1.2532 + \frac{0.2850(2 - 1.2532)}{0.2850 + 5}$$

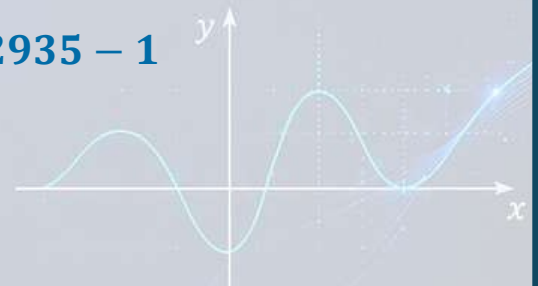
$$= 1.2532 + \frac{0.2128}{5.2850}$$

$$= 1.2935$$

$$f(1.2935) = (1.2935)^3 - 1.2935 - 1$$

$$= 2.1648 - 1.2935 - 1$$

$$= -0.1287$$



Answer



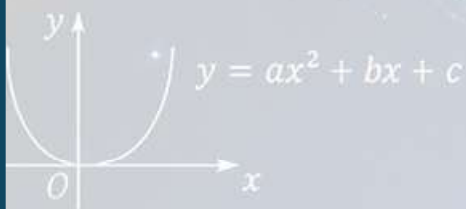
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 68

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 4

Q4. Find the root of $f(x) = x^3 - 2$ using Regula Falsi Method up to 3 iterations.

Solution:-

X	1	2
F(x)	-1	6

Iteration 1

$$x_2 = 1 + \frac{1(2 - 1)}{1 + 6}$$

$$x_2 = 1.142857$$

$$f(1.142857) = 1.142857^3 - 2 = -0.5073$$

Iteration 2

$$x_3 = 1.142857 + \frac{0.5073(2 - 1.142857)}{0.5073 + 6}$$

$$x_3 = 1.2085$$

$$f(1.2085) = 1.2085^3 - 2 = -0.2352$$

Iteration 3

$$x_4 = 1.2085 + \frac{0.2352(2 - 1.2085)}{0.2352 + 6}$$

$$x_4 = 1.2383$$

$$f(1.2383) = 1.2383^3 - 2 = -0.0996$$

$$\boxed{x = 1.2383}$$

Answer

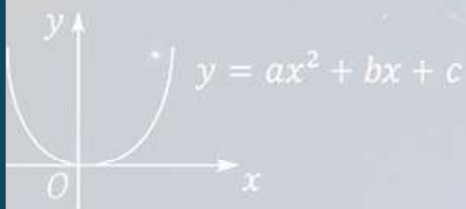
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 69

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





(EXERCISE - 1)



1. Q1. Regula–Falsi Method is used to find:

- (A) Area under a curve
- (B) Root of a non-linear equation
- (C) Matrix inverse
- (D) Derivative

Q2. Regula–Falsi Method is also known as:

- (A) Mid-Point Method
- (B) Fixed Point Method
- (C) False Position Method
- (D) Euler Method

2. Q3. The condition for applying Regula–Falsi Method is:

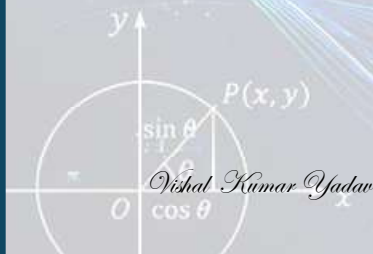
- (A) $f(x_0) + f(x_1) = 0$
- (B) $f(x_0) - f(x_1) = 0$
- (C) $f(x_0) \times f(x_1) < 0$
- (D) $f(x_0) \times f(x_1) > 0$

3. Q4. Regula–Falsi Method is based on:

- (A) Numerical Differentiation
- (B) Linear Interpolation
- (C) Matrix Method
- (D) Taylor Series

4. Q5. In Regula–Falsi Method, the root always lies:

- (A) Outside the interval
- (B) At infinity
- (C) Within the selected interval
- (D) None of these

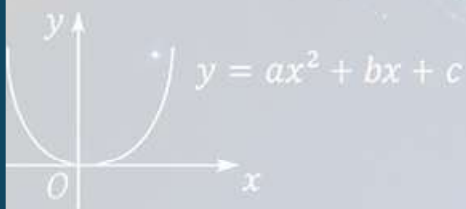


$$\sin^2 \theta + \cos^2 \theta = 1$$

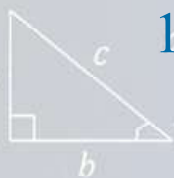
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 70

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 2)



1. Find the root of the non-linear equation $f(x) = x^3 - x - 2 = 0$ using Regula Falsi method.

2. Solve

$f(x) = x^3 - 4x - 9 = 0$ using Regula Falsi method.

3. Find a real root of

$f(x) = x^2 - 4x - 10 = 0$ using Regula Falsi method.

4. Determine the root of

$f(x) = x^3 - 2x - 5 = 0$.

5. Solve

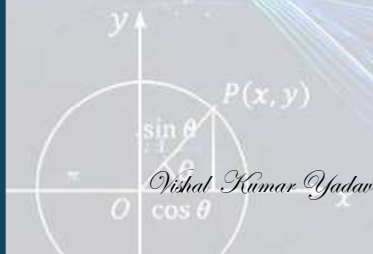
$f(x) = x^3 + x^2 - 3x - 3 = 0$ using Regula Falsi method.

6. Find root of

$f(x) = x^3 - 7x + 6 = 0$.

7. Find the root of

$f(x) = x^2 - x - 6 = 0$.

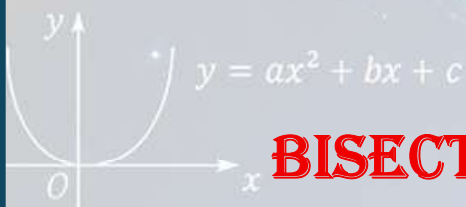


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 71

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



BISECTION (BOLZANO) METHOD

Let $f(x) = 0$ be a non-linear equation. If initial approximation/root is given $[a, b]$, if initial approximations are not given, be tabulated fx and find the value of x .
and find the values $x = a$ and $x = b$
Such that,

$$f(a) \cdot f(b) < 0$$

and using the interval $[a, b]$ as initial approximation, find the value of x by Bi-section.

1st Iteration

$$X_1 = \frac{a+b}{2}$$

If

$$F(x_1) = 0$$

then x_1 is the root of the equation.

$$F(x_1) = 0$$

Find the root of the equation

$$f(x) = x^3 - x - 2$$

by the non-linear equation if initial approximations are not given.

$$f(x) = x^3 - x - 2$$

Since initial approximations are not given, we tabulate $f(x)$

0	1	2
-2	-2	4

At

$$x = 1, x = 2$$

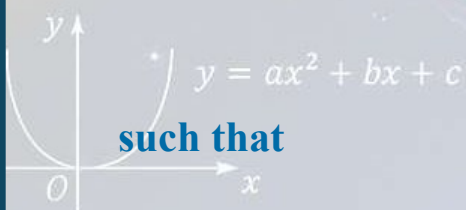
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 72

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





$$f(1) \cdot f(2) < 0$$

Its root lies between (1, 2).
By Bi-Section
1st Iteration

$$x_1 = \frac{1 + 2}{2}$$

$$x_1 = 1.5$$

$$\begin{aligned} f(1.5) &= (1.5)^3 - (1.5) - 2 \\ &= 3.375 - 1.5 - 2 \\ &= -0.125 < 0 \end{aligned}$$

Since

$$f(1.5) \cdot f(2) < 0$$

root lies between

$$(1.5, 2)$$

2nd Iteration

$$x_2 = \frac{1.5 + 2}{2}$$

$$x_2 = 1.75$$

$$\begin{aligned} f(1.75) &= (1.75)^3 - 1.75 - 2 \\ &= 5.359375 - 1.75 - 2 \\ &= 1.609375 > 0 \end{aligned}$$

Since

$$f(1.5) \cdot f(1.75) < 0$$

root lies between

$$(1.5, 1.75)$$

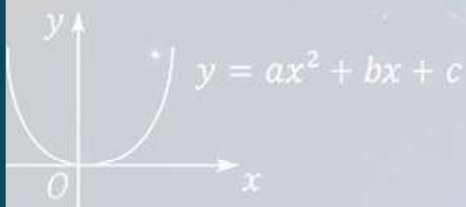


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 73

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Q1. How that equation $xe^x - 1$ has a root in $(\frac{1}{2}, 1)$ and find the approximate to the root within 10^{-1} using bisection method

Solution:-

$$f(x) = xe^x - 1$$

At

$$x = \frac{1}{2} = 0.5$$

$$f(0.5) = 0.5 \times e^{0.5} - 1 \\ = -0.1756393$$

At

$$x = 1$$

$$f(1) = 1 \times e^1 - 1$$

1st iteration

$$\frac{\frac{1}{2} + 1}{2} = 0.75$$

$$f(0.75) = 0.75e^{0.75} - 1 \\ = 0.587750$$

Since,

its root lies b/w $]0.5, 0.75[$

2nd iteration

$$\frac{0.75 + 1}{2} = 0.875$$

$$f(0.875) = 0.875e^{0.875} - 1 \\ = 1.09901$$

Since,

its root lies b/w $]0.75, 0.875[$

3rd iteration

$$\frac{0.75 + 0.875}{2} = 0.8125$$

$$f(0.8125) = 0.8125e^{0.8125} - 1 \\ = 0.83099$$

Since,

its root lies b/w $]0.8125, 0.875[$

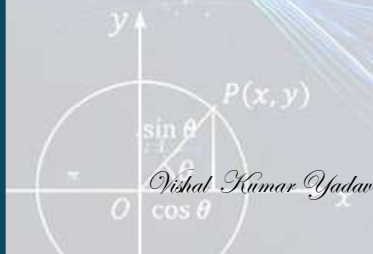
Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

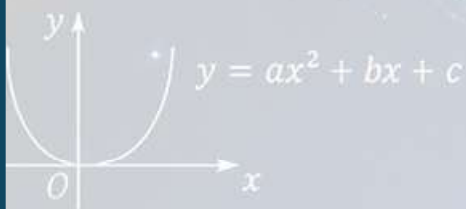
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 74

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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PRACTICAL - 2

Q2. Find the root of the equation $x^3 - 4x - 9 = 0$ using Bisection Method

Solution:-

Let

$$f(x) = x^3 - 4x - 9$$

x	2	3
$F(x)$	-9	6

Since

$$f(2) \cdot f(3) < 0$$

Root lies between 2 and 3.

Iteration 1

$$x = \frac{2 + 3}{2}$$

$$= 2.5$$

$$f(2.5) = (2.5)^3 - 4(2.5) - 9$$

$$= 15.625 - 10 - 9$$

$$= -3.375$$

Root lies between 2.5 and 3.

Iteration 2

$$x = \frac{2.5 + 3}{2}$$

$$= 2.75$$

$$f(2.75) = 20.796875 - 11 - 9$$

$$= 0.796875$$

Root lies between 2.5 and 2.75.

Iteration 3

$$x = \frac{2.5 + 2.75}{2}$$

$$= 2.625$$

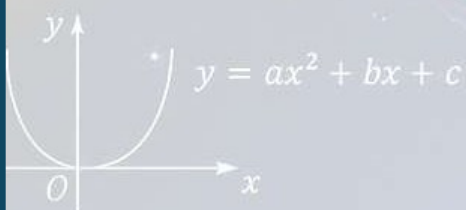


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

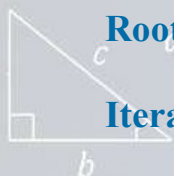
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$f(2.625) = 18.0879 - 10.5 - 9 \\ = -1.4121$$

Root lies between 2.625 and 2.75.



Iteration 4

$$x = \frac{2.625 + 2.75}{2} \\ = 2.6875 \\ f(2.6875) = -0.3391$$

Root lies between 2.6875 and 2.75.

Iteration 5

$$x = \frac{2.6875 + 2.75}{2} \\ = 2.71875 \\ f(2.71875) = 0.2209$$

Root lies between 2.6875 and 2.71875.

Iteration 6

$$x = \frac{2.6875 + 2.71875}{2} \\ = 2.703125 \\ f(2.703125) = -0.0611$$

Root lies between 2.703125 and 2.71875.

Iteration 7

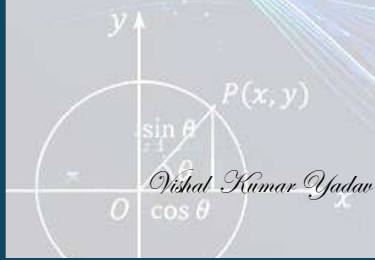
$$x = \frac{2.703125 + 2.71875}{2} \\ = 2.7109375 \\ f(2.7109375) = 0.0793$$

Root lies between 2.703125 and 2.7109375.

$$x = 2.706$$



Answer

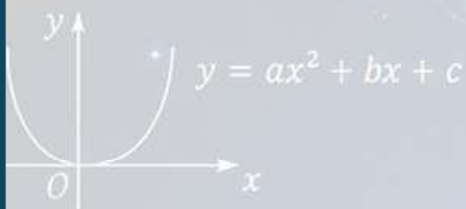


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 76

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 3

Q3. Find the root between 0 and 1 of the equation
 $x^3 + x - 1 = 0$

by Bisection Method

Solution:-

Let

$$f(x) = x^3 + x - 1$$

X	0	1
$F(x)$	-1	1

Since

$$f(0).f(1) < 0$$

Root lies between 0 and 1.

Iteration 1

$$x = \frac{0 + 1}{2}$$

$$= 0.5$$

$$f(0.5) = 0.125 + 0.5 - 1$$

$$= -0.375$$

Root lies between 0.5 and 1.

Iteration 2

$$x_2 = \frac{0.5 + 1}{2}$$

$$= 0.75$$

$$f(0.75) = 0.421875 + 0.75 - 1$$

$$= 0.171875$$

Root lies between 0.5 and 0.75.

Iteration 3



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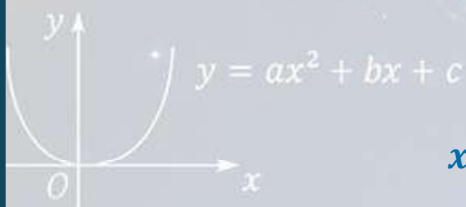


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 77

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

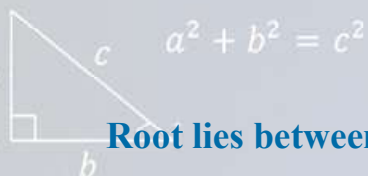


$$x_3 = \frac{0.5 + 0.75}{2}$$

$$= 0.625$$

$$f(0.625) = 0.244141 + 0.625 - 1$$

$$= -0.130859$$



Root lies between 0.625 and 0.75.

Iteration 4

$$x_4 = \frac{0.625 + 0.75}{2}$$

$$= 0.6875$$

$$f(0.6875) = 0.01245$$

Root lies between 0.625 and 0.6875.

Iteration 5

$$x_5 = \frac{0.625 + 0.6875}{2}$$

$$= 0.65625$$

$$f(0.65625) = -0.06113$$

Root lies between 0.65625 and 0.6875.

Iteration 6

$$x_6 = \frac{0.65625 + 0.6875}{2}$$

$$= 0.671875$$

$$f(0.671875) = -0.02483$$

Root lies between 0.671875 and 0.6875.

Iteration 7

$$x_7 = \frac{0.671875 + 0.6875}{2}$$

$$= 0.6796875$$

$$f(0.6796875) = -0.00631$$

Root lies between 0.6796875 and 0.6875.

Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

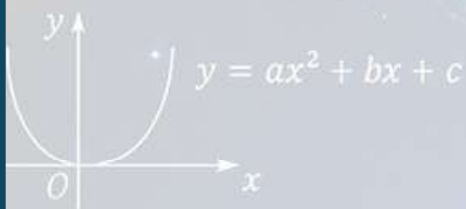
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 78

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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(EXERCISE-1)



Q1. Bisection Method is used to find:

- (A) Area under curve
- (B) Root of a non-linear equation
- (C) Derivative
- (D) Matrix inverse

Q2. The condition for applying Bisection Method is:

- (A) $f(a) + f(b) = 0$
- (B) $f(a) - f(b) = 0$
- (C) $f(a) \times f(b) < 0$
- (D) $f(a) \times f(b) > 0$

Q3. The first approximation in Bisection Method is:

- (A) $x = \frac{a+b}{2}$
- (B) $x = a + b$
- (C) $x = a - b$
- (D) $x = ab$

Q4. Bisection Method is also known as:

- (A) False Position Method
- (B) Mid-Point Method
- (C) Newton-Raphson Method
- (D) Iteration Method

Q5. The main advantage of Bisection Method is:

- (A) Always converges if the interval satisfies the condition
- (B) Requires derivative
- (C) Gives exact root in one step
- (D) Difficult to implement



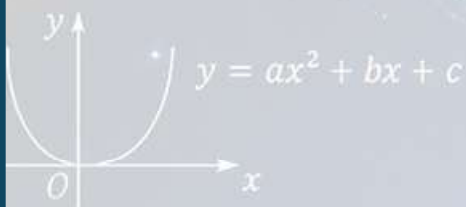
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$$\sin^2 \theta + \cos^2 \theta = 1$$

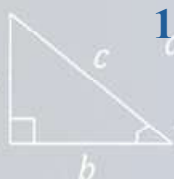
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 79

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE-2)



1. Find the real root of the non-linear equation $f(x) = x^3 - x - 2 = 0$, correct to three decimal places using Bisection method.

2. Find the root of the equation $f(x) = x^3 - 4x - 9 = 0$ using Bisection method, correct to three decimal places.

3. Determine a real root of $f(x) = x^2 - 4x - 10 = 0$ using Bisection method.

4. Find the root of $f(x) = x^3 - 2x - 5 = 0$ by Bisection method.

5. Solve the equation $f(x) = x^3 + x^2 - 3x - 3 = 0$ using Bisection method.

6. Find the root of $f(x) = x^3 - x^2 + 2 = 0$ using Bisection method.

7. Find the real root of $f(x) = x^2 - 3x + 1 = 0$ using Bisection method.



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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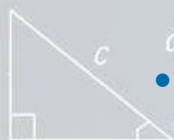
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

NEWTON'S BACKWARD INTERPOLATION

Interpolation



$$a^2 + b^2 = c^2$$

- Interpolation is an art in which we find the polynomial with the help of tabulated values of $f(x)$.

There are two types of interpolation:

- Interpolation at unequal space point.
- Interpolation at equal space point.

Interpolation at unequal space point

Let

x	x_0	x_1	x_2	$\dots$	x_n
$f(x)$	y_0	y_1	y_2	$\dots$	y_n

be the given tabulated values of $f(x)$. If the lengths of any one sub-interval are not equal, then it is called interpolation at unequal space point. Example:

x	1	5	7	8
$f(x)$	2	26	50	65

There are some methods for solving interpolation at unequal space points:

- Lagrange's Method
- Newton Divided Difference
 - Newton Forward Divided Difference
 - Newton Backward Divided Difference

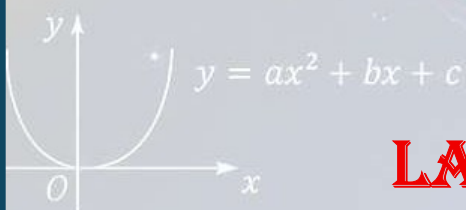


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

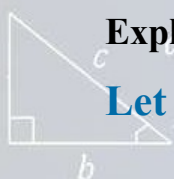
Page no - 81

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



LAGRANGE'S METHOD

Explain Lagrange's method for finding the interpolation polynomial.



Let

$$\begin{array}{ccccccc} x & x_0 & x_1 & x_2 & x_3 & \cdots & x_n \\ f(x) & y_0 & y_1 & y_2 & y_3 & \cdots & y_n \end{array}$$

be the given tabulated values of $f(x)$. Then its Lagrange interpolation polynomial is given by

$$P_n(x) = L_0 y_0 + L_1 y_1 + L_2 y_2 + \cdots + L_n y_n \cdots$$

where,

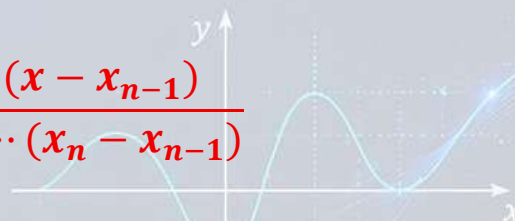
$$L_0 = \frac{(x - x_1)(x - x_2)(x - x_3) \cdots (x - x_n)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3) \cdots (x_0 - x_n)}$$

$$L_1 = \frac{(x - x_0)(x - x_2)(x - x_3) \cdots (x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3) \cdots (x_1 - x_n)}$$

$$L_2 = \frac{(x - x_0)(x - x_1)(x - x_3) \cdots (x - x_n)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3) \cdots (x_2 - x_n)}$$

$\vdots$

$$L_n = \frac{(x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1)(x_n - x_2) \cdots (x_n - x_{n-1})}$$



its put the value in equation (1) and solve we get required polynomial [interpolation polynomial]



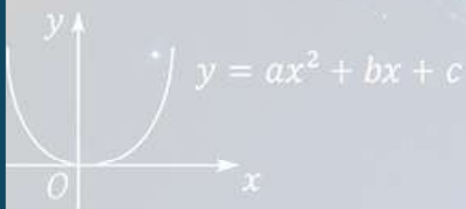
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 82

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =



PRACTICAL - 1

Given the values



x	5	7	11	13	17
y	150	392	1452	2366	5202

Evaluate $f(9)$ using Lagrange's interpolation formula.

Solution:-

By Lagrange's interpolation formula,

$$P_n(x) = L_0y_0 + L_1y_1 + L_2y_2 + L_3y_3 + L_4y_4,$$

where

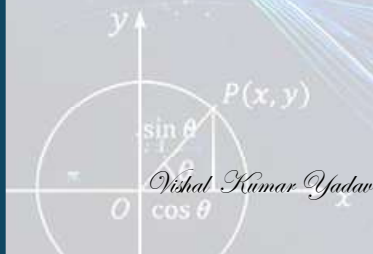
$$L_0 = \frac{(x - x_1)(x - x_2)(x - x_3)(x - x_4)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(x_0 - x_4)}$$

$$= \frac{(9 - 7)(9 - 11)(9 - 13)(9 - 17)}{(5 - 7)(5 - 11)(5 - 13)(5 - 17)}$$

$$= \frac{2(-2)(-4)(-8)}{(-2)(-6)(-8)(-12)}$$

$$= \frac{128}{1152}$$

$$= \frac{1}{9}$$

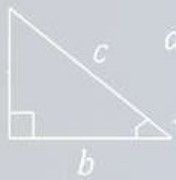
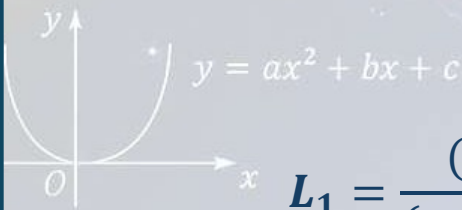


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$a^2 + b^2 = c^2$$

$$L_1 = \frac{(x - x_0)(x - x_2)(x - x_3)(x - x_4)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(x_1 - x_4)}$$

$$= \frac{(9 - 5)(9 - 11)(9 - 13)(9 - 17)}{(7 - 5)(7 - 11)(7 - 13)(7 - 17)}$$

$$= \frac{4(-2)(-4)(-8)}{2(-4)(-6)(-10)}$$

$$= \frac{256}{480}$$

$$= \frac{8}{15}$$

$$L_2 = \frac{(x - x_0)(x - x_1)(x - x_3)(x - x_4)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)(x_2 - x_4)}$$

$$= \frac{(9 - 5)(9 - 7)(9 - 13)(9 - 17)}{(11 - 5)(11 - 7)(11 - 13)(11 - 17)}$$

$$= \frac{4(2)(-4)(-8)}{6(4)(-2)(-6)}$$

$$= \frac{256}{288}$$

$$= \frac{8}{9}$$



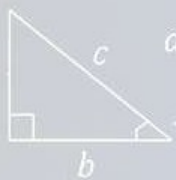
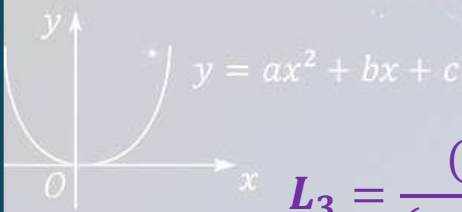
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 84

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$a^2 + b^2 = c^2$$

$$L_3 = \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_4)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)(x_3 - x_4)}$$

$$= \frac{(9 - 5)(9 - 7)(9 - 11)(9 - 17)}{(13 - 5)(13 - 7)(13 - 11)(13 - 17)}$$

$$= \frac{4(2)(-2)(-8)}{8(6)(2)(-4)}$$

$$= \frac{128}{-384}$$

$$= -\frac{1}{3}$$

$$L_4 = \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_3)}{(x_4 - x_0)(x_4 - x_1)(x_4 - x_2)(x_4 - x_3)}$$

$$= \frac{(9 - 5)(9 - 7)(9 - 11)(9 - 13)}{(17 - 5)(17 - 7)(17 - 11)(17 - 13)}$$

$$= \frac{4(2)(-2)(-4)}{12(10)(6)(4)}$$

$$= \frac{64}{2880}$$

$$= \frac{1}{45}$$



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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

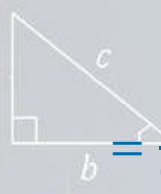
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

Using Lagrange's interpolation formula,



$$P_4(9) = L_0y_0 + L_1y_1 + L_2y_2 + L_3y_3 + L_4y_4$$

$$= \frac{1}{9}(150) + \frac{8}{15}(392) + \frac{8}{9}(1452) - \frac{1}{3}(2366) + \frac{1}{45}(5202).$$

$$= \frac{150}{9} + \frac{3136}{15} + \frac{11616}{9} - \frac{2366}{3} + \frac{5202}{45}$$

$$= \frac{750 + 9408 + 58080 - 35490 + 5202}{45}$$

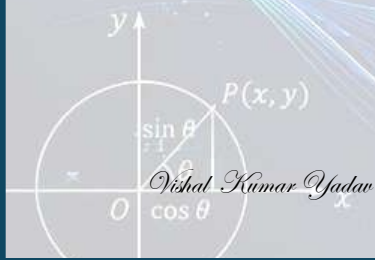
$$= \frac{33750}{45}$$

$$= 750.$$

$$f(9) = 750.$$

CIMAGE
Knowledge . Skills . Values

Answer

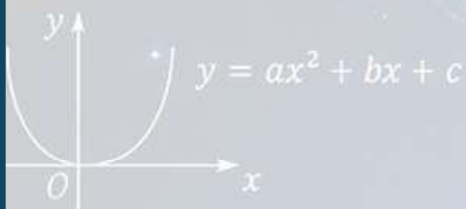


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 2

Given the values

$$a^2 + b^2 = c^2$$

x	5	7	11	13	17
y	150	392	1452	2366	5202

Evaluate $f(9)$ using Lagrange's interpolation formulate.

Solution:-

by Lagrange's

$$P_n(x) = L_0y_0 + L_1y_1 + L_2y_2 + \dots + L_ny_n$$

Where,

$$L_0 \Rightarrow \frac{(x - x_1)(x - x_2)(x - x_3)(x - x_4)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(x_0 - x_4)}$$

$$\Rightarrow \frac{(9 - 7)(9 - 11)(9 - 13)(9 - 17)}{(5 - 7)(5 - 11)(5 - 13)(5 - 17)}$$

$$\Rightarrow \frac{2 \times (-2) \times (-4) \times (-8)}{(-2) \times (-6) \times (-8) \times (-12)}$$

$$\Rightarrow \frac{128}{1152}$$

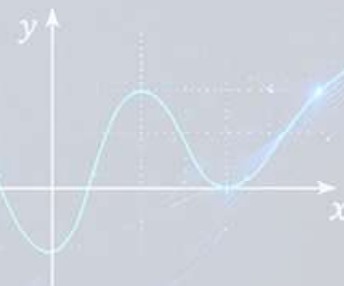
$$\Rightarrow \frac{1}{9} \Rightarrow 16.66$$

$$L_1 \Rightarrow \frac{(x - x_0)(x - x_2)(x - x_3)(x - x_4)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(x_1 - x_4)}$$

$$\Rightarrow \frac{(9 - 5)(9 - 11)(9 - 13)(9 - 17)}{(7 - 5)(7 - 11)(7 - 13)(7 - 17)}$$

$$\Rightarrow \frac{4 \times (-2) \times (-4) \times (-8)}{2 \times (-4) \times (-6) \times (-10)}$$

$$\Rightarrow \frac{256}{480} \Rightarrow \frac{8}{15} \Rightarrow 209.06$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

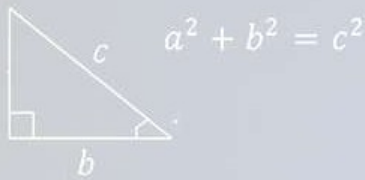
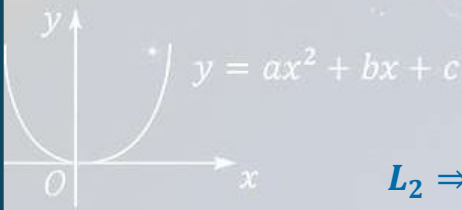
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



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$$L_2 \Rightarrow \frac{(x - x_0)(x - x_1)(x - x_3)(x - x_4)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)(x_2 - x_4)}$$

$$\Rightarrow \frac{(9 - 5)(9 - 7)(9 - 13)(9 - 17)}{(11 - 5)(11 - 7)(11 - 13)(11 - 17)}$$

$$\Rightarrow \frac{4 \times 2 \times (-4) \times (-8)}{6 \times 4 \times (-2) \times (-6)}$$

$$\Rightarrow \frac{256}{288}$$

$$\Rightarrow \frac{8}{9} \Rightarrow 1290.66$$

$$L_3 \Rightarrow \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_4)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)(x_3 - x_4)}$$

$$\Rightarrow \frac{(9 - 5)(9 - 7)(9 - 11)(9 - 17)}{(13 - 5)(13 - 7)(13 - 11)(13 - 17)}$$

$$\Rightarrow \frac{4 \times 2 \times (-2) \times (-8)}{8 \times 6 \times 2 \times (-4)}$$

$$\Rightarrow \frac{128}{-384}$$

$$\Rightarrow -\frac{1}{3} \Rightarrow -788.66$$

$$L_4 \Rightarrow \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_3)}{(x_4 - x_0)(x_4 - x_1)(x_4 - x_2)(x_4 - x_3)}$$

$$\Rightarrow \frac{(9 - 5)(9 - 7)(9 - 11)(9 - 13)}{(17 - 5)(17 - 7)(17 - 11)(17 - 13)}$$

$$\Rightarrow \frac{4 \times 2 \times (-2) \times (-4)}{12 \times 10 \times 6 \times 4}$$

$$\Rightarrow \frac{64}{2880}$$

$$\Rightarrow \frac{1}{45} \Rightarrow 115.6$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 88

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =



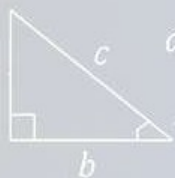
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$$y = ax^2 + bx + c$$

Using Lagrange's Formula,

$$P_n(9) \Rightarrow L_0y_0 + L_1y_1 + L_2y_2 + L_3y_3 + L_4y_4$$



$$a^2 + b^2 = c^2$$

$$\Rightarrow 16.66 \times 150 + 209.06 \times 392 + 1290.66 \times 1452 \\ + (-788.66) \times 2366 + 115.6 \times 5202$$

$$\Rightarrow 2499 + 81951.52 + 1874038.32 \\ + 1865969.56 + 601351.2$$

$$\Rightarrow 4425809.6$$

Answer



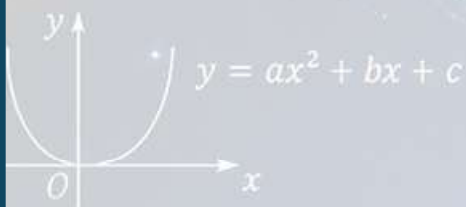
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





PRACTICAL - 3

Q.3 Using Lagrange's Interpolation Formula



Given,

x	30	32	35	38
$y = \tan x$	0.5774	0.6249	0.7002	0.7813

find $\tan 33^\circ$.

Solution:-

The Lagrange interpolation formula is

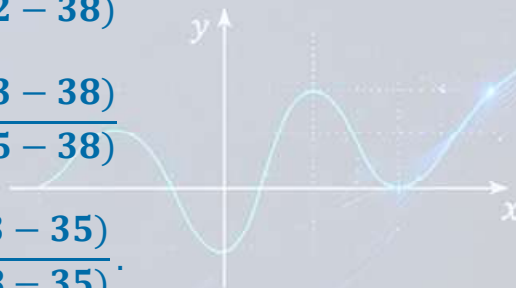
$$y(x) = \sum_{i=0}^n y_i L_i(x),$$

where

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}.$$

Substituting $x = 33$,

$$\begin{aligned} \tan 33^\circ &= 0.5774 \frac{(33 - 32)(33 - 35)(33 - 38)}{(30 - 32)(30 - 35)(30 - 38)} \\ &+ 0.6249 \frac{(33 - 30)(33 - 35)(33 - 38)}{(32 - 30)(32 - 35)(32 - 38)} \\ &+ 0.7002 \frac{(33 - 30)(33 - 32)(33 - 38)}{(35 - 30)(35 - 32)(35 - 38)} \\ &+ 0.7813 \frac{(33 - 30)(33 - 32)(33 - 35)}{(38 - 30)(38 - 32)(38 - 35)}. \end{aligned}$$



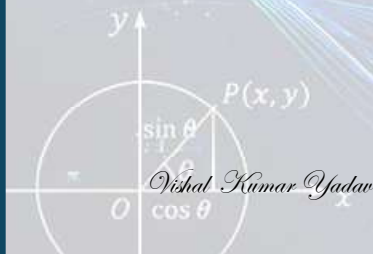
$$\sin^2 \theta + \cos^2 \theta = 1$$

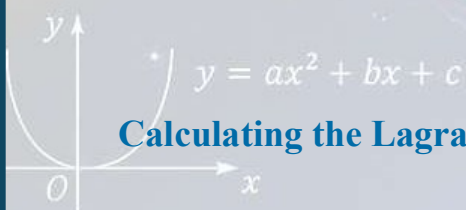
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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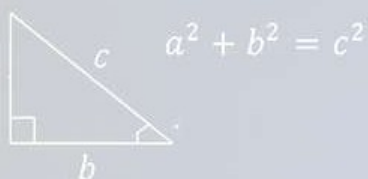
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$





Calculating the Lagrange coefficients,



$$L_0 = \frac{(1)(-2)(-5)}{(-2)(-5)(-8)} = -0.125,$$

$$L_1 = \frac{(3)(-2)(-5)}{(2)(-3)(-6)} = 0.833333,$$

$$L_2 = \frac{(3)(1)(-5)}{(5)(3)(-3)} = 0.333333,$$

$$L_3 = \frac{(3)(1)(-2)}{(8)(6)(3)} = -0.041667.$$

Hence,

$$\begin{aligned} \tan 33^\circ &= 0.5774(-0.125) + 0.6249(0.833333) \\ &\quad + 0.7002(0.333333) - 0.7813(0.041667) \end{aligned}$$

$$\approx -0.072175 + 0.520750 + 0.233400 - 0.032554$$

$$\approx 0.64942.$$

$$\boxed{\tan 33^\circ \approx 0.6494}$$

Answer



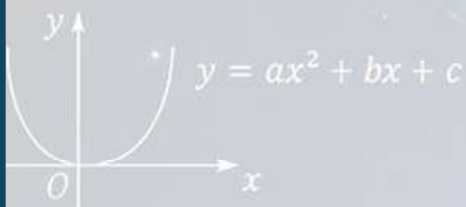
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





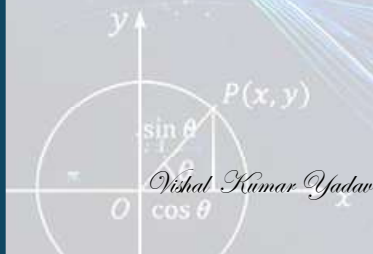
EXERCISE - 1

(Objectives)



$$a^2 + b^2 = c^2$$

1. Lagrange interpolation is mainly used when:
 - a. The intervals between x -values are equal
 - b. The intervals between x -values are unequal
 - c. Only one data point is given
 - d. None of the above
2. The Lagrange interpolation polynomial passes through:
 - a. Some of the given points
 - b. All of the given points
 - c. Only the first and last points
 - d. None of the above
3. The degree of the Lagrange interpolation polynomial is:
 - a. Equal to the number of data points
 - b. One less than the number of data points
 - c. Twice the number of data points
 - d. Independent of data points
4. The Lagrange basis polynomial $L_i(x)$ is defined as:
 - a. Product of terms excluding x_i
 - b. Sum of terms including x_i
 - c. Ratio of differences including x_i
 - d. None of the above
5. Lagrange interpolation is particularly useful when:
 - a. Data points are equally spaced
 - b. Data points are unequally spaced
 - c. Only one data point is given
 - d. None of the above

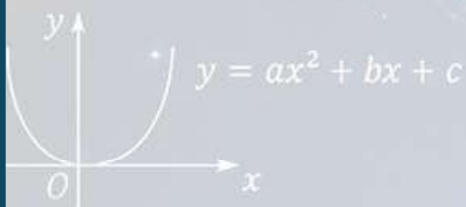


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

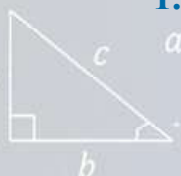
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXERCISE - 2

1. Given the data:



$$a^2 + b^2 = c^2$$

x	$f(x)$
1	2
2	4
3	9

Use Lagrange interpolation to estimate $f(2.5)$.

2. The following values are given:

x	y
0	1
1	3
2	7

Find the value of y when $x = 1.5$.

3. The population of a town is given below:

Year	Population
2000	20000
2002	22000
2005	27000

Estimate the population in the year 2003 using Lagrange interpolation.

4. The values of a function are given as:

x	$f(x)$
2	4
4	16
5	25

Find $f(3)$ using Lagrange interpolation.

5. The following data is given:

x	y
1	1
3	9
6	36

Estimate the value of y when $x = 4$.



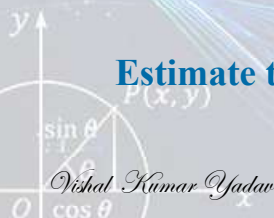
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

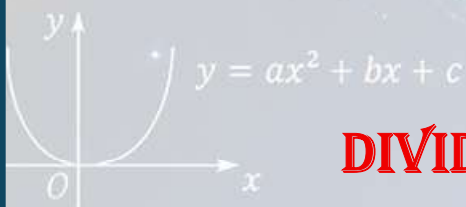
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



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DIVIDED DIFFERENCE METHOD

1st Divided Difference



$$a^2 + b^2 = c^2$$

$$[a, b] = \frac{f(b) - f(a)}{b - a}$$

2nd Divided Difference

$$[a, b, c] = \frac{[b, c] - [a, b]}{c - a}$$

3rd Divided Difference

$$[a, b, c, d] = \frac{[b, c, d] - [a, b, c]}{d - a}$$

Similarly, higher-order divided differences can be obtained.

Table of Divided Differences

x	$f(x)$	$[a, b]$	$[a, b, c]$	$[a, b, c, d, \dots]$
x_0	y_0			
		$[x_0, x_1]$		
x_1	y_1		$[x_0, x_1, x_2]$	
		$[x_1, x_2]$		$[x_0, x_1, x_2, \dots, x_n]$
x_2	y_2	$\vdots$	$[x_{n-2}, x_{n-1}, x_n]$	
$\vdots$	$\vdots$	$[x_{n-1}, x_n]$		
x_n	y_n			



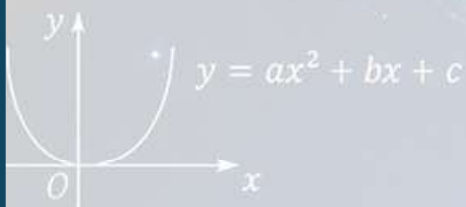
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Find the 3rd divided difference [2,4,9,10] of

$$f(x) = x^3 - 2x$$

Solution:-

1st Divided Difference

Since

$$f(2) = 2^3 - 2(2) = 4, \quad f(4) = 4^3 - 2(4) = 56,$$

$$[2, 4] = \frac{f(4) - f(2)}{4 - 2} = \frac{56 - 4}{2} = 26.$$

$$f(9) = 9^3 - 2(9) = 711,$$

$$[4, 9] = \frac{f(9) - f(4)}{9 - 4} = \frac{711 - 56}{5} = 131.$$

Further,

$$f(10) = 10^3 - 2(10) = 980,$$

therefore

$$[9, 10] = \frac{f(10) - f(9)}{10 - 9} = \frac{980 - 711}{1} = 269.$$

2nd Divided Difference

$$[2, 4, 9] = \frac{[4, 9] - [2, 4]}{9 - 2} = \frac{131 - 26}{7} = 15.$$

$$[4, 9, 10] = \frac{[9, 10] - [4, 9]}{10 - 4} = \frac{269 - 131}{6} = 23.$$

3rd Divided Difference

$$[2, 4, 9, 10] = \frac{[4, 9, 10] - [2, 4, 9]}{10 - 2} = \frac{23 - 15}{8} = 1.$$

$$\boxed{[2, 4, 9, 10] = 1}$$

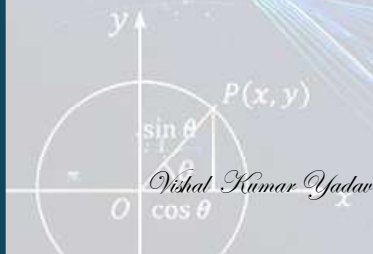
Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

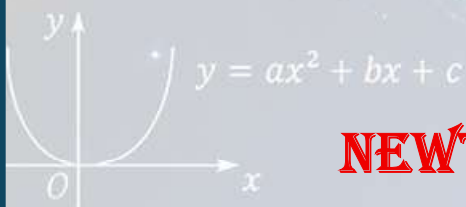
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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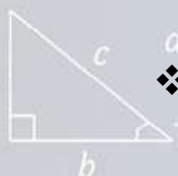
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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NEWTON DIVIDED DIFFERENCE INTERPOLATION



$$a^2 + b^2 = c^2$$

❖ Explain Newton divided difference for finding interpolating polynomial.

Let,

$$f(x) = \begin{matrix} x & x_0 & x_1 & \cdots & x_n \\ y & y_0 & y_1 & \cdots & y_n \end{matrix}$$

be the given tabulated values of $f(x)$. Then the Newton divided difference table is given by

x	$f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\cdots$	$\Delta^n y$
x_0	y_0					
		$[x_0, x_1]$				
x_1	y_1		$[x_0, x_1, x_2]$			
		$[x_1, x_2]$		$[x_0, x_1, x_2, x_3]$		
x_2	y_2		$[x_1, x_2, x_3]$	$\vdots$	$\cdots$	$[x_0, x_1, \dots, x_n]$
		$[x_2, x_3]$	$\vdots$	$[x_{n-3}, x_{n-2}, x_{n-1}, x_n]$		
x_3	y_3	$\vdots$	$[x_{n-2}, x_{n-1}, x_n]$			
$\vdots$	$\vdots$	$[x_{n-1}, x_n]$				
x_n	y_n					

Thus, the forward divided difference interpolating polynomial is given by

$$f(x) = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] + \cdots + (x - x_0)(x - x_1) \cdots (x - x_{n-1})[x_0, x_1, \dots, x_n]$$

y

$\sin \theta$

$\cos \theta$

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$$\sin^2 \theta + \cos^2 \theta = 1$$

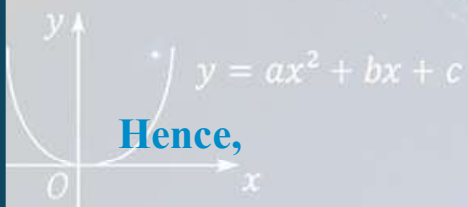
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cos \theta$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



Hence,



The Newton backward divided difference interpolation formula is given by ;

$$f(x) = y_n + (x - x_n)[x_{n-1}, x_n] + (x - x_n)(x - x_{n-1})[x_{n-2}, x_{n-1}, x_n] + \dots + (x - x_n)(x - x_{n-1}) \dots (x - x_1)[x_0, x_1, x_2, \dots, x_n].$$

Hence,

$$P_n(x) = y_n + (x - x_n)[x_{n-1}, x_n] + (x - x_n)(x - x_{n-1})[x_{n-2}, x_{n-1}, x_n] + \dots + (x - x_n)(x - x_{n-1}) \dots (x - x_1)[x_0, x_1, x_2, \dots, x_n].$$

CIMAGE
Knowledge · Skill · Success



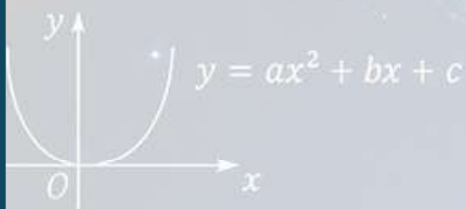
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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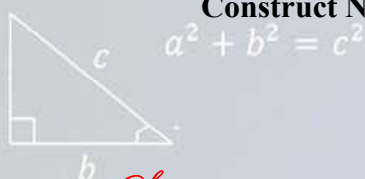
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

Construct Newton's Divided Difference Table for the Following Data

x	5	7	11	13	17
y	150	392	1452	2366	5202



Solution:-

$$x_0 = 5, y_0 = 150$$

$$x_1 = 7, y_1 = 392$$

$$x_2 = 11, y_2 = 1452$$

$$x_3 = 13, y_3 = 2366$$

$$x_4 = 17, y_4 = 5202$$

First Divided Differences

$$[5, 7] = \frac{392 - 150}{7 - 5} = \frac{242}{2} = 121$$

$$[7, 11] = \frac{1452 - 392}{11 - 7} = \frac{1060}{4} = 265$$

$$[11, 13] = \frac{2366 - 1452}{13 - 11} = \frac{914}{2} = 457$$

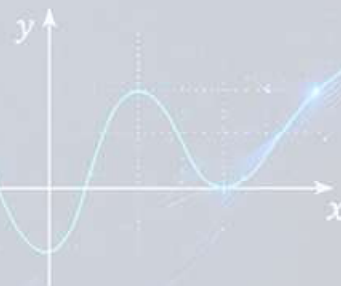
$$[13, 17] = \frac{5202 - 2366}{17 - 13} = \frac{2836}{4} = 709$$

Second Divided Differences

$$[5, 7, 11] = \frac{265 - 121}{11 - 5} = \frac{144}{6} = 24$$

$$[7, 11, 13] = \frac{457 - 265}{13 - 7} = \frac{192}{6} = 32$$

$$[11, 13, 17] = \frac{709 - 457}{17 - 11} = \frac{252}{6} = 42$$



$$\sin^2 \theta + \cos^2 \theta = 1$$

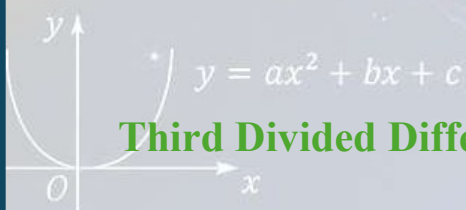
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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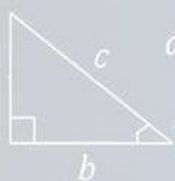
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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Third Divided Differences



$$a^2 + b^2 = c^2$$

$$[5, 7, 11, 13] = \frac{32 - 24}{13 - 5} = \frac{8}{8} = 1$$

$$[7, 11, 13, 17] = \frac{42 - 32}{17 - 7} = \frac{10}{10} = 1$$

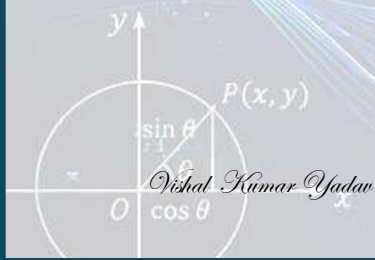
Fourth Divided Difference

$$[5, 7, 11, 13, 17] = \frac{1 - 1}{17 - 5} = \frac{0}{12} = 0$$

Answer

NEWTON'S DIVIDED DIFFERENCE TABLE

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
5	150	121	24	1	0
7	392	265	32	1	
11	1452	457	42		
13	2366	709			
17	5202				



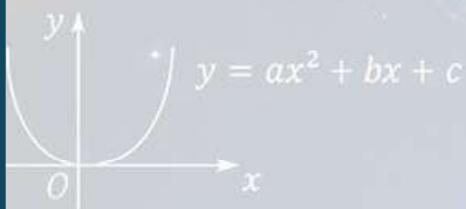
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

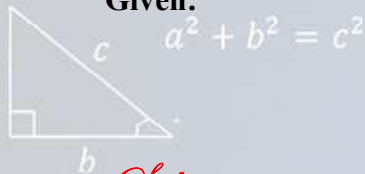
Page no - 99

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 3

Given:



x	0	1	3	4
$f(x)$	1	4	40	85

Solution:-

Construct the Newton divided difference table:

x_i	$f(x_i)$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
0	1	$\frac{4-1}{1-0} = 3$	$\frac{18-3}{3-0} = 5$	$\frac{9-5}{4-0} = \frac{1}{4}$
1	4	$\frac{40-4}{3-1} = 18$	$\frac{45-18}{4-1} = 9$	
3	40	$\frac{85-40}{4-3} = 45$		
4	85			

Using Newton's divided difference interpolation formula,

$$f(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2).$$

Substituting the divided differences,

$$f(x) = 1 + 3(x - 0) + 5(x - 0)(x - 1) + \frac{1}{4}(x - 0)(x - 1)(x - 3).$$

$$f(x) = 1 + 3x + 5x(x - 1) + \frac{1}{4}x(x - 1)(x - 3).$$

Expanding,

$$5x(x - 1) = 5x^2 - 5x,$$

$$\frac{1}{4}x(x - 1)(x - 3) = \frac{1}{4}(x^3 - 4x^2 + 3x) = \frac{1}{4}x^3 - x^2 + \frac{3}{4}x.$$

$$\begin{aligned} f(x) &= 1 + 3x + 5x^2 - 5x + \frac{1}{4}x^3 - x^2 + \frac{3}{4}x \\ &= \frac{1}{4}x^3 + 4x^2 - \frac{5}{4}x + 1. \end{aligned}$$

$$f(x) = \frac{1}{4}x^3 + 4x^2 - \frac{5}{4}x + 1$$

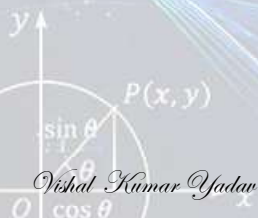
Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

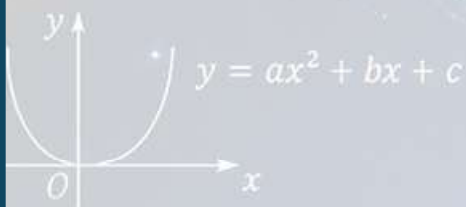
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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PRACTICAL - 4

From the following table, find $f(x)$ and hence $f(6)$ using divided difference formula

x	1	2	7	8
$f(x)$	1	5	5	4

Solution:-

Divided Difference Table

x	$f(x)$	First D.D.	Second D.D.	Third D.D.
1	1	$\frac{5-1}{2-1} = 4$	$\frac{0-\frac{4}{5}}{7-1} = -\frac{2}{15}$	$\frac{-\frac{1}{6}-(-\frac{2}{15})}{8-1} = \frac{1}{105}$
2	5	$\frac{5-5}{7-2} = 0$	$\frac{-1-0}{8-2} = -\frac{1}{6}$	
7	5	$\frac{4-5}{8-7} = -1$		
8	4			

Using Newton's divided difference formula,

$$f(x) = f(1) + (x-1)f[1,2] + (x-1)(x-2)f[1,2,7] + (x-1)(x-2)(x-7)f[1,2,7,8].$$

Substituting the divided differences,

$$f(x) = 1 + 4(x-1) - \frac{2}{15}(x-1)(x-2) + \frac{1}{105}(x-1)(x-2)(x-7).$$

$$f(x) = \frac{x^3}{105} - \frac{23x^2}{105} + \frac{272x}{105} - \frac{31}{21}.$$

$$\begin{aligned} f(6) &= 1 + 4(6-1) - \frac{2}{15}(6-1)(6-2) + \frac{1}{105}(6-1)(6-2)(6-7) \\ &= 1 + 20 - \frac{40}{15} - \frac{20}{105} \\ &= 21 - \frac{8}{3} - \frac{4}{21} \\ &= \frac{382}{21}. \end{aligned}$$

$$f(x) = 1 + 4(x-1) - \frac{2}{15}(x-1)(x-2) + \frac{1}{105}(x-1)(x-2)(x-7)$$

$$f(6) = \frac{382}{21} \approx 18.19.$$

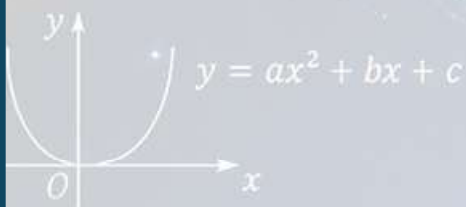
Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

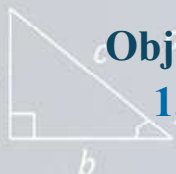
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 101

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

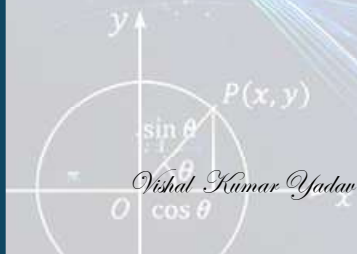


EXERCISE- 1



Objective Questions on Newton's Divided Difference

- Newton's divided difference interpolation formula is mainly used when:
 - The intervals between x -values are equal
 - The intervals between x -values are unequal
 - Only one data point is given
 - None of the above
- The divided difference operator is denoted by:
 - Δ
 - ∇
 - Square brackets $[]$
 - D
- The Newton's divided difference formula is based on:
 - Forward differences
 - Backward differences
 - Central differences
 - Divided differences
- The divided difference of order zero is equal to:
 - First difference
 - The function value itself
 - Second difference
 - None of the above
- The Newton's divided difference polynomial of degree n requires:
 - n data points
 - $n + 1$ data points
 - $2n$ data points
 - $n - 2$ data points

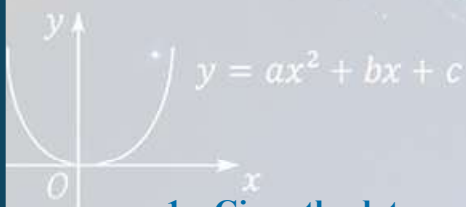


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

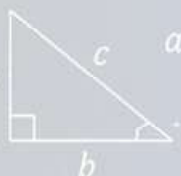
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXERCISE- 2

1. Given the data:



$$a^2 + b^2 = c^2$$

x	$f(x)$
1	2
2	4
4	16
7	49

Use Newton's divided difference interpolation to estimate $f(3)$.

2. The following values are given:

x	y
5	25
6	36
9	81
11	121

Find the value of y when $x = 7$.

3. The population of a town is given below:

Year	Population
2000	20000
2002	22000
2005	27000
2008	34000

Estimate the population in the year 2004 using Newton's divided difference interpolation.

4. The values of a function are given as:

x	$f(x)$
2	4
3	9
6	36
8	64

Find $f(5)$ using Newton's divided difference interpolation.

5. The following data is given:

x	y
1	1
4	16
5	25
7	49

Estimate the value of y when $x = 6$.

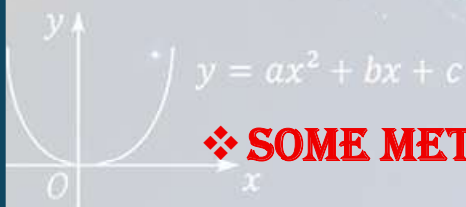


$$\sin^2 \theta + \cos^2 \theta = 1$$

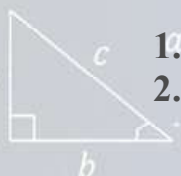
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 103

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



❖ SOME METHODS FOR SOLVING INTERPOLATION AT EQUAL SPACE POINTS



1. Newton Forward Interpolation
2. Newton Backward Interpolation

NEWTON'S FORWARD INTERPOLATION

Newton's Forward Interpolation formula is used to estimate the value of a function when the required value lies near the beginning of an equally spaced data table.

FORMULA

$$P_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \dots + \frac{p(p-1)\dots(p-n+1)}{n!}\Delta^n y_0$$

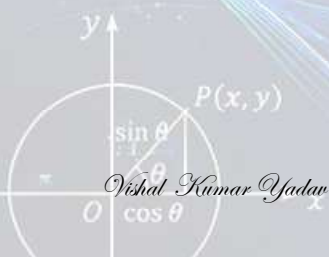
where

$$p = \frac{x_1 - x_0}{h}$$

Δy_0 = first forward difference

$\Delta^2 y_0, \Delta^3 y_0, \dots$

are higher-order forward differences.



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

❖ STATE AND PROVE NEWTON FORWARD INTERPOLATION FORMULA

Let $y = f(x)$ be a function that takes the values $y_0, y_1, y_2, \dots, y_n$ corresponding to the equidistant independent variables $x_0, x_1, x_2, \dots, x_n$.

Let the step size (the difference between consecutive x values) be h , such that $x_i - x_{i-1} = h$.

If we want to evaluate the function at some point x near the beginning of the dataset, we define a parameter p such that:

$$p = \frac{x - x_0}{h}$$

Newton's Forward Interpolation formula approximates the function $f(x)$ using an n -th degree polynomial $P_n(x)$:

$$P_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots + \frac{p(p-1)\dots(p-n+1)}{n!}\Delta^n y_0$$

Where Δ represents the forward difference operator (e.g., $\Delta y_0 = y_1 - y_0$, $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$, etc.).

Newton forward difference table

PROOF

Step 1: Assume a general polynomial form

We begin by assuming that the given data can be approximated by a polynomial of degree n in the following specific format:

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0)(x - x_1)\dots(x - x_{n-1})$$

Our goal is to find the values of the constants $a_0, a_1, a_2, \dots, a_n$. We do this by setting $P_n(x_i) = y_i$ for each data point.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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y

$$y = ax^2 + bx + c$$

Step 2: Solve for a_0

Substitute $x = x_0$ into the polynomial. All terms containing $(x - x_0)$ will become zero.

$$a^2 + b^2 = c^2$$

$$P_n(x_0) = a_0$$

Since $P_n(x_0) = y_0$, we get:

$$a_0 = y_0$$

Step 3: Solve for a_1

Substitute $x = x_1$ into the polynomial. All terms from the third term onwards contain $(x - x_1)$ and will become zero.

$$P_n(x_1) = a_0 + a_1(x_1 - x_0) = y_1$$

Since the points are equidistant, we know $x_1 - x_0 = h$. Substituting $a_0 = y_0$ and $x_1 - x_0 = h$:

$$y_0 + a_1 h = y_1$$

$$a_1 = \frac{y_1 - y_0}{h}$$

Using the forward difference operator ($\Delta y_0 = y_1 - y_0$), we get:

$$a_1 = \frac{\Delta y_0}{h}$$

Step 4: Solve for a_2

Substitute $x = x_2$ into the polynomial.

$$P_n(x_2) = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1) = y_2$$

We know that $x_2 - x_0 = 2h$ and $x_2 - x_1 = h$. Substituting our known values for a_0 and a_1 :

$$y_0 + \left(\frac{y_1 - y_0}{h}\right)(2h) + a_2(2h)(h) = y_2$$

$$y_0 + 2(y_1 - y_0) + 2h^2 a_2 = y_2$$

$$2y_1 - y_0 + 2h^2 a_2 = y_2$$

$$2h^2 a_2 = y_2 - 2y_1 + y_0$$

Recognizing that $y_2 - 2y_1 + y_0$ is the second forward difference $\Delta^2 y_0$:

y

sin θ cos θ

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O

x

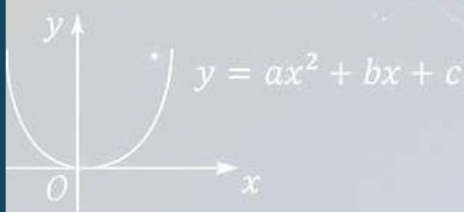
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =



$$a_2 = \frac{\Delta^2 y_0}{2h^2} = \frac{\Delta^2 y_0}{2! h^2}$$

Step 5: Generalize for a_n

By continuing this pattern for $x_3, x_4, \dots, x_n$, we can establish the general formula for the n -th constant:

$$a_n = \frac{\Delta^n y_0}{n! h^n}$$

Step 6: Substitute constants back into the polynomial

Replacing $a_0, a_1, \dots, a_n$ in our original polynomial equation:

$$P_n(x) = y_0 + \frac{\Delta y_0}{h}(x - x_0) + \frac{\Delta^2 y_0}{2! h^2}(x - x_0)(x - x_1) + \dots + \frac{\Delta^n y_0}{n! h^n}(x - x_0) \dots (x - x_{n-1})$$

Step 7: Transform the equation using p

To make the formula easier to use for interpolation, we introduce the variable $p = \frac{x - x_0}{h}$. This means $x - x_0 = ph$.

Now, we express all $(x - x_i)$ terms in terms of p and h :

- $x - x_0 = ph$
- $x - x_1 = x - (x_0 + h) = (x - x_0) - h = ph - h = h(p - 1)$
- $x - x_2 = x - (x_0 + 2h) = (x - x_0) - 2h = ph - 2h = h(p - 2)$
- $x - x_{n-1} = h(p - (n - 1))$

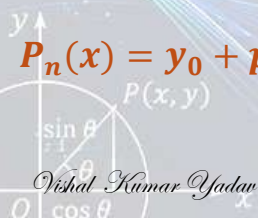
Step 8: Final substitution

Substitute these p expressions into the polynomial from Step 6:

- First degree term: $\frac{\Delta y_0}{h} \cdot (ph) = p\Delta y_0$
- Second degree term: $\frac{\Delta^2 y_0}{2! h^2} \cdot (ph) \cdot h(p - 1) = \frac{\Delta^2 y_0}{2! h^2} \cdot h^2 p(p - 1) = \frac{p(p-1)}{2!} \Delta^2 y_0$
- n -th degree term: $\frac{\Delta^n y_0}{n! h^n} \cdot (ph) \cdot h(p - 1) \dots h(p - n + 1) = \frac{p(p-1)\dots(p-n+1)}{n!} \Delta^n y_0$

Putting it all together, we arrive at the final proved formula:

$$P_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \dots + \frac{p(p-1)\dots(p-n+1)}{n!} \Delta^n y_0$$



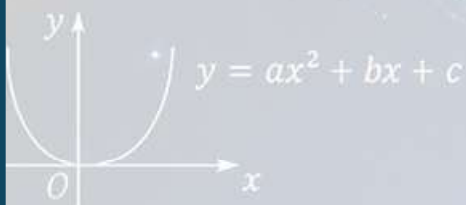
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

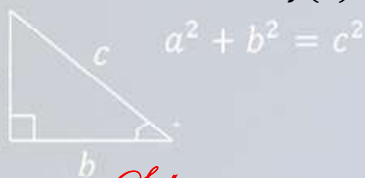
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL - 1

The function $f(x)$ has the exact values shown in the table.



x	0	5	10	15	20	25
$f(x)$	6	10	13	17	23	21

Solution:-

Using Newton's Forward Difference Interpolation Method, estimate the value of $f(30)$.

x	$f(x)$	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$	$\Delta^5 y_0$
0	6					
		4				
5	10		-1			
		3		2		
10	13		1		-1	
		4		1		-10
15	17		2		-11	
		6		-10		
20	23		-8			
		-2				
25	21					

$h = 5$

Given,

For $x = 30$,

$$u = \frac{x - x_0}{h}$$

$$u = \frac{30 - 0}{5} = 6$$

Newton Forward Formula

$$f(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!}\Delta^4 y_0 + \frac{u(u-1)(u-2)(u-3)(u-4)}{5!}\Delta^5 y_0$$

$$\begin{aligned} f(30) &= 6 + (6)(4) + \frac{6(5)}{2}(-1) + \frac{6(5)(4)}{6}(2) \\ &\quad + \frac{6(5)(4)(3)}{24}(-1) + \frac{6(5)(4)(3)(2)}{120}(-10) \\ &= 6 + 24 - 15 + 40 - 15 - 60 \end{aligned}$$

$$f(30) = -20$$

Answer

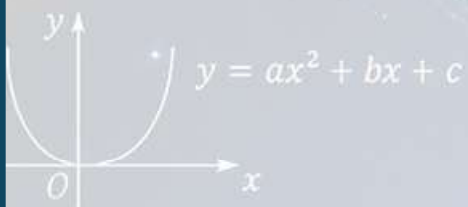
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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PRACTICAL - 2

Construct Newton's forward interpolation table for the following data:

x	4	6	8	10
y	1	3	8	10

Solution:-

First forward differences:

$$\Delta y: \begin{cases} 3 - 1 = 2, \\ 8 - 3 = 5, \\ 10 - 8 = 2. \end{cases}$$

Second forward differences:

$$\Delta^2 y: \begin{cases} 5 - 2 = 3, \\ 2 - 5 = -3. \end{cases}$$

Third forward difference:

$$\Delta^3 y: \{-3 - 3 = -6.\}$$

Hence, the Newton's forward difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
4	1	2	3	-6
6	3	5	-3	
8	8	2		
10	10			

Answer



$$\sin^2 \theta + \cos^2 \theta = 1$$

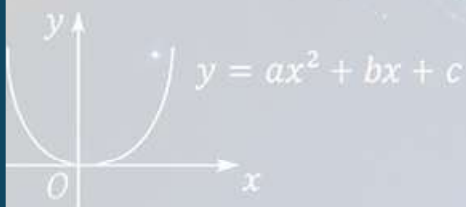
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 109

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

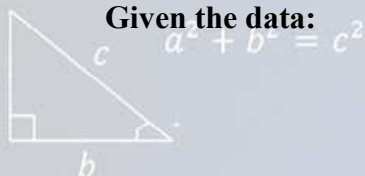
$x =$





PRACTICAL - 3

Given the data:



x	1	3	5
$f(x)$	4	-2	10

Estimate $f(6)$ using Newton's Forward Interpolation.

Solution:-

Since the x -values are equally spaced,

$$h = 3 - 1 = 2.$$

Take

$$x_0 = 1, \quad y_0 = 4.$$

Construct the forward difference table

x	y	Δy	$\Delta^2 y$
1	4	-6	18
3	-2	12	
5	10		

where

$$\Delta y_0 = -2 - 4 = -6,$$

$$\Delta y_1 = 10 - (-2) = 12,$$

$$\Delta^2 y_0 = 12 - (-6) = 18.$$

For $x = 6$,

$$u = \frac{x - x_0}{h} = \frac{6 - 1}{2} = 2.5.$$

Apply Newton's Forward Interpolation Formula

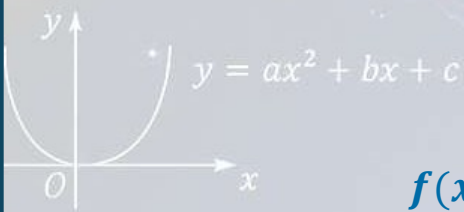


$$\sin^2 \theta + \cos^2 \theta = 1$$

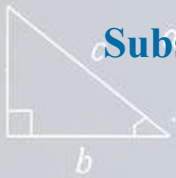
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 110

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$f(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0.$$



Substituting the values,

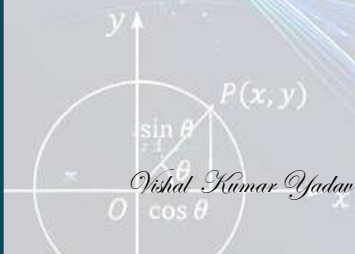
$$\begin{aligned} f(6) &= 4 + 2.5(-6) + \frac{2.5(2.5-1)}{2} (18) \\ &= 4 - 15 + \frac{2.5 \times 1.5}{2} \times 18 \\ &= 4 - 15 + 1.875 \times 18 \\ &= 4 - 15 + 33.75 \\ &= 22.75. \end{aligned}$$

Hence,

$$f(6) = 22.75 = \frac{91}{4}.$$

Answer

CIMAGE
Knowledge . Skill . Success



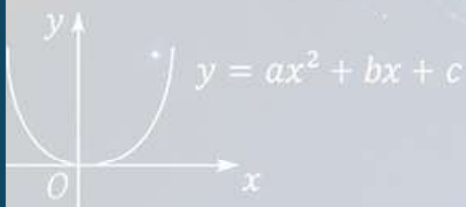
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

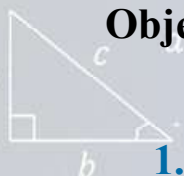
Page no - 111

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

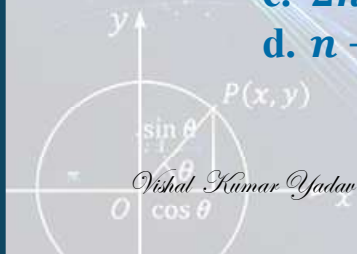
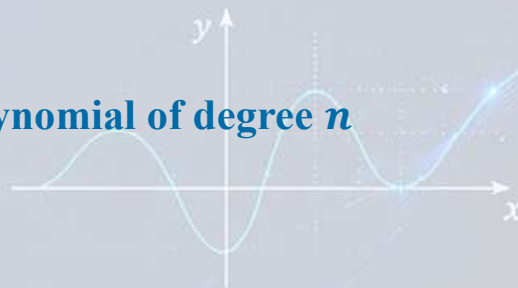


EXERCISE - 1

Objective Questions on Newton's Forward Interpolation



1. Newton's forward interpolation is mainly used when:
 - a. The value of x lies near the end of the table
 - b. The value of x lies near the beginning of the table
 - c. The value of x lies in the middle of the table
 - d. None of the above
2. The Newton's forward interpolation formula is based on:
 - a. Central differences
 - b. Backward differences
 - c. Forward differences
 - d. Divided differences
3. If the interval between successive values of x is constant, it is denoted by:
 - a. h
 - b. Δx
 - c. k
 - d. δ
4. In Newton's forward interpolation, the first term of the formula is:
 - a. y_0
 - b. y_1
 - c. y_n
 - d. Δy_0
5. The Newton's forward interpolation polynomial of degree n uses how many data points?
 - a. n
 - b. $n + 1$
 - c. $2n$
 - d. $n - 2$

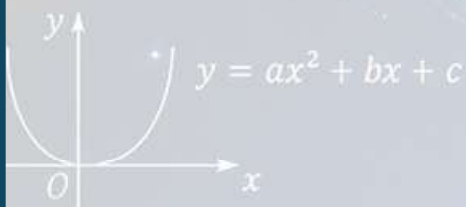


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 112

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXERCISE - 2

Numerical Problems on Newton's Forward Interpolation

1. The following table gives the values of $f(x)$:

x	0	1	2	3
$f(x)$	1	2	4	8

Use Newton's forward interpolation to estimate $f(2.5)$.

2. Given the data:

x	y
10	100
20	400
30	900
40	1600

Find the value of y when $x = 25$.

3. The population of a town is given below:

<i>Year</i>	<i>Population</i>
2000	20000
2005	24000
2010	29000
2015	35000

Estimate the population in the year 2008 using Newton's forward interpolation.

4. The values of a function are given as:

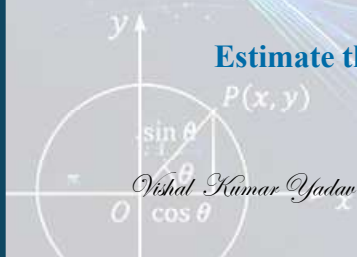
x	$f(x)$
1	1
2	8
3	27
4	64

Find $f(2.5)$ using Newton's forward interpolation.

5. The following data is given:

x	y
5	12
6	13
7	15
8	18

Estimate the value of y when $x = 6.5$.



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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 113

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

FORMULA

$$P_n(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots + \frac{p(p+1) \dots (p+n-1)}{n!} \nabla^n y_n$$

Where

y_n = last value of y

$$p = \frac{x - x_n}{h}$$

∇y_n = first backward difference

$\nabla^2 y_n, \nabla^3 y_n, \dots, \nabla^n y_n$ = higher order backward differences.

STATE AND PROVE NEWTON'S BACKWARD INTERPOLATION FORMULA

Let $y = f(x)$ be a function that takes the values $y_0, y_1, y_2, \dots, y_n$ corresponding to the equidistant independent variables $x_0, x_1, x_2, \dots, x_n$.

Let the step size (the difference between consecutive x values) be h , such that $x_i - x_{i-1} = h$.

If we want to evaluate the function at some point x near the end of the dataset, we define a parameter p such that:

$$p = \frac{x - x_n}{h}$$

Newton's Backward Interpolation formula approximates the function $f(x)$ using an n -th degree polynomial $P_n(x)$:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 114

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

$$P_n(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \dots$$



$$a^2 + b^2 = c^2 = \frac{p(p+1) \dots (p+n-1)}{n!} \nabla^n y_n$$

Where ∇ represents the backward difference operator (e.g., $\nabla y_n = y_n - y_{n-1}$, $\nabla^2 y_n = \nabla y_n - \nabla y_{n-1}$, etc.).

Newton backward difference table

PROOF

Step 1: Assume a general polynomial form

We begin by assuming that the given data can be approximated by a polynomial of degree n . Since we are interpolating backwards from the end of the dataset, we structure the polynomial around

x_n, x_{n-1}, x_{n-2} , etc.:

$$P_n(x) = a_0 + a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) + \dots + a_n(x - x_n)(x - x_{n-1}) \dots (x - x_1)$$

Our goal is to find the values of the constants $a_0, a_1, a_2, \dots, a_n$. We do this by setting $P_n(x_i) = y_i$ for each data point, starting from x_n and working backwards.

Step 2: Solve for a_0

Substitute $x = x_n$ into the polynomial. All terms containing $(x - x_n)$ will become zero.

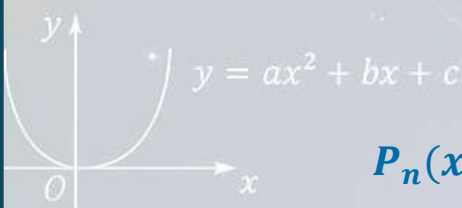
$$P_n(x_n) = a_0$$

Since $P_n(x_n) = y_n$, we get:

$$a_0 = y_n$$

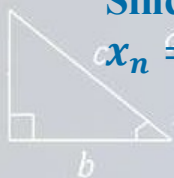
Step 3: Solve for a_1

Substitute $x = x_{n-1}$ into the polynomial. All terms from the third term onwards contain $(x - x_{n-1})$ and will become zero.



$$P_n(x_{n-1}) = a_0 + a_1(x_{n-1} - x_n) = y_{n-1}$$

Since the points are equidistant with step size h , we know $x_{n-1} - x_n = -h$. Substituting $a_0 = y_n$:



$$y_n + a_1(-h) = y_{n-1}$$

$$a_1 h = y_n - y_{n-1}$$

$$a_1 = \frac{y_n - y_{n-1}}{h}$$

Using the backward difference operator ($\nabla y_n = y_n - y_{n-1}$), we get:

$$a_1 = \frac{\nabla y_n}{h}$$

Step 4: Solve for a_2

Substitute $x = x_{n-2}$ into the polynomial.

$$P_n(x_{n-2}) = a_0 + a_1(x_{n-2} - x_n) + a_2(x_{n-2} - x_n)(x_{n-2} - x_{n-1}) = y_{n-2}$$

We know that $x_{n-2} - x_n = -2h$ and $x_{n-2} - x_{n-1} = -h$.

Substituting our known values for a_0 and a_1 :

$$y_n + \left(\frac{y_n - y_{n-1}}{h}\right)(-2h) + a_2(-2h)(-h) = y_{n-2}$$

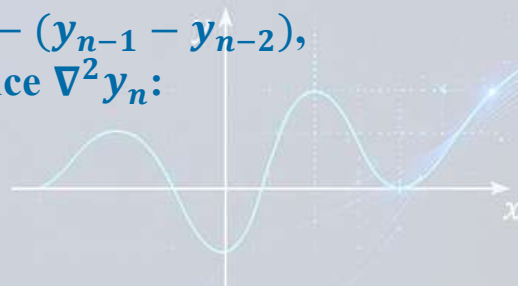
$$y_n - 2(y_n - y_{n-1}) + 2h^2 a_2 = y_{n-2}$$

$$-y_n + 2y_{n-1} + 2h^2 a_2 = y_{n-2}$$

$$2h^2 a_2 = y_{n-2} - 2y_{n-1} + y_n$$

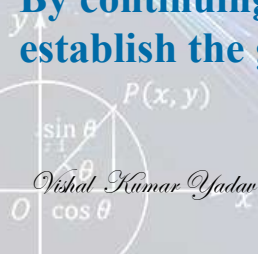
Rearranging the right side gives $(y_n - y_{n-1}) - (y_{n-1} - y_{n-2})$, which is exactly the second backward difference $\nabla^2 y_n$:

$$a_2 = \frac{\nabla^2 y_n}{2h^2} = \frac{\nabla^2 y_n}{2! h^2}$$



Step 5: Generalize for a_n

By continuing this pattern backwards for $x_{n-3}, x_{n-4}, \dots, x_0$, we can establish the general formula for the k -th constant:



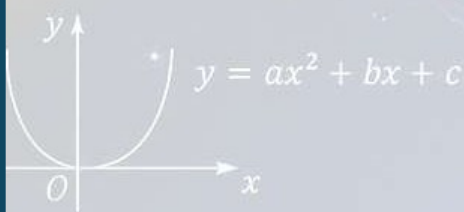
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 116

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



$$a_k = \frac{\nabla^k y_n}{k! h^k}$$

Step 6: Substitute constants back into the polynomial
Replacing $a_0, a_1, \dots, a_n$ in our original polynomial equation:

$$P_n(x) = y_n + \frac{\nabla y_n}{h}(x - x_n) + \frac{\nabla^2 y_n}{2! h^2}(x - x_n)(x - x_{n-1}) + \dots + \frac{\nabla^n y_n}{n! h^n}(x - x_n) \dots (x - x_1)$$

Step 7: Transform the equation using p

To make the formula useful for interpolation, we introduce the variable $p = \frac{x - x_n}{h}$. This means $x - x_n = ph$.

Now, we express all the product terms in terms of p and h :

- $x - x_n = ph$
- $x - x_{n-1} = x - (x_n - h) = (x - x_n) + h = ph + h = h(p + 1)$
- $x - x_{n-2} = x - (x_n - 2h) = (x - x_n) + 2h = ph + 2h = h(p + 2)$
- $x - x_{n-k} = h(p + k)$

Step 8: Final substitution

Substitute these p expressions into the polynomial from Step 6:

- First degree term: $\frac{\nabla y_n}{h} \cdot (ph) = p \nabla y_n$
- Second degree term: $\frac{\nabla^2 y_n}{2! h^2} \cdot (ph) \cdot h(p + 1) = \frac{\nabla^2 y_n}{2! h^2} \cdot h^2 p(p + 1) = \frac{p(p+1)}{2!} \nabla^2 y_n$
- n -th degree term: $\frac{\nabla^n y_n}{n! h^n} \cdot (ph) \cdot h(p + 1) \dots h(p + n - 1) = \frac{p(p+1) \dots (p+n-1)}{n!} \nabla^n y_n$

Putting it all together, we arrive at the final proved formula:

$$P_n(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots + \frac{p(p+1) \dots (p+n-1)}{n!} \nabla^n y_n$$

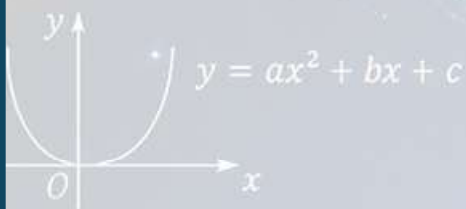
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 117

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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PRACTICAL - 1

Construct Newton's backward interpolation table For the Following data.

x	80	85	90	95	100
y	5026	5674	6363	7088	7854

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
80	5026				
		648			
85	5674		41		
		689		-5	
90	6363		36		10
		725		5	
95	7088		41		
		766			
100	7854				

Solution:-

Given:

$x(^{\circ})$	0	10	20	30	40
$\sin x$	0	0.17365	0.34202	0.50000	0.64279

Since 38° is close to 40° , we use Newton's Backward Interpolation Formula.

$$h = 10, \quad p = \frac{x - x_n}{h} = \frac{38 - 40}{10} = -0.2.$$

Backward Difference Table:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
0	0				
10	0.17365	0.17365			
20	0.34202	0.16837	-0.00528		
30	0.50000	0.15798	-0.01039	-0.00511	
40	0.64279	0.14279	-0.01519	-0.00480	0.00031

Newton's backward interpolation formula is



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$$\sin^2 \theta + \cos^2 \theta = 1$$

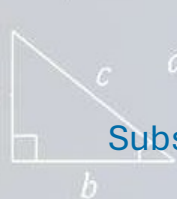
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 118

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

Substituting the values,

$$\begin{aligned} \sin 38^\circ &= 0.64279 + (-0.2)(0.14279) \\ &\quad + \frac{(-0.2)(0.8)}{2}(-0.01519) \\ &\quad + \frac{(-0.2)(0.8)(1.8)}{6}(-0.00480) \\ &\quad + \frac{(-0.2)(0.8)(1.8)(2.8)}{24}(0.00031). \end{aligned}$$

Calculating,

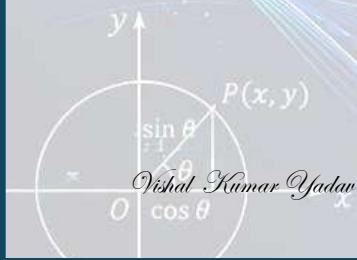
$$\begin{aligned} \sin 38^\circ &= 0.64279 - 0.028558 + 0.0012152 + 0.0002304 - 0.0000104 \\ &= 0.61567. \end{aligned}$$

Hence,

$$\boxed{\sin 38^\circ \approx 0.61567.}$$

Answer

CIMAGE
Knowledge . Skill . Success

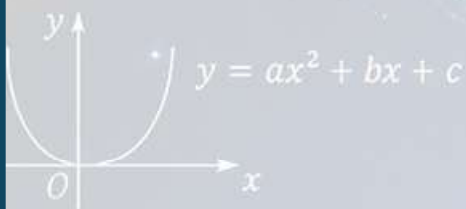


$$\sin^2 \theta + \cos^2 \theta = 1$$

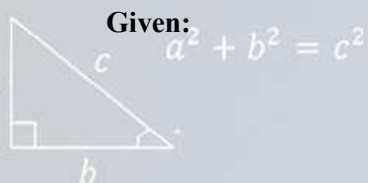
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 119

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



PRACTICAL- 2



x	1	2	3	4	5	6	7	8
$y = f(x)$	1	8	27	64	125	216	343	512

Find $f(7.5)$ using Newton's Backward Interpolation Formula.

Solution:-

Backward Difference Table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
1	1			
2	8			7
3	27	19	12	
4	64	37	18	6
5	125	61	24	6
6	216	91	30	6
7	343	127	36	6
8	512	169	42	6

From the last row,

$$y_n = 512, \quad \nabla y_n = 169, \quad \nabla^2 y_n = 42, \quad \nabla^3 y_n = 6.$$

- Since the third differences are constant, higher-order differences are zero.

Newton's Backward Formula

$$f(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n,$$

$$p = \frac{x - x_n}{h} = \frac{7.5 - 8}{1} = -0.5.$$

Substituting the values,

$$\begin{aligned} f(7.5) &= 512 + (-0.5)(169) + \frac{(-0.5)(0.5)}{2}(42) + \frac{(-0.5)(0.5)(1.5)}{6}(6). \\ &= 512 - 84.5 - 5.25 - 0.375 \\ &= 421.875. \end{aligned}$$

$$\boxed{f(7.5) = 421.875}$$

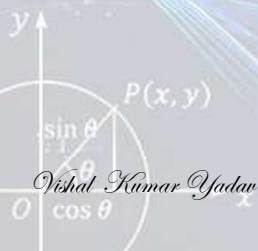
Answer

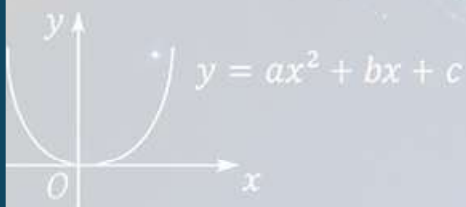
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 120

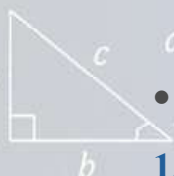
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





EXERCISE - 1

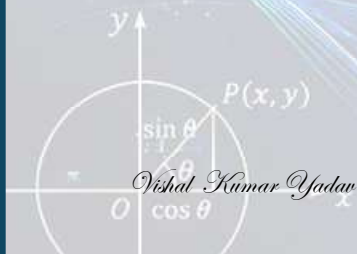
(Objectives)



$$a^2 + b^2 = c^2$$

• Objective Questions on Newton's Backward Interpolation

- Newton's backward interpolation is mainly used when:
 - The value of x lies near the beginning of the table
 - The value of x lies near the end of the table
 - The value of x lies in the middle of the table
 - None of the above
- Newton's backward interpolation formula is based on:
 - Forward differences
 - Backward differences
 - Central differences
 - Divided differences
- In Newton's backward interpolation, the first term of the formula is:
 - y_0
 - y_n
 - Δy_0
 - ∇y_n
- The backward difference operator is denoted by:
 - Δ
 - ∇
 - δ
 - D
- The Newton's backward interpolation polynomial of degree n requires:
 - n data points
 - $n + 1$ data points
 - $2n$ data points
 - $n - 2$ data points



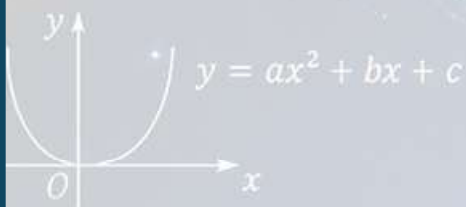
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

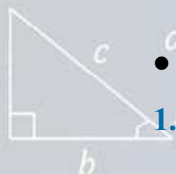
Page no - 121

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EXERCISE - 2

(Subjectives)



$$a^2 + b^2 = c^2$$

• Numerical Problems on Newton's Backward Interpolation

1. The following table gives values of $f(x)$:

x	$f(x)$
0	1
1	2
2	4
3	8

Use Newton's backward interpolation to estimate $f(2.5)$.

2. Given the data:

x	y
10	100
20	400
30	900
40	1600

Find the value of y when $x = 35$.

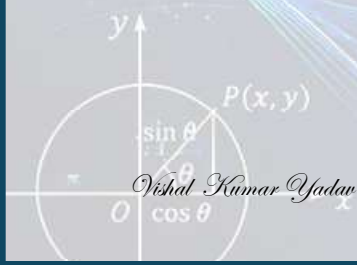
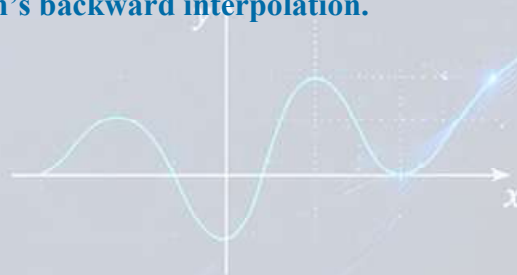
3. The population of a town is given below:

Year	Population
2000	20000
2005	24000
2010	29000
2015	35000

Estimate the population in the year 2013 using Newton's backward interpolation.

4. The values of a function are given as:

x	$f(x)$
1	1
2	8
3	27
4	64



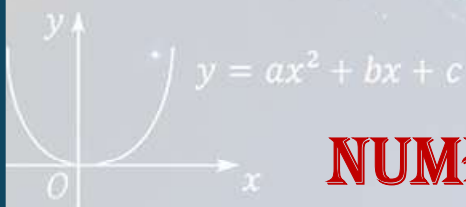
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 122

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



NUMERICAL INTEGRATION

➤ Numerical Integration is the process to calculate the value of definite integration

$$\int_a^b f(x) dx$$

- Some method for solving numerical integration.

i) Trapezoidal Rule

ii) Simpson's 1/3 Rule

iii) Simpson's 3/8 Rule

TRAPEZOIDAL RULE

(THEORY)

❖ Explain trapezoidal rule for solving definite integration (Numerical Integration)

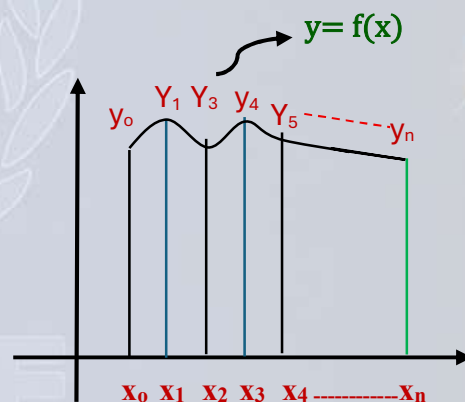
➤ Now,

we divide the graph of $f(x)$ into (n) number of small trapezium with height of is trapezium

small

$$h = \frac{(b-a)}{n} \text{ where,}$$

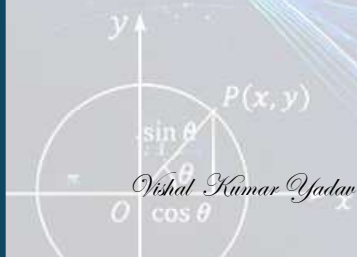
n = no of sub interval



$$\begin{aligned} \int_a^b f(x) dx &= \text{Area}(\text{trap}(x_0 y_0 y_1 x_1)) + \text{Area}(\text{trap}(x_1 y_1 y_2 x_2)) + \dots + \\ &\text{Area}(\text{trap}(x_{n-1} y_{n-1} y_n x_n)) \end{aligned}$$

$$\frac{h}{2}(y_0 + y_1) + \frac{h}{2}(y_1 + y_2) + \frac{h}{2}(y_2 + y_3) + \dots + \frac{h}{2}(y_{n-1} + y_n)$$

$$\Rightarrow \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + y_3 \dots y_{n-1})]$$

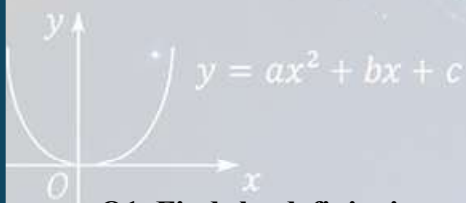


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

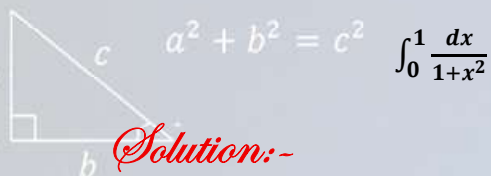
Page no - 123

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 1)

Q1. Find the definite integration by using trapezoidal rule :-



Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}$$

$$h = \frac{1}{4}$$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
F(x)	1	$\frac{16}{4}$	$\frac{4}{5}$	$\frac{16}{25}$	$\frac{1}{2}$

y_0

y_1

y_2

y_3

y_4

By - Trapezoidal rule :-

$$\circ \int_0^1 \frac{dx}{1+x^2} = \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)]$$

$$= \frac{1}{4 \times 2} [(1 + \frac{1}{2}) + 2(\frac{16}{17} + \frac{4}{5} + \frac{16}{25})]$$

$$= \frac{1}{8} [(\frac{3}{2} + \frac{32}{17} + \frac{8}{5} + \frac{32}{25})]$$

$$= \frac{1}{8}(6.2623)$$

$$= 0.7826$$

Answer



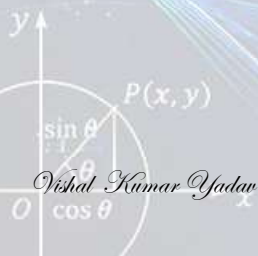
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

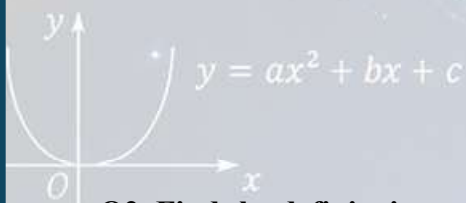
Page no - 124

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =

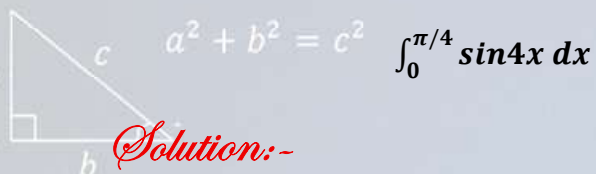


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(PRACTICAL - 2)

Q2. Find the definite integration by using trapezoidal rule :-



Given that,

$$F(x) = \sin 4x \, dx$$

$$a = 0, \quad b = \pi/4, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{\pi/4-0}{4} = 0.196$$

$$h = 0.196$$

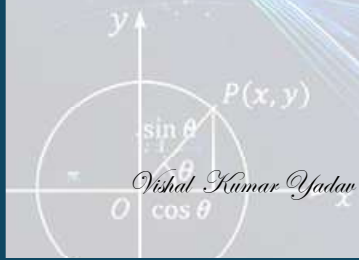
x	0	0.196	0.392	0.588	0.785
F(x)	0	0.706	1	0.710	0
	y_0	y_1	y_2	y_3	y_4

By - Trapezoidal rule :-

$$\begin{aligned}
 \circ \int_0^1 \frac{dx}{1+x^2} &= \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)] \\
 &= \frac{0.196}{2} [(0 + 0) + 2(0.706 + 1 + 0.710)] \\
 &= 0.098 [(0 + 1.412 + 2 + 1.42)] \\
 &= 0.098(4.832) \\
 &= 0.473536
 \end{aligned}$$



Answer



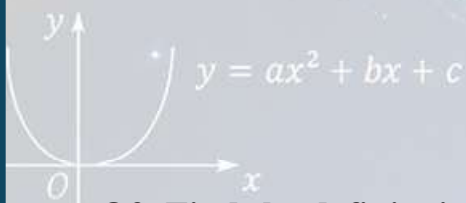
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

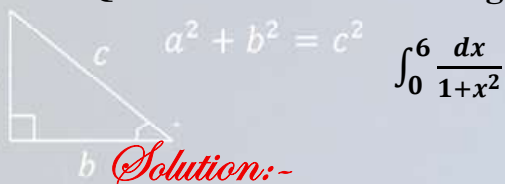
Page no - 125

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 3)

Q3. Find the definite integration by using trapezoidal rule :-



Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 6, \quad n = 4$$

let,

$$h = \frac{b-a}{n} = \frac{6-0}{4} = 1.5$$

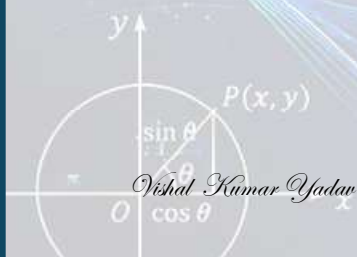
$$h = 1.5$$

x	0	1.5	3	4.5	6
F(x)	1	0.3077	0.1	0.0471	0.027
	y_0	y_1	y_2	y_3	y_4

By - Trapezoidal rule :-

$$\begin{aligned}
 \circ \int_0^6 \frac{dx}{1+x^2} &= \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)] \\
 &= \frac{1.5}{2} [(1 + 0.027) + 2(0.3077 + 0.1 + 0.0471)] \\
 &= 0.75 [(1.027 + 0.9096)] \\
 &= 0.75(1.9366) \\
 &= 1.4524
 \end{aligned}$$

Answer



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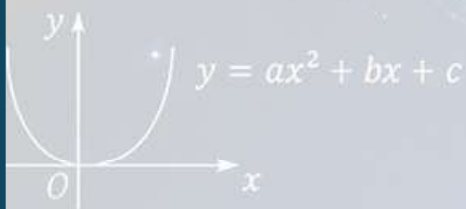


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 126

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 1)

(Objectives)

❖ Trapezoidal Rule – Multiple Choice Questions :-

1. What is the primary use of the Trapezoidal Rule?

- A) Differentiation
- B) Integration
- C) Factorization
- D) Interpolation

2. The Trapezoidal Rule approximates the area under a curve by dividing it into:

- A) Triangles
- B) Rectangles
- C) Trapezoids
- D) Circles

3. In the Composite Trapezoidal Rule, the step size h is given by:

- A) $(a + b)/n$
- B) $(b - a)/n$
- C) ab/n
- D) $(a - b)/n$

4. Which of the following is the formula for the Single Trapezoidal Rule?

- A) $(b - a)[f(a) + f(b)]$
- B) $\frac{1}{2}(b - a)[f(a) + f(b)]$
- C) $(b + a)[f(a) - f(b)]$
- D) $\frac{1}{2}(a + b)[f(a) + f(b)]$

5. In the Composite Trapezoidal Rule, the first and last ordinates are counted:

- A) Twice
- B) Three Times
- C) Once
- D) Four Times



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

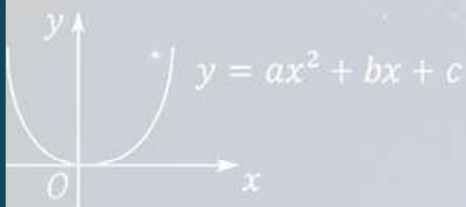
$x =$

$P(x, y)$

$\sin \theta$

$\cos \theta$

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(EXERCISE - 2)

(Subjectives)

❖ Trapezoidal Rule – Numerical Questions :-

Q1. Find the definite integration by using Trapezoidal Rule:

$$\int_0^1 \frac{dx}{1+x^2}$$

Q2. Find the definite integration by using Trapezoidal Rule:

$$\int_0^2 (x^2 + 1) dx$$

Q3. Find the definite integration by using Trapezoidal Rule:

$$\int_0^4 x^2 dx$$

Q4. Find the definite integration by using Trapezoidal Rule:

$$\int_1^3 (x^3 + 2x) dx$$

Q5. Find the definite integration by using Trapezoidal Rule:

$$\int_0^1 e^x dx$$

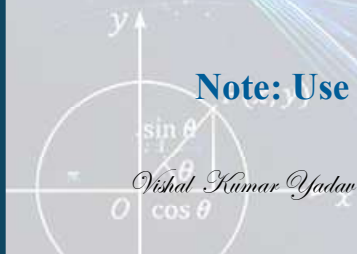
Q6. Find the definite integration by using Trapezoidal Rule:

$$\int_0^\pi \sin x dx$$

Q7. Find the definite integration by using Trapezoidal Rule:

$$\int_1^2 \frac{dx}{x}$$

Note: Use $n = 5$ in all numerical problems.



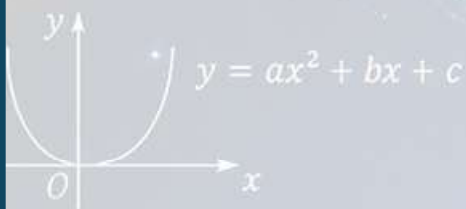
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 128

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

SIMPSON'S $\frac{1}{3}$ RULE

(THEORY)

❖ Explain Simpson's $\frac{1}{3}$ rule for solving definite integration (Numerical Integration)

➤ Let,

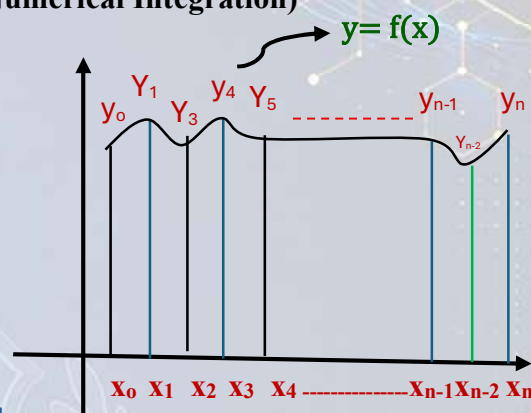
$f(x)$ be the real value continuous function defined on closed interval $[a, b]$. We have to find

$$\int_a^b f(x) dx$$

We divide the graph of $f(x)$ into

n (number) of partitions of parabola, where

n is the number corresponding to each pair of sub intervals.



$$\int_a^b f(x) dx = \text{Area}(\text{Para}(x_0, y_0, y_1, y_2, x_2)) + \text{Area}(\text{Para}(x_2, y_2, y_3, y_4, x_4)) + \text{Area}(\text{Para}(x_4, y_4, y_5, y_6, x_6)) + \dots + \text{Area}(\text{Para}(x_{n-2}, y_{n-2}, y_{n-1}, y_n, x_n))$$

$$= \frac{h}{3} (y_0 + 4y_1 + y_2) + \frac{h}{3} (y_2 + 4y_3 + y_4) + \frac{h}{3} (y_4 + 4y_5 + y_6) + \dots + \frac{h}{3} (y_{n-2} + 4y_{n-1} + y_n)$$

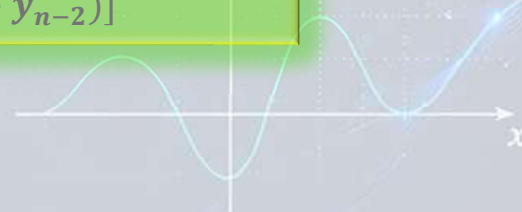
$$= \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2})]$$

Hence,

$$\int_a^b f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2})]$$

where,

$$h = \frac{b - a}{n}$$



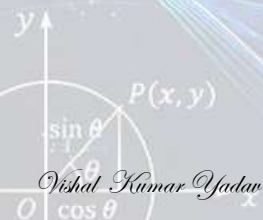
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

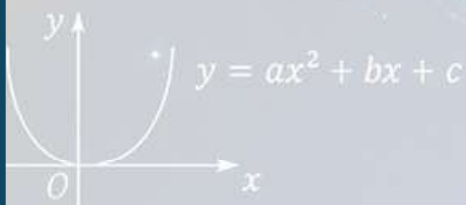
Page no - 129

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$

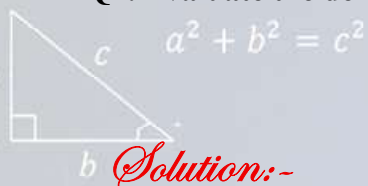


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(PRACTICAL - 1)

Q1. Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule :-



$$\int_0^1 \frac{dx}{1+x^2}$$

Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By - Simpson's $\frac{1}{3}$ rule :-

$$\circ \int_0^1 \frac{dx}{1+x^2} = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3 \times 6} [(1 + \frac{1}{2}) + 4(\frac{36}{37} + \frac{4}{5} + \frac{36}{61}) + 2(\frac{9}{10} + \frac{9}{13})]$$

$$= \frac{1}{18} [(\frac{3}{2} + \frac{144}{37} + \frac{16}{5} + \frac{144}{61} + \frac{18}{10} + \frac{18}{13})]$$

$$= \frac{1}{18} (14.137163)$$

$$= 0.785397945234$$

Answer



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

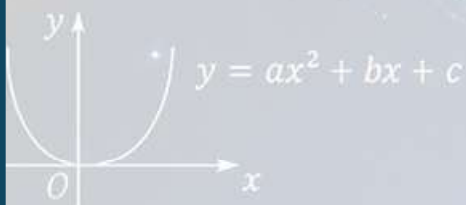
Page no - 130

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$



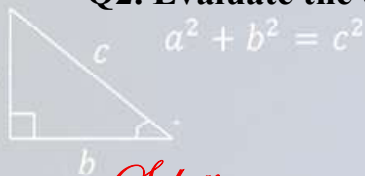
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$$y = ax^2 + bx + c$$

(PRACTICAL - 2)

Q2. Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule :-



$$a^2 + b^2 = c^2$$

$$\int_0^6 \frac{dx}{1+x^2}$$

Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 6, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{6-0}{6} = 1$$

$$h = 1$$

x	0	1	2	3	4	5	6
F(x)	1	0.5	0.2	0.1	0.0588	0.0385	0.027

$$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$$

By - Simpson's $\frac{1}{3}$ rule :-

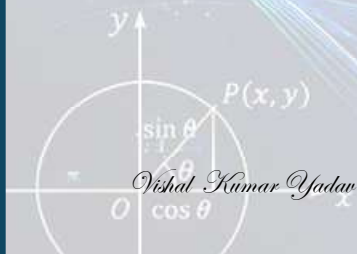
$$\circ \int_0^1 \frac{dx}{1+x^2} = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3} [(1 + 0.027) + 4(0.5 + 0.1 + 0.0385) + 2(0.2 + 0.0588)]$$

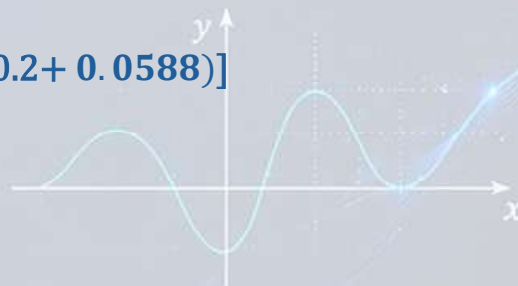
$$= \frac{1}{3} (4.0986)$$

$$= 1.3662$$

Answer



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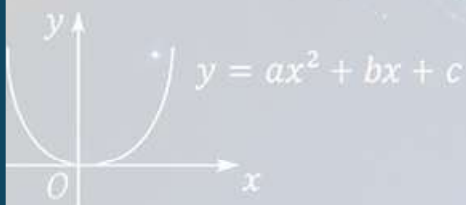


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

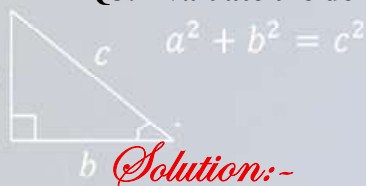
Page no - 131

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 3)

Q3. Evaluate the definite integration by using Simpson's $\frac{1}{3}$ rule :-



$$\int_0^1 \frac{dx}{1+x^2}$$

Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$

By - Simpson's $\frac{1}{3}$ rule :-

$$\circ \int_0^1 \frac{dx}{1+x^2} = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

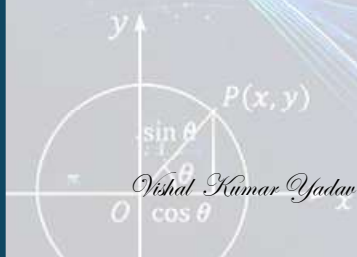
$$= \frac{1}{3 \times 6} [(1 + \frac{1}{2}) + 4(\frac{36}{37} + \frac{4}{5} + \frac{36}{61}) + 2(\frac{9}{10} + \frac{9}{13})]$$

$$= \frac{1}{18} [(\frac{3}{2} + \frac{144}{37} + \frac{16}{5} + \frac{144}{61} + \frac{18}{10} + \frac{18}{13})]$$

$$= \frac{1}{18} (14.137163)$$

$$= 0.785397945234$$

Answer



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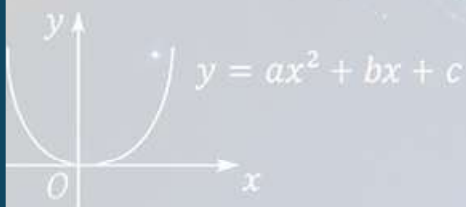


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 1)

(Objectives)

❖ **Simpson's 1/3 Rule – Multiple Choice Questions :-**

Q1. In Simpson's 1/3 Rule, the number of sub-intervals n must be:

- A) Odd number
- B) Prime number
- C) Even number
- D) Multiple of 3

Q2. The factor outside the bracket in Simpson's 1/3 Rule is:

- A) $\frac{h}{2}$
- B) $\frac{h}{3}$
- C) $\frac{3h}{8}$
- D) $\frac{h}{4}$

Q3. Simpson's 1/3 Rule is used for:

- A) Numerical Differentiation
- B) Numerical Integration
- C) Matrix Multiplication
- D) Interpolation Only

Q4. In Simpson's 1/3 Rule, the coefficient of the odd ordinates is:

- A) 1
- B) 2
- C) 3
- D) 4

Q5. The width of each sub-interval in Simpson's 1/3 Rule is given by:

- A) $h = \frac{n}{b-a}$
- B) $h = \frac{a+b}{n}$
- C) $h = \frac{b-a}{n}$
- D) $h = \frac{ab}{n}$



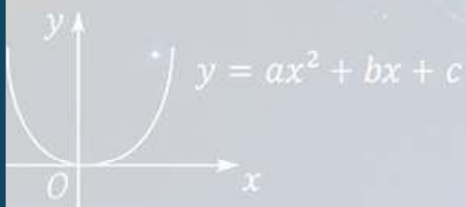
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 133

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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(EXERCISE - 2)

(Subjectives)

❖ **Simpson's 1/3 Rule – Numerical Questions :-**

Q1. Explain Simpson's 1/3 Rule for Numerical Integration and derive its formula.

Q2. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \frac{1}{1+x} dx$$

Q3. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 x^2 dx$$

Q4. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 e^x dx$$

Q5. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \sin x dx$$

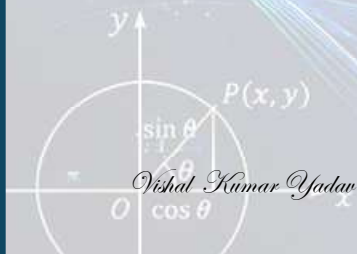
Q6. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \sqrt{1+x} dx$$

Q7. Find the definite integration by using Simpson's 1/3 Rule:

$$\int_0^2 \frac{1}{1+x^2} dx$$

Note: Use $n = 2$ in all numerical problems.

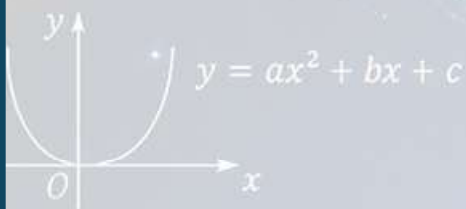


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 134

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

SIMPSON'S $\frac{3}{8}$ RULE

(THEORY)

❖ Explain Simpson's $\frac{3}{8}$ rule for solving definite integration (Numerical Integration)

➤ Let,

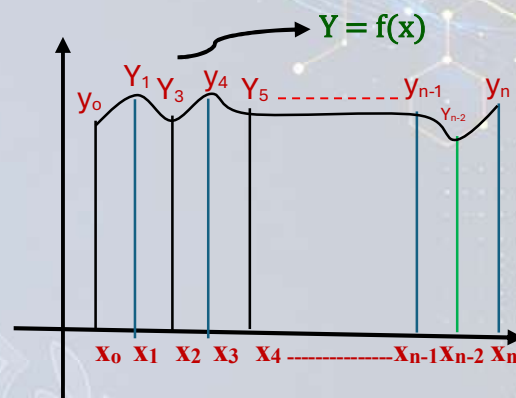
$f(x)$ be the real value continuous function defined on closed interval $[a, b]$. We have to find

$$\int_a^b f(x) dx$$

We divide the graph of $f(x)$ into

n (number) of partitions of parabola, where

n is the number corresponding to each pair of sub intervals.



$$\begin{aligned} \int_a^b f(x) dx &= \text{Area(Para}(x_0, y_0, y_1, y_2, y_3, x_3)) + \text{Area(Para}(x_3, y_3, y_4, y_5, y_6, x_6)) \\ &+ \text{Area(Para}(x_6, y_6, y_7, y_8, y_9, x_9)) + \dots + \text{Area} \\ &(\text{Para}(x_{n-2}, y_{n-1}, y_{n-1}, y_n, x_n)) \end{aligned}$$

$$\begin{aligned} &= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) + \frac{3h}{8} (y_2 + 3y_4 + 3y_5 + y_6) + \frac{3h}{8} (y_6 + 3y_7 + 3y_8 + y_9) + \dots \\ &+ \frac{3h}{8} (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n) \end{aligned}$$

Hence,

$$\begin{aligned} \int_a^b f(x) dx &= \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_4 + y_7 + \dots + y_{n-1}) \\ &+ 3(y_2 + y_5 + y_8 + \dots + y_{n-2}) \\ &+ 2(y_3 + y_6 + y_9 + \dots + y_{n-3})] \end{aligned}$$

where,

$$h = \frac{b-a}{n}$$

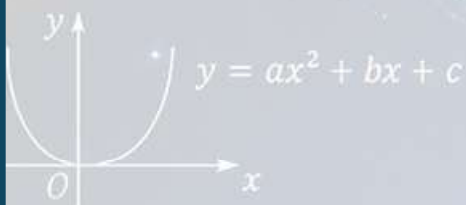
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 135

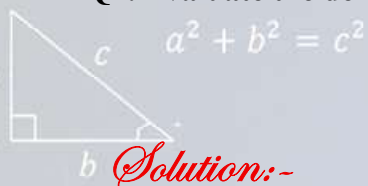
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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(PRACTICAL - 1)

Q1. Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule :-



$$\int_0^1 \frac{dx}{1+x^2}$$

Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

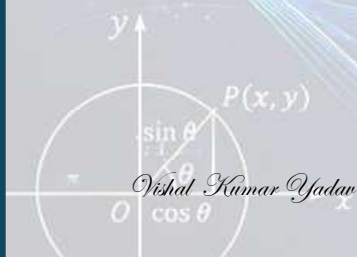
$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$

By - Simpson's $\frac{1}{3}$ rule :-

$$\begin{aligned}
 \circ \quad \int_0^1 \frac{dx}{1+x^2} &= \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3)] \\
 &= \frac{3 \times 1/6}{8} [(1 + \frac{1}{2}) + 3(\frac{36}{37} + \frac{9}{13}) + 3(\frac{9}{10} + \frac{36}{61}) + 2(\frac{4}{5})] \\
 &= \frac{1}{16} [(\frac{3}{2} + 3(1.66) + 3(1.49) + 2(\frac{4}{5}))] \\
 &= \frac{1}{16} (\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47) \\
 &= 0.7843
 \end{aligned}$$

Answer



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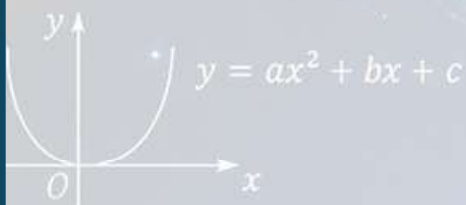
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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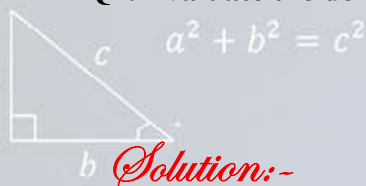
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =



(PRACTICAL - 2)

Q2. Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule :-



$$\int_0^1 \frac{dx}{1+x^2}$$

Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$

By - Simpson's $\frac{1}{3}$ rule :-

$$\circ \int_0^1 \frac{dx}{1+x^2} = \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3)]$$

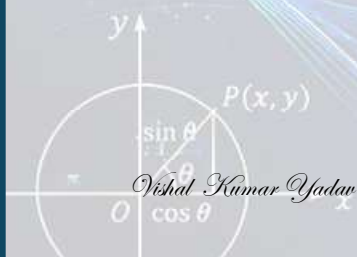
$$= \frac{3 \times 1/6}{8} [(1 + \frac{1}{2}) + 3(\frac{36}{37} + \frac{9}{13}) + 3(\frac{9}{10} + \frac{36}{61}) + 2(\frac{4}{5})]$$

$$= \frac{1}{16} [(\frac{3}{2} + 3(1.66) + 3(1.49) + 2(\frac{4}{5}))]$$

$$= \frac{1}{16} (\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47)$$

$$= 0.7843$$

Answer

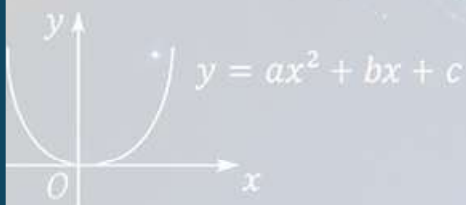


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

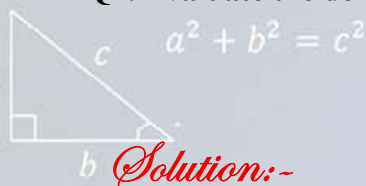
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 3)

Q1. Evaluate the definite integration by using Simpson's $\frac{3}{8}$ rule :-



$$\int_0^1 \frac{dx}{1+x^2}$$

Solution:-

Given that,

$$F(x) = \frac{1}{1+x^2}$$

$$a = 0, \quad b = 1, \quad n = 6$$

let,

$$h = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

$$h = \frac{1}{6}$$

x	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$	1
F(x)	1	$\frac{36}{37}$	$\frac{9}{10}$	$\frac{4}{5}$	$\frac{9}{13}$	$\frac{36}{61}$	$\frac{1}{2}$

By - Simpson's $\frac{1}{3}$ rule :-

$$\circ \int_0^1 \frac{dx}{1+x^2} = \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3)]$$

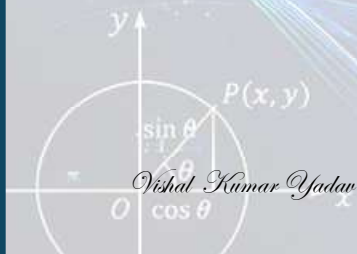
$$= \frac{3 \times 1/6}{8} [(1 + \frac{1}{2}) + 3(\frac{36}{37} + \frac{9}{13}) + 3(\frac{9}{10} + \frac{36}{61}) + 2(\frac{4}{5})]$$

$$= \frac{1}{16} [(\frac{3}{2} + 3(1.66) + 3(1.49) + 2(\frac{4}{5}))]$$

$$= \frac{1}{16} (\frac{3}{2} + \frac{8}{5} + 4.98 + 4.47)$$

$$= 0.7843$$

Answer



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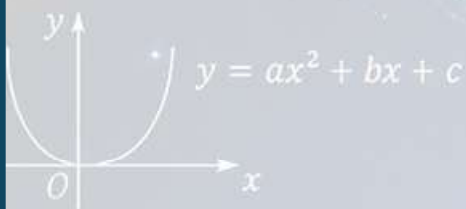


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 1)

(Objectives)

❖ $a^2 + b^2 = c^2$
Simpson's 3/8 Rule – Multiple Choice Questions :-

Q1. In Simpson's 3/8 Rule, the coefficient of y_1 is:

- A) 1
- B) 2
- C) 3
- D) 4

Q2. The factor outside the bracket in Simpson's 3/8 Rule is:

- A) $\frac{h}{3}$
- B) $\frac{3h}{8}$
- C) $\frac{h}{2}$
- D) $\frac{8h}{3}$

Q3. Simpson's 3/8 Rule is a method of:

- A) Numerical Differentiation
- B) Numerical Integration
- C) Matrix Analysis
- D) Interpolation Only

Q4. For Simpson's 3/8 Rule, the number of sub-intervals n should be:

- A) Even
- B) Odd
- C) Multiple of 2
- D) Multiple of 3

Q5. The general formula of Simpson's 3/8 Rule begins with:

- A) $\frac{h}{2}$
- B) $\frac{h}{3}$
- C) $\frac{3h}{8}$
- D) $\frac{8h}{3}$



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

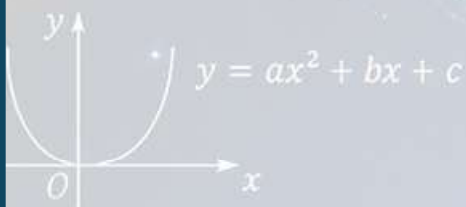
$x =$

$P(x, y)$

$\sin \theta$

$\cos \theta$

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(EXERCISE - 2)

(Subjectives)

❖ **Simpson's 3/8 Rule – Numerical Questions :-**

Q1. Explain Simpson's 3/8 Rule for Numerical Integration and derive its formula.

Q2. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \frac{1}{1+x} dx$$

Q3. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 x^2 dx$$

Q4. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 e^x dx$$

Q5. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \sin x dx$$

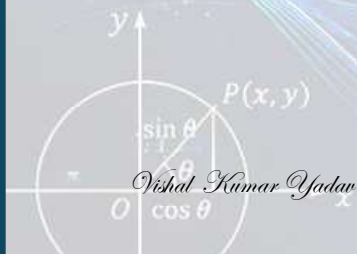
Q6. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \sqrt{1+x} dx$$

Q7. Find the definite integration by using Simpson's 3/8 Rule:

$$\int_0^3 \frac{1}{1+x^2} dx$$

Note: Use $n = 3$ in all numerical problems.

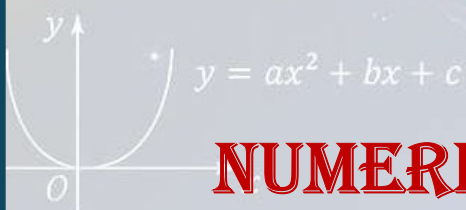


$$\sin^2 \theta + \cos^2 \theta = 1$$

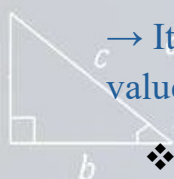
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



NUMERICAL DIFFERENTIATION



→ It is the process to finding the derivative of $f(x)$ with the help of given tabulated value of $f(x)$.

❖ Find the first and second derivative of following table:-

x	1	2	3	4	5
F(x)	3	11	31	69	131

x	F(x)			
1	3			
2	11	8	12	
3	31	20	18	6
4	69	38	24	6
5	131	62		

$$P_n(x) = 3 + (x-1) \times 8 + (x-1)(x-2) \times 12 + (x-1)(x-2)(x-3) \times 6$$

$$= 3 + 8x - 8 + 12x^2 - 36x + 24 + 6x^3 - 36x^2 + 66x - 36$$

$$f(x) = 6x^3 - 24x^2 + 38x - 17$$

$$f'(x) = 18x^2 - 48x + 38$$

$$f''(x) = 36x - 48$$



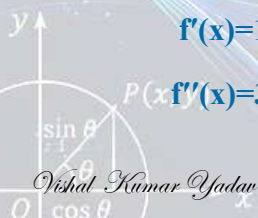
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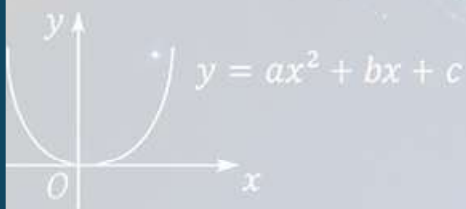
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 141

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

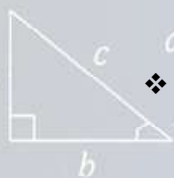
x =





EULER'S METHOD

(THEORY)



$$a^2 + b^2 = c^2$$

❖ Explain Euler's method for solving ODE differential equation.

Let, $\frac{dx}{dy} = f(x, y)$ be the initial value problem with initial condition

$y(x_0) = y_0$ and step length (h).

$f(x)$ be the given soln of given diffn eqn we will be expanding

$f(x)$ in the neighbourhood of x_0

by using Taylor's series expansion

$$F(x) = F(x_0) + (x - x_0)F'(x_0) + (x - x_0)^2 \frac{F''(x_0)}{2!} + (x - x_0)^3 \frac{F'''(x_0)}{3!} + \dots$$

Put,

$$x - x_0 = h$$

$$x = x_0 + h$$

$$F(x_0 + h) = F(x_0) + hF'(x_0) + \frac{h^2}{2!} F''(x_0) + \frac{h^3}{3!} F'''(x_0) + \dots$$

Let,

$$y = F(x) \Rightarrow y_0 = F(x_0)$$

$$F'(x) = f(x, y) \Rightarrow F'(x_0) = f(x_0, y_0)$$

$$F''(x) = f'(x, y) \Rightarrow F''(x_0) = f'(x_0, y_0)$$

$$\vdots \quad \vdots \quad \vdots$$

$$\vdots \quad \vdots \quad \vdots$$

$$\vdots \quad \vdots \quad \vdots$$

$$F^n(x) = f^{n-1}(x, y) \Rightarrow F^n(x_0) = f^{n-1}(x_0, y_0)$$

Put this value in eqn (i)

$$y_1 = y_0 + h f(x_0, y_0) + \frac{h^2}{2!} f'(x_0, y_0) + \frac{h^3}{3!} f''(x_0, y_0) + \dots$$

Neglecting the terms containing more than 1° (degree) in h.

$$y_1 = y_0 + h f(x_0, y_0) \quad y_2 = y_1 + h f(x_1, y_1) \quad y_3 = y_2 + h f(x_2, y_2)$$

$$\vdots \quad \vdots \quad \vdots$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$x_{n+1} = x_n + h$$

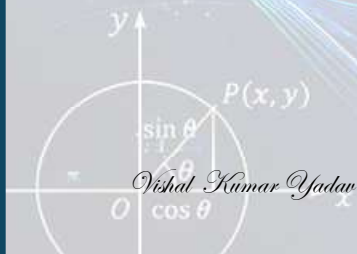


$$\sin^2 \theta + \cos^2 \theta = 1$$

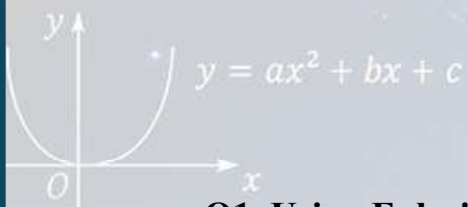
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

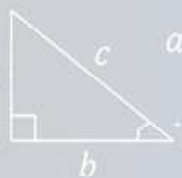


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(PRACTICAL - 1)

Q1. Using Euler's method, find $y(0.4)$ for the differential equation,



$$a^2 + b^2 \frac{dy}{dx} = x + y$$

**with initial condition $y(0) = 1$
and step length $h = 0.2$.**

Solution:-

Given

$$f(x, y) = x + y$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.2$$

By Euler's

$$y_{n+1} = y_n + h f(x_n, y_n)$$

1st approximation

$$\text{Put } n = 0$$

$$y_1 = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.2 \cdot f(0, 1)$$

$$= 1 + 0.2 \times (0 + 1)$$

$$= 1.2$$

$$\rightarrow x_1 = x_0 + h$$

$$= 0 + 0.2$$

$$= 0.2$$

$$\boxed{y(0.2) = 1.2}$$



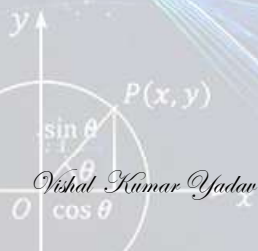
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Page no - 143

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$



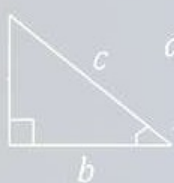
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$$y = ax^2 + bx + c$$

2nd approximation

Put $n = 1$



$$\begin{aligned} y_2 &= y_1 + h f(x_1, y_1) \\ a^2 + b^2 &= c^2 \\ &= 1.2 + 0.2 f(0.2, 1.2) \\ &= 1.2 + 0.2 [0.2 + 1.2] \\ &= 1.2 + 0.2 \times 1.4 \\ &= 1.2 + 0.28 \\ &= 1.48 \end{aligned}$$

$$\begin{aligned} \rightarrow x_2 &= x_1 + h \\ &= 0.2 + 0.2 \\ &= 0.4 \end{aligned}$$

$$y(0.4) = 1.48$$

3rd approximation

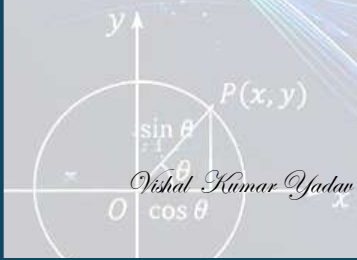
Put $n = 2$

$$\begin{aligned} y_3 &= y_2 + h f(x_2, y_2) \\ &= 1.48 + 0.2 f(0.4, 1.48) \\ &= 1.48 + 0.2 \times (0.4 + 1.48) \\ &= 1.48 + 0.2 \times 1.88 \\ &= 1.48 + 0.376 \\ &= 1.856 \end{aligned}$$

$$\begin{aligned} \rightarrow x_3 &= x_2 + h \\ &= 0.4 + 0.2 \\ &= 0.6 \end{aligned}$$

$$y(0.6) = 1.856$$

Final Answer



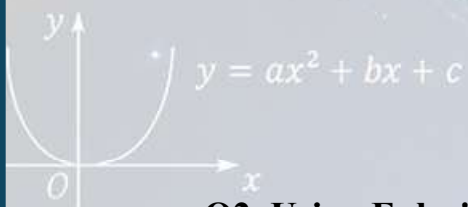
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

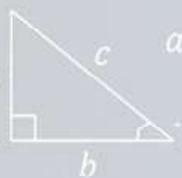
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 2)

Q2. Using Euler's method, find $y(0.4)$ for the differential equation,



$$a^2 + b^2 \frac{dy}{dx} = x + y$$

**with initial condition $y(0) = 1$
and step length $h = 0.2$.**

Solution:-

Given

$$f(x, y) = x + y$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.2$$

By Euler's

$$y_{n+1} = y_n + h f(x_n, y_n)$$

1st approximation

$$\text{Put } n = 0$$

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) \\ &= 1 + 0.2 \cdot f(0, 1) \\ &= 1 + 0.2 \times (0 + 1) \\ &= 1.2 \end{aligned}$$

$$\begin{aligned} \rightarrow x_1 &= x_0 + h \\ &= 0 + 0.2 \\ &= 0.2 \end{aligned}$$

$$\boxed{y(0.2) = 1.2}$$



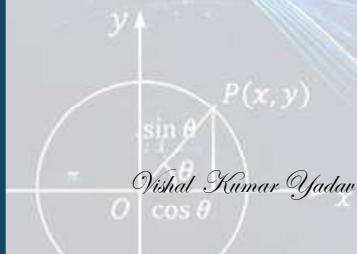
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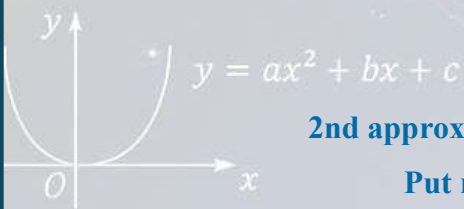
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

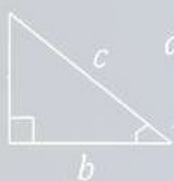
$$x =$$





2nd approximation

Put $n = 1$



$$\begin{aligned} y_2 &= y_1 + h f(x_1, y_1) \\ &= 1.2 + 0.2 f(0.2, 1.2) \\ &= 1.2 + 0.2 [0.2 + 1.2] \\ &= 1.2 + 0.2 \times 1.4 \\ &= 1.2 + 0.28 \\ &= 1.48 \end{aligned}$$

$$\begin{aligned} \rightarrow x_2 &= x_1 + h \\ &= 0.2 + 0.2 \\ &= 0.4 \end{aligned}$$

$$y(0.4) = 1.48$$

3rd approximation

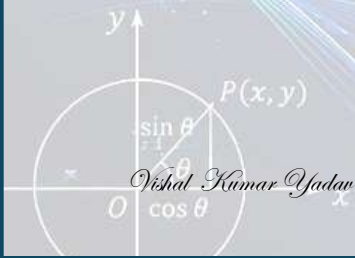
Put $n = 2$

$$\begin{aligned} y_3 &= y_2 + h f(x_2, y_2) \\ &= 1.48 + 0.2 f(0.4, 1.48) \\ &= 1.48 + 0.2 \times (0.4 + 1.48) \\ &= 1.48 + 0.2 \times 1.88 \\ &= 1.48 + 0.376 \\ &= 1.856 \end{aligned}$$

$$\begin{aligned} \rightarrow x_3 &= x_2 + h \\ &= 0.4 + 0.2 \\ &= 0.6 \end{aligned}$$

$$y(0.6) = 1.856$$

Final Answer



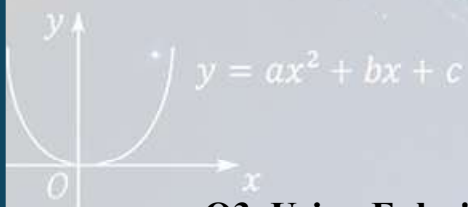
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

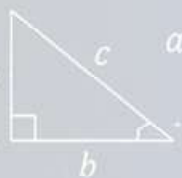
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 3)

Q3. Using Euler's method, find $y(0.4)$ for the differential equation,



$$a^2 + b^2 \frac{dy}{dx} = x + y$$

**with initial condition $y(0) = 1$
and step length $h = 0.2$.**

Solution:-

Given

$$f(x, y) = x + y$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.2$$

By Euler's

$$y_{n+1} = y_n + h f(x_n, y_n)$$

1st approximation

Put $n = 0$

$$y_1 = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.2 \cdot f(0, 1)$$

$$= 1 + 0.2 \times (0 + 1)$$

$$= 1.2$$

$$\rightarrow x_1 = x_0 + h$$

$$= 0 + 0.2$$

$$= 0.2$$

$$\boxed{y(0.2) = 1.2}$$

2nd approximation



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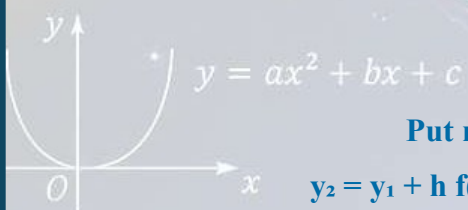


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

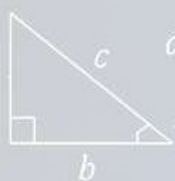
Page no - 147

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



Put $n = 1$

$$\begin{aligned}
 y_2 &= y_1 + h f(x_1, y_1) \\
 &= 1.2 + 0.2 f(0.2, 1.2) \\
 &= 1.2 + 0.2 [0.2 + 1.2] \\
 &= 1.2 + 0.2 \times 1.4 \\
 &= 1.2 + 0.28 \\
 &= 1.48
 \end{aligned}$$



$$\begin{aligned}
 \rightarrow x_2 &= x_1 + h \\
 &= 0.2 + 0.2 \\
 &= 0.4
 \end{aligned}$$

$$y(0.4) = 1.48$$

3rd approximation

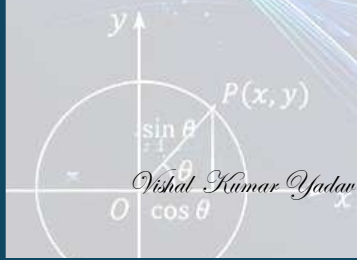
Put $n = 2$

$$\begin{aligned}
 y_3 &= y_2 + h f(x_2, y_2) \\
 &= 1.48 + 0.2 f(0.4, 1.48) \\
 &= 1.48 + 0.2 \times (0.4 + 1.48) \\
 &= 1.48 + 0.2 \times 1.88 \\
 &= 1.48 + 0.376 \\
 &= 1.856
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow x_3 &= x_2 + h \\
 &= 0.4 + 0.2 \\
 &= 0.6
 \end{aligned}$$

$$y(0.6) = 1.856$$

Final Answer



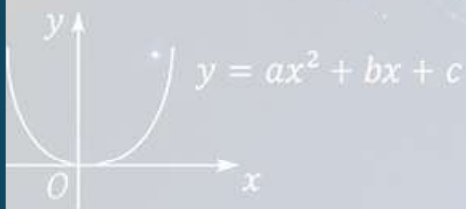
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 148

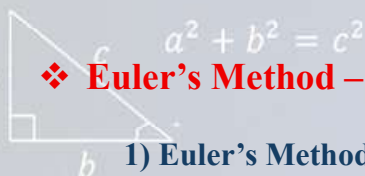
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 1)

(Objectives)

❖ Euler's Method – Multiple Choice Questions :-



1) Euler's Method is mainly used to

- A) Solve differential equations numerically
- B) Solve algebraic equations
- C) Find matrix determinants
- D) Perform integration only

2) Which formula is used in Euler's Method

- A) $y_{n+1} = y_n + h f(x_n, y_n)$
- B) $y_{n+1} = y_n - h f(x_n, y_n)$
- C) $y_{n+1} = y_n + h^2$
- D) $y_{n+1} = x_n + y_n$

3) In Euler's Method, h represents

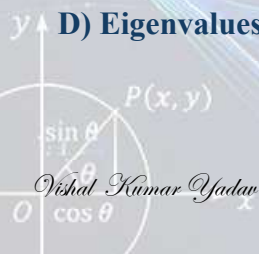
- A) Step size
- B) Initial value
- C) Final answer
- D) Error value

4) Euler's Method provides

- A) Approximate solution
- B) Exact solution
- C) Infinite solution
- D) Graphical solution only

5) Which of the following is required to apply Euler's Method

- A) Initial condition
- B) Matrix multiplication
- C) Laplace transform
- D) Eigenvalues



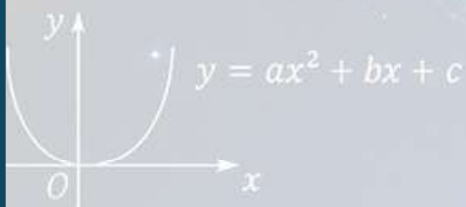
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 149

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 2)

(Subjectives)

❖ **Euler's Method – Numerical Questions :-**

Q1. Using Euler's method, find $y(0.4)$ for the differential equation, $dy/dx = x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q2. Using Euler's method, find $y(0.3)$ for the differential equation, $dy/dx = x - y$, with initial condition $y(0) = 2$ and step length $h = 0.1$.

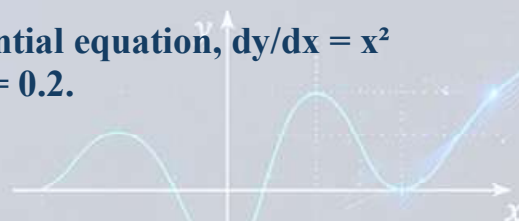
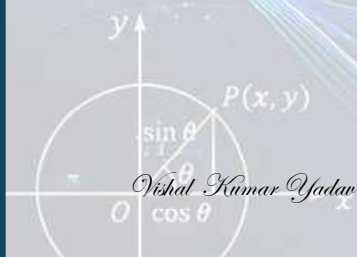
Q3. Using Euler's method, find $y(0.6)$ for the differential equation, $dy/dx = x^2 + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q4. Using Euler's method, find $y(0.4)$ for the differential equation, $dy/dx = y - x$, with initial condition $y(0) = 0$ and step length $h = 0.2$.

Q5. Using Euler's method, find $y(0.5)$ for the differential equation, $dy/dx = x + 2y$, with initial condition $y(0) = 1$ and step length $h = 0.1$.

Q6. Using Euler's method, find $y(0.8)$ for the differential equation, $dy/dx = x^2 - y$, with initial condition $y(0) = 2$ and step length $h = 0.2$.

Q7. Using Euler's method, find $y(0.6)$ for the differential equation, $dy/dx = 2x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

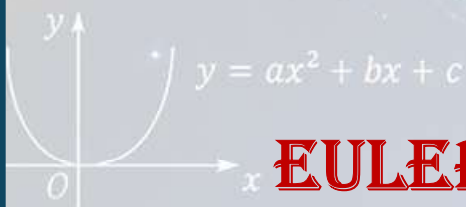


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

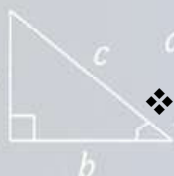
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



EULER'S MODIFIED METHOD

(THEORY)



$$a^2 + b^2 = c^2$$

- ❖ Explain Euler's Modified Method for solving ODE differential equation.

Let, $\frac{dx}{dy} = f(x, y)$ be the initial value problem with initial condition

$y(x_0) = y_0$ and step length (h).

$f(x)$ be the given soln of given diffn eqn we will be expending

$f(x)$ in the neighbourhood of x_0

by using Taylor's series expansion

By Euler's

$$y_1 = y_0 + h f(x_0, y_0)$$

By Euler's Modified

$$y_1^1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^0)]$$

$$y_1^2 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^1)]$$

$$y_1^3 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^2)]$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

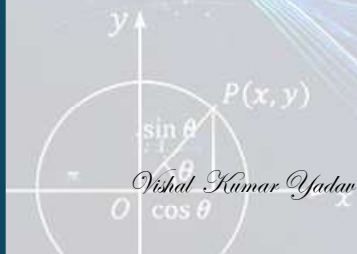
$$y_{n+1}^1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^n)]$$

- General formula of x

"Euler's modified method"

$$y_{n+1}^1 = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^n)]$$

- Euler's formula and Euler's modified formula jointly are called Euler's correct formula.



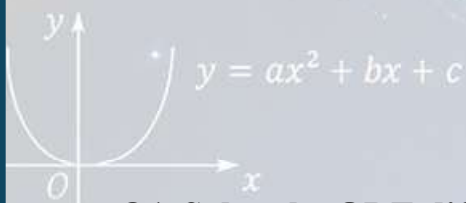
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 151

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(PRACTICAL - 1)

Q1. Solve the ODE differential equation $dx/dy=x+y$ with initial condition $y_0=1$ and step length $h=0.1$ find $y(0.1)$ by using Euler's modified method.

Solution:-

Given

$$f(x, y) = x + y$$

$$y(0) = 1$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x_0 & y_0 \end{array}$$

$$h = 0.1$$

By Euler's

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) \\ &= 1 + 0.1(0, 1) \\ &= 1 + 0.1(0 + 1) \end{aligned}$$

$$y_1 = 1.1$$

$$\triangleright x_1 = x_0 + h$$

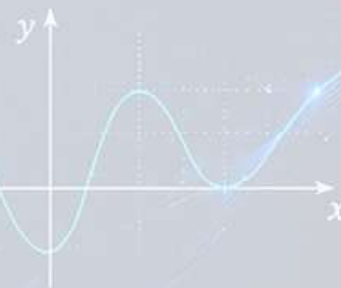
$$= 0 + 0.1$$

$$= 0.1$$

By Euler's Modified:

1st approximation:

$$\begin{aligned} y_1^1 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^0)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.1] \\ &= 1 + \frac{0.1}{2} \times 2.2 \end{aligned}$$



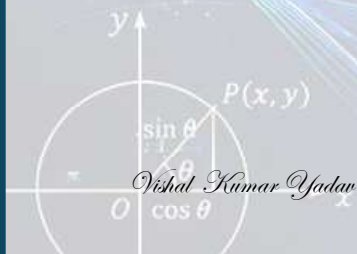
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

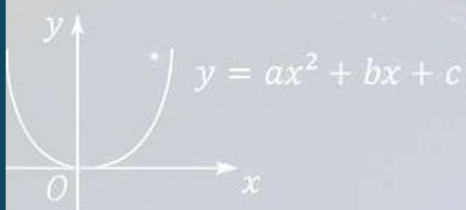
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$

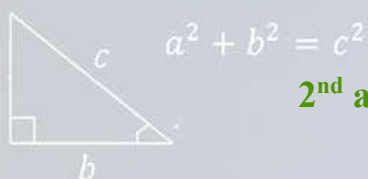


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$$= 1 + 0.11$$

$$y_0^1 = 1.11$$



2nd approximation:

$$\begin{aligned} y_1^2 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^1)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.11)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.11] \\ &= 1 + \frac{0.1}{2} \times 2.21 \end{aligned}$$

$$y_1^2 = 1.1105$$

3rd approximation:

$$\begin{aligned} y_1^3 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^2)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1105)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.1105] \\ &= 1 + \frac{0.1}{2} \times 2.2105 \end{aligned}$$

$$y_1^3 = 1.110525$$

Final Answer

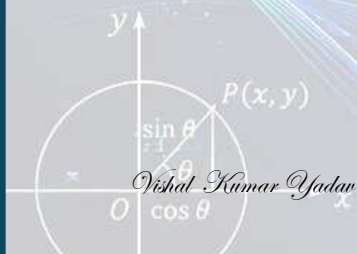


$$\sin^2 \theta + \cos^2 \theta = 1$$

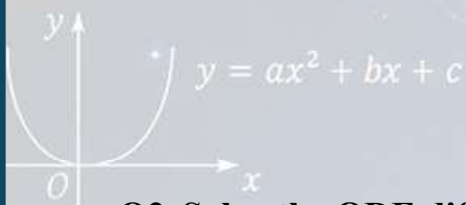
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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(PRACTICAL - 2)

Q2. Solve the ODE differential equation $dx/dy=x+y$ with initial condition $y_0=1$ and step length $h=0.1$ find $y(0.1)$ by using Euler's modified method.

Solution:-

Given

$$f(x, y) = x + y$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

By Euler's

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) \\ &= 1 + 0.1(0, 1) \\ &= 1 + 0.1(0 + 1) \end{aligned}$$

$$y_1 = 1.1$$

$$\triangleright x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$= 0.1$$

By Euler's Modified:

1st approximation:

$$\begin{aligned} y_0^1 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^0)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.1] \\ &= 1 + \frac{0.1}{2} \times 2.2 \end{aligned}$$



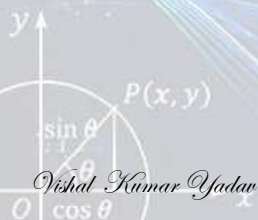
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

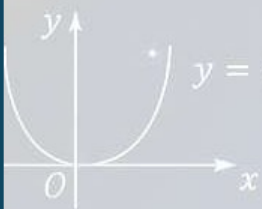
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$



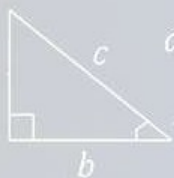
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$$y = ax^2 + bx + c$$

$$= 1 + 0.11$$

$$y_0^1 = 1.11$$



$$a^2 + b^2 = c^2$$

2nd approximation:

$$\begin{aligned} y_1^2 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^1)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.11)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.11] \\ &= 1 + \frac{0.1}{2} \times 2.21 \end{aligned}$$

$$y_1^2 = 1.1105$$

3rd approximation:

$$\begin{aligned} y_1^3 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^2)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1105)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.1105] \\ &= 1 + \frac{0.1}{2} \times 2.2105 \end{aligned}$$

$$y_1^3 = 1.110525$$

Final Answer

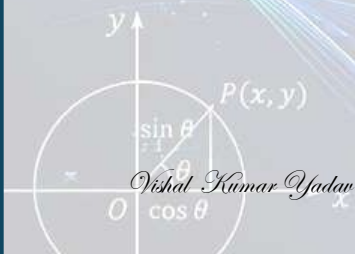


$$\sin^2 \theta + \cos^2 \theta = 1$$

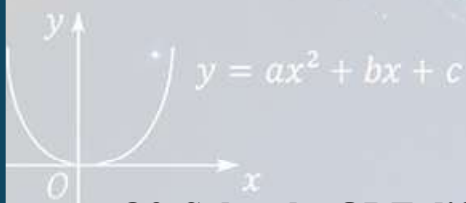
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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(PRACTICAL - 3)

Q3. Solve the ODE differential equation $dx/dy=x+y$ with initial condition $y_0=1$ and step length $h=0.1$ find $y(0.1)$ by using Euler's modified method.

Solution:-

Given

$$f(x, y) = x + y$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

By Euler's

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) \\ &= 1 + 0.1(0, 1) \\ &= 1 + 0.1(0 + 1) \end{aligned}$$

$$y_1 = 1.1$$

$$\triangleright x_1 = x_0 + h$$

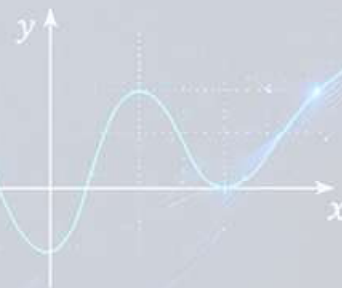
$$= 0 + 0.1$$

$$= 0.1$$

By Euler's Modified:

1st approximation:

$$\begin{aligned} y_1^1 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^0)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.1] \\ &= 1 + \frac{0.1}{2} \times 2.2 \end{aligned}$$



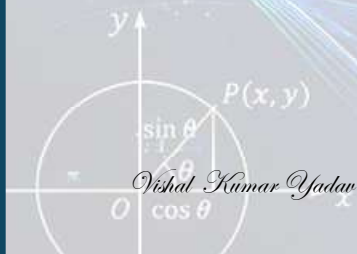
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

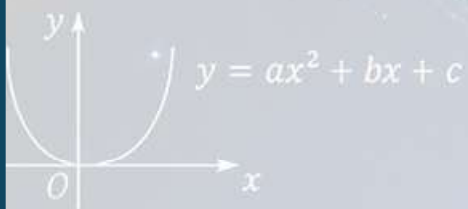
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$

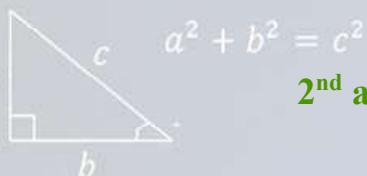


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$$= 1 + 0.11$$

$$y_0^1 = 1.11$$



2nd approximation:

$$\begin{aligned} y_1^2 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^1)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.11)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.11] \\ &= 1 + \frac{0.1}{2} \times 2.21 \end{aligned}$$

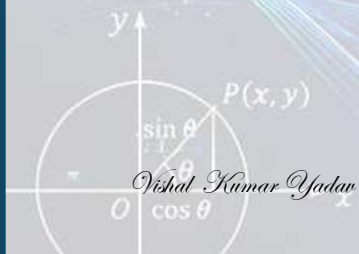
$$y_1^2 = 1.1105$$

3rd approximation:

$$\begin{aligned} y_1^3 &= y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^2)] \\ &= 1 + \frac{0.1}{2} [f(0, 1) + f(0.1, 1.1105)] \\ &= 1 + \frac{0.1}{2} [0 + 1 + 0.1 + 1.1105] \\ &= 1 + \frac{0.1}{2} \times 2.2105 \end{aligned}$$

$$y_1^3 = 1.110525$$

Final Answer

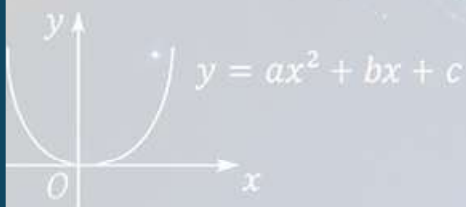


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

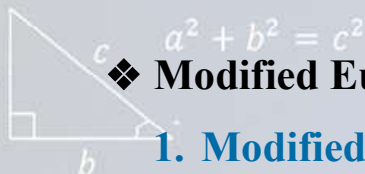
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 1)

(Objectives)



❖ **Modified Euler's Method – Multiple Choice Questions :-**

- Modified Euler's Method is also known as**
 - Heun's Method
 - Newton's Method
 - Simpson's Method
 - Bisection Method
- Modified Euler's Method is mainly used to**
 - Improve the accuracy of Euler's Method
 - Solve quadratic equations
 - Find determinants
 - Calculate eigenvalues
- Which formula is used in the correction step of Modified Euler's Method**
 - $y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$
 - $y_{n+1} = y_n + h f(x_n, y_n)$
 - $y_{n+1} = y_n - h f(x_n, y_n)$
 - $y_{n+1} = x_n + y_n$
- Modified Euler's Method consists of**
 - Predictor and Corrector steps
 - Differentiation only
 - Integration only
 - Matrix operations
- Modified Euler's Method gives**
 - More accurate approximation than Euler's Method
 - Exact solution always
 - Infinite solutions
 - Graphical solutions only



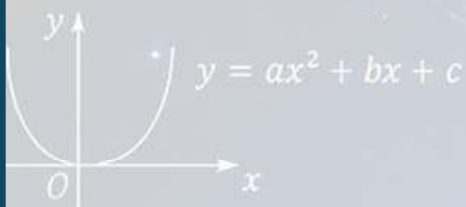
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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 158

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 1)

(Subjectives)

❖ **Modified Euler's Method – Numerical Questions :-**

Q1. Using Modified Euler's method, find $y(0.4)$ for the differential equation, $dy/dx = x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

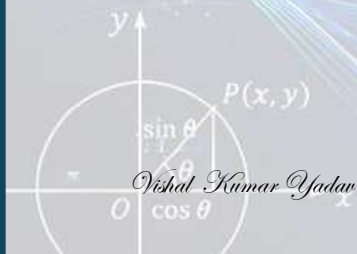
Q2. Using Modified Euler's method, find $y(0.3)$ for the differential equation, $dy/dx = x - y$, with initial condition $y(0) = 2$ and step length $h = 0.1$.

Q3. Using Modified Euler's method, find $y(0.6)$ for the differential equation, $dy/dx = x^2 + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q4. Using Modified Euler's method, find $y(0.4)$ for the differential equation, $dy/dx = y - x$, with initial condition $y(0) = 0$ and step length $h = 0.2$.

Q5. Using Modified Euler's method, find $y(0.5)$ for the differential equation, $dy/dx = x + 2y$, with initial condition $y(0) = 1$ and step length $h = 0.1$.

Q6. Using Modified Euler's method, find $y(0.8)$ for the differential equation, $dy/dx = x^2 - y$, with initial condition $y(0) = 2$ and step length $h = 0.2$.



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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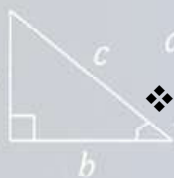
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

RUNGE-KUTTA METHOD (2ND ORDER)

(THEORY)



$$a^2 + b^2 = c^2$$

❖ Explain Runge-Kutta Method for second order:-

Let,

$\frac{dx}{dy} = f(x, y)$ be the initial value problem with initial condition

$$y(x_0) = y_0$$

and,

step length (h).

then,

its Runge-Kutta second order formula is given by:-

$$y_{n+1} = y_n + \frac{h}{2} f(K_1 + K_2)$$

where,

$$K_1 = f(x_n, y_n)$$

$$K_2 = f(x_n + h, y_n + hK_1)$$



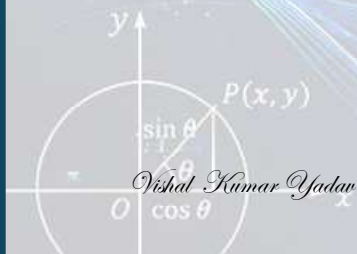
$$\sin^2 \theta + \cos^2 \theta = 1$$

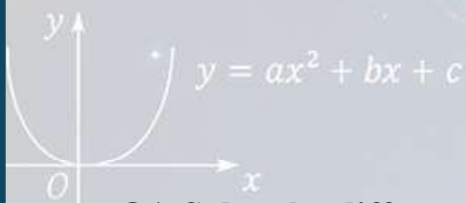
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =





(PRACTICAL - 1)

Q1. Solve the differential equation $dx/dy = y(2-y)$ with initial condition $y_0=1$ and step length $h = 0.1$ find $y(0.2)$ by using Runge-Kutta Method fo 2nd order .

Solution:-

Given

$$f(x, y) = y(2-y)$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

$$y(0.2) = ?$$

By R.K. Method (2nd order)

1st approximation:

$$y_1 = y_0 + \frac{h}{2} f(K_1 + K_2) \text{ ---- (i)}$$

$$K_1 = f(x_0, y_0)$$

$$= f(0, 1)$$

$$\{ \text{ie, } f(x, y) = y(2 - y) \}$$

$$= 1(2 - 1)$$

$$K_1 = 1$$

$$K_2 = f(x_0 + h, y_0 + hK_1)$$

$$= f(0 + 0.1, 1 + 0.1 * 1)$$



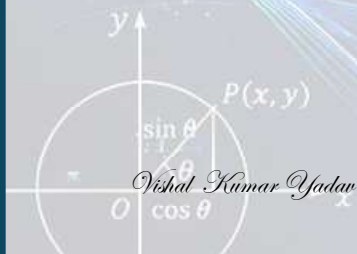
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

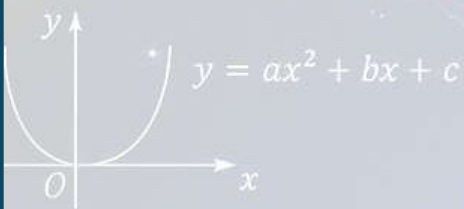
Page no - 161

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$

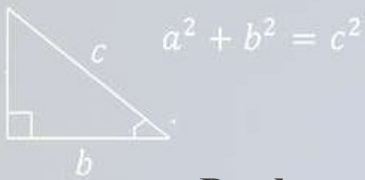


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$$= 1.1(2 - 1.1)$$

$$K_1 = 0.99$$



Put k_1 and k_2 in equation (i)

$$y_1 = y_0 + \frac{h}{2} f(K_1 + K_2)$$

$$= 1 + \frac{0.1}{2} f(1 + 0.99)$$

$$= 1 + 0.05 * 1.99$$

$$Y_1 = 1.0995$$

$$\Rightarrow x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$x_1 = 0.1$$

$$y(0.1) = 1.0995$$

2nd approximation Knowledge . Skill . Success

$$Y_2 = y_0 + \frac{h}{2} f(K_1 + K_2) \text{ ---- (ii)}$$

$$k_1 = f(x_1, y_1)$$

$$= f(0.1, 1.0995)$$

$$\{ f(x, y) = y(2 - y) \}$$

$$= 1.0995(2 - 1.0995)$$

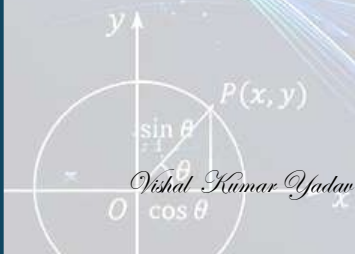


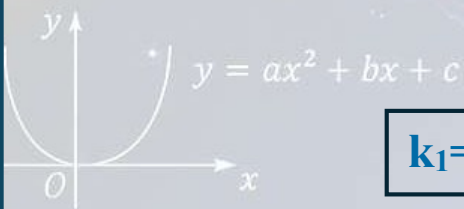
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

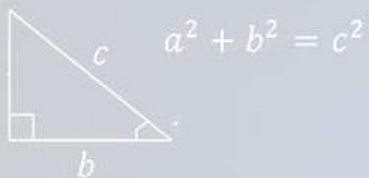
Page no - 162

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





$$k_1 = 0.99009975$$



$$\begin{aligned} k_2 &= f(x_1+h, y_1+hk_1) \\ &= f(0.1+0.1, 1.0995+0.1 \times 0.99009975) \\ &= f(0.2, 1.198509975) \\ &= 1.198509975(2-1.198509975) \end{aligned}$$

$$\{ f(x, y) = y(2 - y) \}$$

$$k_2 = 0.9605937898$$

Put k_1 and k_2 in eqn (ii)

$$\begin{aligned} y_2 &= y_0 + \frac{h}{2} [k_1 + k_2] \\ &= 1.0995 + 0.1 [0.99009975 + 0.9605937898] \end{aligned}$$

$$y_2 = 1.197034677$$

$$\begin{aligned} \Rightarrow x_2 &= x_1 + h \\ &= 0.1 + 0.1 \\ &= 0.2 \end{aligned}$$

$$y(0.2) = 1.197034677$$



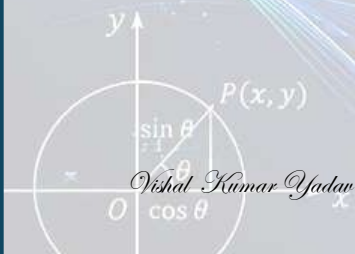
Final Answer

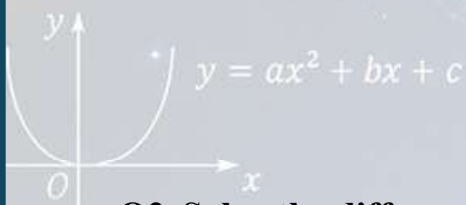
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 163

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





(PRACTICAL - 2)

Q2. Solve the differential equation $dx/dy = y(2-y)$ with initial condition $y_0=1$ and step length $h = 0.1$ find $y(0.2)$ by using Runge-Kutta Method fo 2nd order .

Solution:-

Given

$$f(x, y) = y(2-y)$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

$$y(0.2) = ?$$

By R.K. Method (2nd order)

1st approximation:

$$y_1 = y_0 + \frac{h}{2} f(K_1 + K_2) \text{ ---- (i)}$$

$$K_1 = f(x_0, y_0)$$

$$= f(0, 1)$$

$$\{ \text{ie, } f(x, y) = y(2 - y) \}$$

$$= 1(2 - 1)$$

$$K_1 = 1$$

$$K_2 = f(x_0 + h, y_0 + hK_1)$$

$$= f(0 + 0.1, 1 + 0.1 * 1)$$



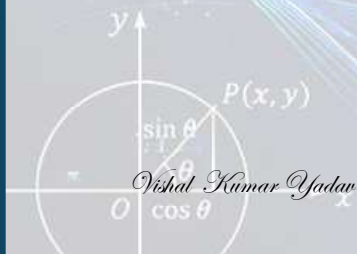
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

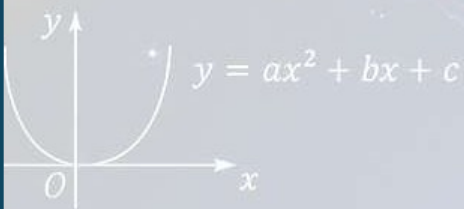
Page no - 164

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$

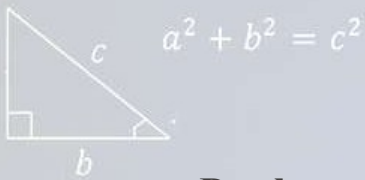


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$$= 1.1(2 - 1.1)$$

$$K_1 = 0.99$$



Put k_1 and k_2 in equation (i)

$$y_1 = y_0 + \frac{h}{2} f(K_1 + K_2)$$

$$= 1 + \frac{0.1}{2} f(1 + 0.99)$$

$$= 1 + 0.05 * 1.99$$

$$Y_1 = 1.0995$$

$$\Rightarrow x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$x_1 = 0.1$$

$$y(0.1) = 1.0995$$

2nd approximation Knowledge . Skill . Success

$$Y_2 = y_0 + \frac{h}{2} f(K_1 + K_2) \text{ ---- (ii)}$$

$$k_1 = f(x_1, y_1)$$

$$= f(0.1, 1.0995)$$

$$\{ f(x, y) = y(2 - y) \}$$

$$= 1.0995(2 - 1.0995)$$

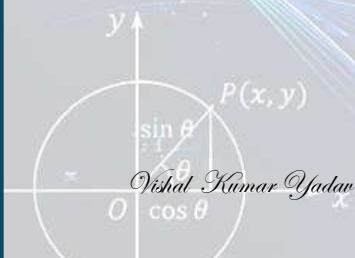


$$\sin^2 \theta + \cos^2 \theta = 1$$

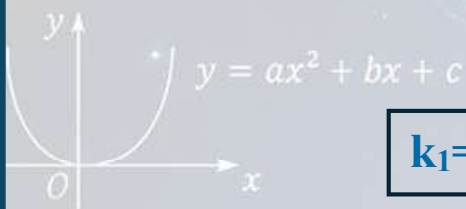
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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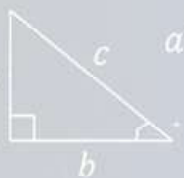
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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$$k_1 = 0.99009975$$



$$a^2 + b^2 = c^2$$

$$k_2 = f(x_1 + h, y_1 + hk_1)$$

$$= f(0.1 + 0.1, 1.0995 + 0.1 \times 0.99009975)$$

$$= f(0.2, 1.198509975)$$

$$= 1.198509975(2 - 1.198509975)$$

$$\{ f(x, y) = y(2 - y) \}$$

$$k_2 = 0.9605937898$$

Put k_1 and k_2 in eqn (ii)

$$y_2 = y_0 + \frac{h}{2} [k_1 + k_2]$$

$$= 1.0995 + 0.1 [0.99009975 + 0.9605937898]$$

$$y_2 = 1.197034677$$

$$\Rightarrow x_2 = x_1 + h$$

$$= 0.1 + 0.1$$

$$= 0.2$$

$$y(0.2) = 1.197034677$$



Final Answer

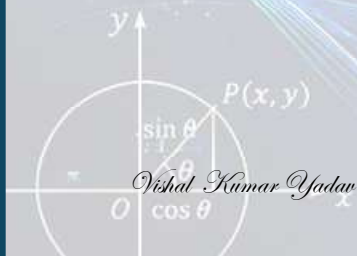
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

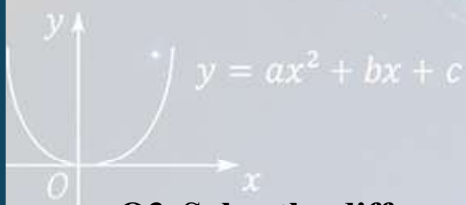
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$



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(PRACTICAL - 3)

Q3. Solve the differential equation $dx/dy = y(2-y)$ with initial condition $y_0=1$ and step length $h = 0.1$ find $y(0.2)$ by using Runge-Kutta Method fo 2nd order .

Solution:-

Given

$$f(x, y) = y(2-y)$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

$$y(0.2) = ?$$

By R.K. Method (2nd order)

1st approximation:

$$y_1 = y_0 + \frac{h}{2} f(K_1 + K_2) \text{ ---- (i)}$$

$$\begin{aligned} K_1 &= f(x_0, y_0) \\ &= f(0, 1) \end{aligned}$$

$$\{f(x, y) = y(2 - y)\}$$

$$= 1(2 - 1)$$

$$K_1 = 1$$

$$\begin{aligned} K_2 &= f(x_0 + h, y_0 + hK_1) \\ &= f(0 + 0.1, 1 + 0.1 \cdot 1) \end{aligned}$$



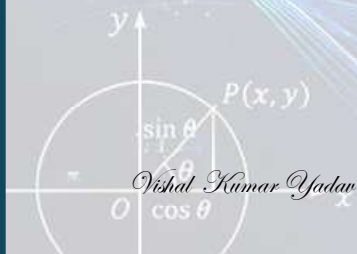
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

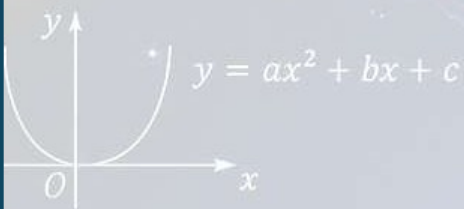
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x =$$

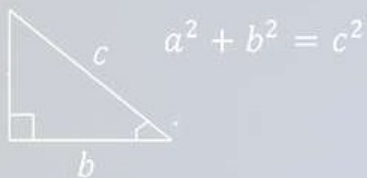


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$$= 1.1(2 - 1.1)$$

$$K_1 = 0.99$$



Put k_1 and k_2 in equation (i)

$$y_1 = y_0 + \frac{h}{2} f(K_1 + K_2)$$

$$= 1 + \frac{0.1}{2} f(1 + 0.99)$$

$$= 1 + 0.05 * 1.99$$

$$Y_1 = 1.0995$$

$$\Rightarrow x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$x_1 = 0.1$$

$$y(0.1) = 1.0995$$

2nd approximation Knowledge . Skill . Success

$$Y_2 = y_0 + \frac{h}{2} f(K_1 + K_2) \text{ ---- (ii)}$$

$$k_1 = f(x_1, y_1)$$

$$= f(0.1, 1.0995)$$

$$\{ f(x, y) = y(2 - y) \}$$

$$= 1.0995(2 - 1.0995)$$

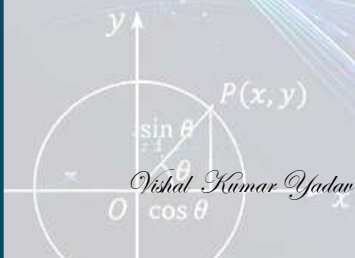


$$\sin^2 \theta + \cos^2 \theta = 1$$

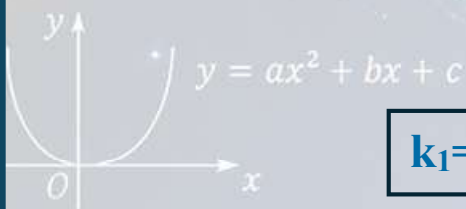
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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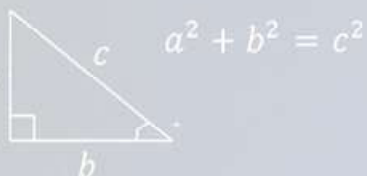
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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$$k_1 = 0.99009975$$



$$\begin{aligned} k_2 &= f(x_1+h, y_1+hk_1) \\ &= f(0.1+0.1, 1.0995+0.1 \times 0.99009975) \\ &= f(0.2, 1.198509975) \\ &= 1.198509975(2-1.198509975) \end{aligned}$$

$$\{ f(x, y) = y(2-y) \}$$

$$k_2 = 0.9605937898$$

Put k_1 and k_2 in eqn (ii)

$$\begin{aligned} y_2 &= y_0 + \frac{h}{2} [k_1 + k_2] \\ &= 1.0995 + 0.1 [0.99009975 + 0.9605937898] \end{aligned}$$

$$y_2 = 1.197034677$$

$$\begin{aligned} \Rightarrow x_2 &= x_1 + h \\ &= 0.1 + 0.1 \\ &= 0.2 \end{aligned}$$

$$y(0.2) = 1.197034677$$



Final Answer

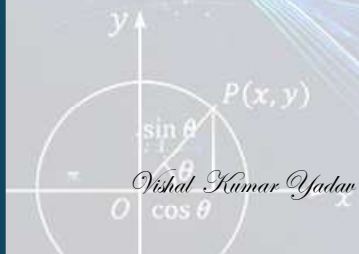
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

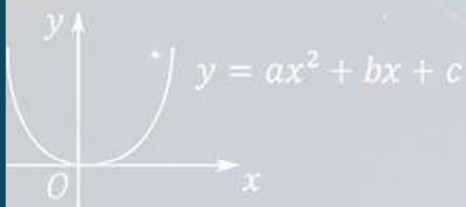
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$

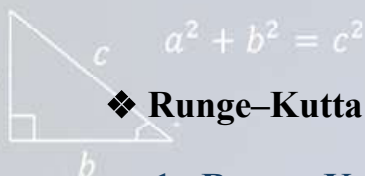


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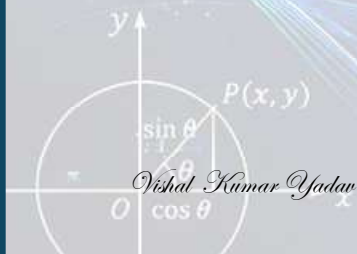
(EXERCISE - 1)

(Objectives)



❖ Runge–Kutta Method (Second Order) – Multiple Choice Questions :-

- Runge–Kutta Second Order (RK2) method is used to
 - Solve differential equations numerically
 - Solve quadratic equations
 - Find matrix determinants
 - Perform matrix inversion
- RK2 method is more accurate than
 - Euler's Method
 - Gaussian Elimination
 - Newton–Raphson Method
 - Bisection Method
- In RK2 method, k_1 is calculated as
 - $h f(x_n, y_n)$
 - $h + f(x_n, y_n)$
 - $x_n + y_n$
 - $h^2 f(x_n, y_n)$
- The RK2 method is also known as
 - Improved Euler Method
 - Simpson's Rule
 - Trapezoidal Rule
 - Gauss Elimination Method
- RK2 method gives
 - Better approximation than Euler's Method
 - Exact solution always
 - Infinite solutions
 - Graphical solutions only

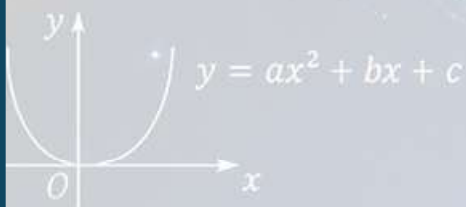


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 2)

(Subjectives)



❖ Runge–Kutta Method (Second Order) – Numerical Questions :-

Q1. Using the Runge–Kutta Second Order (RK2) method, find $y(0.4)$ for the differential equation, $dy/dx = x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q2. Using the Runge–Kutta Second Order (RK2) method, find $y(0.3)$ for the differential equation, $dy/dx = x - y$, with initial condition $y(0) = 2$ and step length $h = 0.1$.

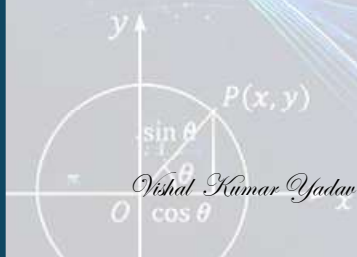
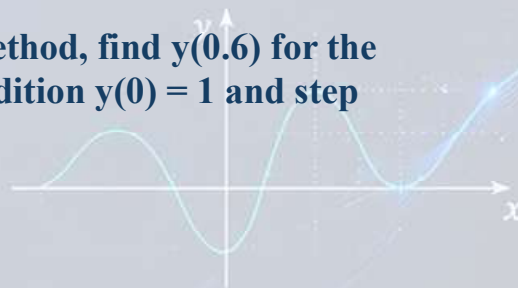
Q3. Using the Runge–Kutta Second Order (RK2) method, find $y(0.6)$ for the differential equation, $dy/dx = x^2 + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q4. Using the Runge–Kutta Second Order (RK2) method, find $y(0.4)$ for the differential equation, $dy/dx = y - x$, with initial condition $y(0) = 0$ and step length $h = 0.2$.

Q5. Using the Runge–Kutta Second Order (RK2) method, find $y(0.5)$ for the differential equation, $dy/dx = x + 2y$, with initial condition $y(0) = 1$ and step length $h = 0.1$.

Q6. Using the Runge–Kutta Second Order (RK2) method, find $y(0.8)$ for the differential equation, $dy/dx = x^2 - y$, with initial condition $y(0) = 2$ and step length $h = 0.2$.

Q7. Using the Runge–Kutta Second Order (RK2) method, find $y(0.6)$ for the differential equation, $dy/dx = 2x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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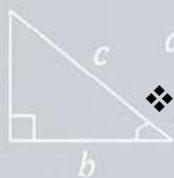
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

RUNGE-KUTTA METHOD (4ND ORDER)

(THEORY)



$$a^2 + b^2 = c^2$$

❖ Explain Runge-Kutta Method for fourth order:-

Let,

$\frac{dx}{dy} = f(x, y)$ be the initial value problem with initial condition

$$y_0 = x_0$$

and,

$$\text{step length } h = x_n - x_{n-1}$$

then,

its Runge-Kutta fourth order formula is given by:-

$$y_{n+1} = y_n + \frac{h}{6} f(K_1 + 2K_2 + 2K_3 + K_4)$$

where,

$$K_1 = f(x_n, y_n)$$

$$K_2 = f(x_n + \frac{h}{2}, y_n + \frac{K_1}{2})$$

$$K_3 = f(x_n + \frac{h}{2}, y_n + \frac{K_2}{2})$$

$$K_4 = f(x_n + h, y_n + K_3)$$



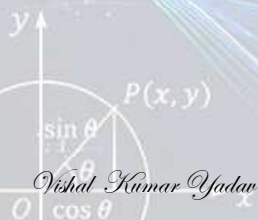
$$\sin^2 \theta + \cos^2 \theta = 1$$

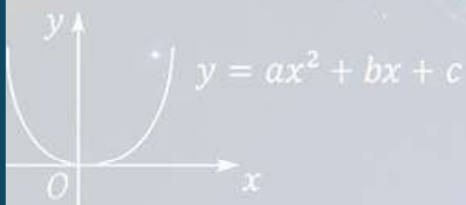
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

x =





(PRACTICAL - 1)

Q1. Solve the differential equation $\frac{dx}{dy} = \frac{1}{x+y}$ with initial condition $y(0)=1$ find $y(0.1)$ and step length ($h = 0.1$) by using Runge-Kutta Method for 4th order .

Solution:-

Given

$$f(x, y) = \frac{1}{x+y}$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

$$y(0.1) = ?$$

By R.K. Method (4th order)

1st approximation:

$$y_1 = y_0 + \frac{h}{6} f(K_1 + 2K_2 + 2K_3 + K_4) \text{ --- (i)}$$

$$\begin{aligned} K_1 &= f(x_0, y_0) \\ &= f(0, 1) \end{aligned}$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0+1}$$

$$K_1 = 1$$



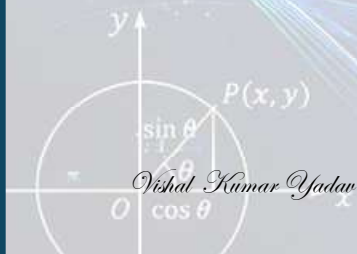
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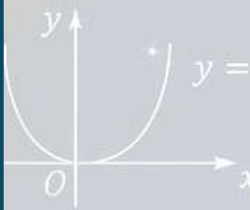
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

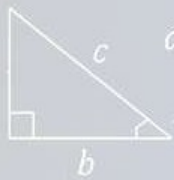
$$x =$$



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$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

$$K_2 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$

$$= f\left(0 + \frac{0.1}{2}, 1 + \frac{1}{2}\right)$$

$$= f(0.05, 1.5)$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0.05+1.5}$$

$$= \frac{1}{1.55}$$

$$K_2 = 0.6451612903$$

$$K_3 = f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

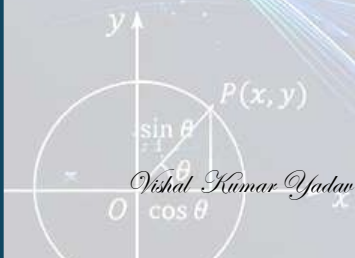
$$= f\left(0 + \frac{0.1}{2}, 1 + \frac{0.6451612903}{2}\right)$$

$$= f(0.05, 1.32....)$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0.05+1.32...}$$

$$K_2 = 0.7285546416$$



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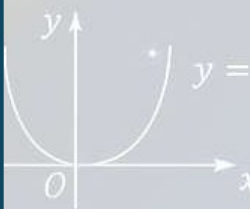


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

$$K_4 = f(x_n + h, y_n + k_3)$$

$$= f(0 + 0.1, 1 + 0.72 \dots)$$

$$= f(0.1, 1.72 \dots)$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0.1 + 1.72 \dots}$$

$$K_2 = 0.5468824807$$

Put K_1, K_2, K_3 and K_4 in equation (i)

$$y_1 = y_0 + \frac{h}{6} f(K_1 + 2K_2 + 2K_3 + K_4)$$

$$= 1 + \frac{0.1}{6} f(1 + 2(0.64 \dots) + 2(0.72 \dots) + 0.54 \dots)$$

$$y_1 = 1.0715719058$$

$$\Rightarrow x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$x_1 = 0.1$$

$$y(0.1) = 1.0715719058$$

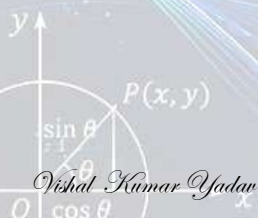
Final Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

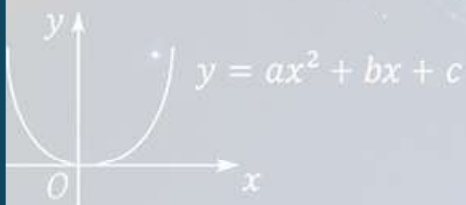
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



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(PRACTICAL - 2)

Q2. Solve the differential equation $\frac{dx}{dy} = \frac{1}{x+y}$ with initial condition $y(0)=1$ find $y(0.1)$ and step length ($h = 0.1$) by using Runge-Kutta Method for 4th order .

Solution:-

Given

$$f(x, y) = \frac{1}{x+y}$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

$$y(0.1) = ?$$

By R.K. Method (4th order)

1st approximation:

$$y_1 = y_0 + \frac{h}{6} f(K_1 + 2K_2 + 2K_3 + K_4) \text{ --- (i)}$$

$$\begin{aligned} K_1 &= f(x_0, y_0) \\ &= f(0, 1) \end{aligned}$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0+1}$$

$$K_1 = 1$$



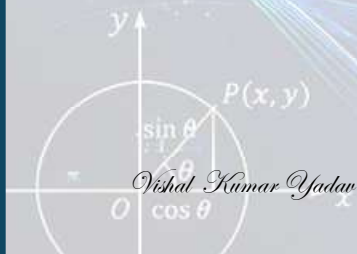
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

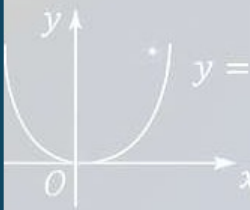
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

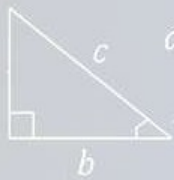
$$x =$$



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$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

$$\begin{aligned} K_2 &= f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right) \\ &= f\left(0 + \frac{0.1}{2}, 1 + \frac{1}{2}\right) \\ &= f(0.05, 1.5) \end{aligned}$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$\begin{aligned} &= \frac{1}{0.05+1.5} \\ &= \frac{1}{1.55} \end{aligned}$$

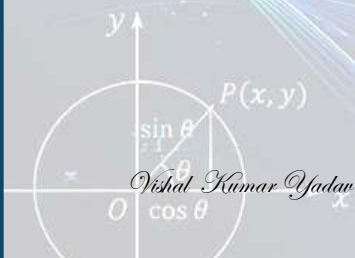
$$K_2 = 0.6451612903$$

$$\begin{aligned} K_3 &= f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right) \\ &= f\left(0 + \frac{0.1}{2}, 1 + \frac{0.6451612903}{2}\right) \\ &= f(0.05, 1.32....) \end{aligned}$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0.05+1.32...}$$

$$K_2 = 0.7285546416$$



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$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

$$K_4 = f(x_n + h, y_n + k_3)$$

$$= f(0 + 0.1, 1 + 0.72 \dots)$$

$$= f(0.1, 1.72 \dots)$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0.1 + 1.72 \dots}$$

$$K_2 = 0.5468824807$$

Put K_1, K_2, K_3 and K_4 in equation (i)

$$y_1 = y_0 + \frac{h}{6} f(K_1 + 2K_2 + 2K_3 + K_4)$$

$$= 1 + \frac{0.1}{6} f(1 + 2(0.64 \dots) + 2(0.72 \dots) + 0.54 \dots)$$

$$y_1 = 1.0715719058$$

$$\Rightarrow x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$x_1 = 0.1$$

$$y(0.1) = 1.0715719058$$

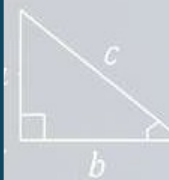
Final Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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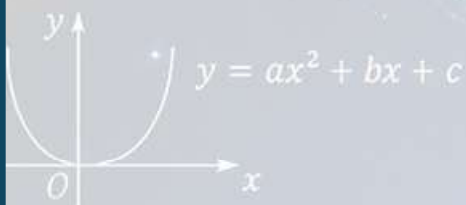
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$a^2 + b^2 = c^2$$



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(PRACTICAL - 1)

Q1. Solve the differential equation $\frac{dx}{dy} = \frac{1}{x+y}$ with initial condition $y(0)=1$ find $y(0.1)$ and step length ($h = 0.1$) by using Runge-Kutta Method for 4th order .

Solution:-

Given

$$f(x, y) = \frac{1}{x+y}$$

$$y(0) = 1$$

$$\begin{matrix} \downarrow & \downarrow \\ x_0 & y_0 \end{matrix}$$

$$h = 0.1$$

$$y(0.1) = ?$$

By R.K. Method (4th order)

1st approximation:

$$y_1 = y_0 + \frac{h}{6} f(K_1 + 2K_2 + 2K_3 + K_4) \text{ --- (i)}$$

$$\begin{aligned} K_1 &= f(x_0, y_0) \\ &= f(0, 1) \end{aligned}$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0+1}$$

$$K_1 = 1$$



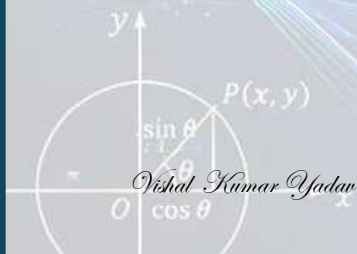
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

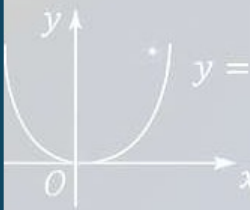
Page no - 179

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

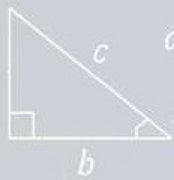
$$x =$$



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$$= f\left(0 + \frac{0.1}{2}, 1 + \frac{0.6451612903}{2}\right)$$

$$= f(0.05, 1.32....)$$

$$\{f(x, y) = \frac{1}{x+y}\}$$

$$= \frac{1}{0.05+1.32...}$$

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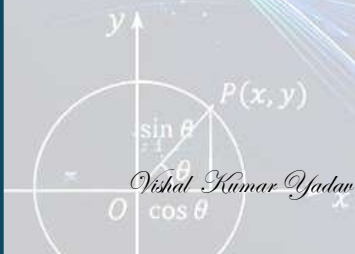


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

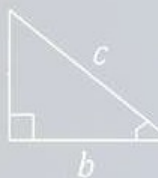
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$





$$y = ax^2 + bx + c$$



$$a^2 + b^2 = c^2$$

$$K_4 = f(x_n + h, y_n + k_3)$$

$$= f(0 + 0.1, 1 + 0.72 \dots)$$

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$$\{f(x, y) = \frac{1}{x+y}\}$$

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$$\Rightarrow x_1 = x_0 + h$$

$$= 0 + 0.1$$

$$x_1 = 0.1$$

$$y(0.1) = 1.0715719058$$

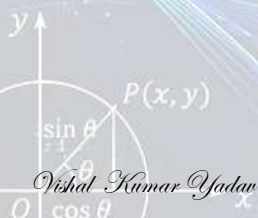
Final Answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

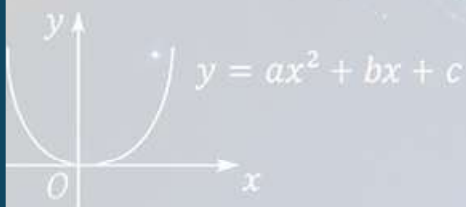
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Page no - 181

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

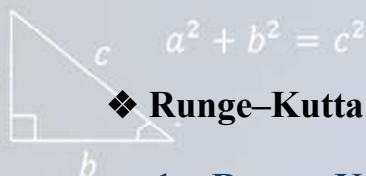


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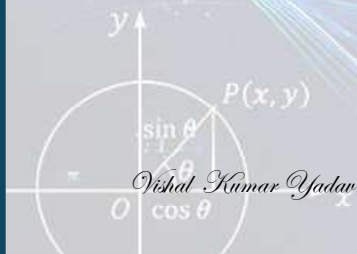
(EXERCISE - 1)

(Objectives)



❖ Runge–Kutta Method (Second Order) – Multiple Choice Questions :-

- Runge–Kutta Fourth Order (RK4) method is mainly used to
 - Solve differential equations numerically
 - Solve quadratic equations
 - Find determinants of matrices
 - Perform matrix multiplication
- RK4 method uses how many intermediate slopes in one step?
 - Four
 - Two
 - Three
 - One
- In RK4 method, the first slope is represented by
 - k_1
 - k_2
 - k_3
 - k_4
- RK4 method is preferred because it provides
 - Higher accuracy than Euler's and RK2 methods
 - Exact solution for every equation
 - Graphical solution only
 - Matrix solution only
- The final value in RK4 method is obtained by using
 - A weighted average of k_1 , k_2 , k_3 and k_4
 - Only k_1
 - Only k_2 and k_3
 - Only k_4

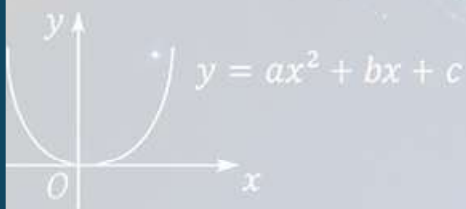


$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



(EXERCISE - 2)

(Subjectives)



❖ Runge–Kutta Method (Second Order) – Numerical Questions :-

Q1. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.4)$ for the differential equation, $dy/dx = x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q2. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.3)$ for the differential equation, $dy/dx = x - y$, with initial condition $y(0) = 2$ and step length $h = 0.1$.

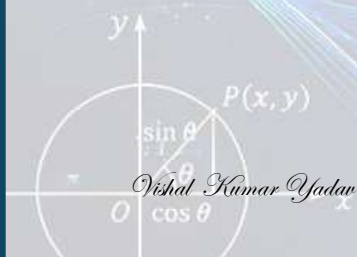
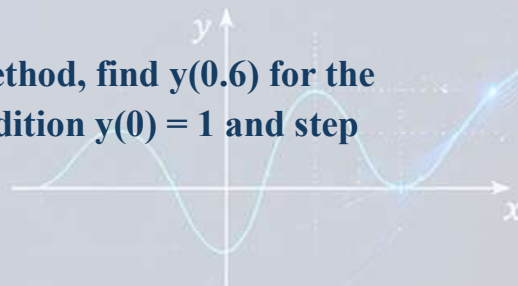
Q3. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.6)$ for the differential equation, $dy/dx = x^2 + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.

Q4. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.4)$ for the differential equation, $dy/dx = y - x$, with initial condition $y(0) = 0$ and step length $h = 0.2$.

Q5. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.5)$ for the differential equation, $dy/dx = x + 2y$, with initial condition $y(0) = 1$ and step length $h = 0.1$.

Q6. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.8)$ for the differential equation, $dy/dx = x^2 - y$, with initial condition $y(0) = 2$ and step length $h = 0.2$.

Q7. Using the Runge–Kutta Fourth Order (RK4) method, find $y(0.6)$ for the differential equation, $dy/dx = 2x + y$, with initial condition $y(0) = 1$ and step length $h = 0.2$.



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$y = ax^2 + bx + c$$

TAYLOR SERIES METHOD

(THEORY)

❖ Explain Taylor Series Method:-

Let,

$\frac{dy}{dx} = f(x,y)$ be the initial value problem with initial condition $y(x_0) = y_0$ and step length (h), then Taylor Series expansion is given by:

1st Approximation:

Put $n=0$

$$y(x_0+h) = y_0 + \frac{h}{1!}f'(x_0, y_0) + \frac{h^2}{2!}f''(x_0, y_0) + \frac{h^3}{3!}f'''(x_0, y_0) + \dots$$

2nd Approximation:

Put $n=1$

$$y(x_1+h) = y_1 + \frac{h}{1!}f'(x_1, y_1) + \frac{h^2}{2!}f''(x_1, y_1) + \frac{h^3}{3!}f'''(x_1, y_1) + \dots$$

3rd Approximation:

Put $n=2$

$$y(x_2+h) = y_2 + \frac{h}{1!}f'(x_2, y_2) + \frac{h^2}{2!}f''(x_2, y_2) + \frac{h^3}{3!}f'''(x_2, y_2) + \dots$$

$$\begin{matrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \end{matrix}$$



$$y(x_n+h) = y_n + \frac{h}{1!}f'(x_n, y_n) + \frac{h^2}{2!}f''(x_n, y_n) + \frac{h^3}{3!}f'''(x_n, y_n) + \dots$$

$P(x, y)$



$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

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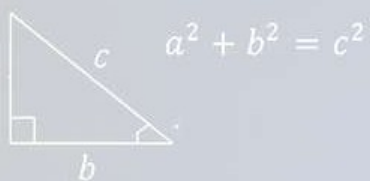
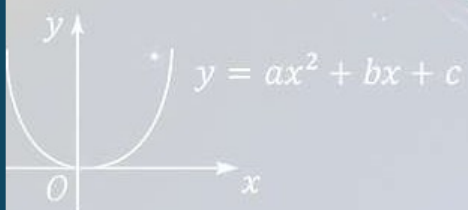
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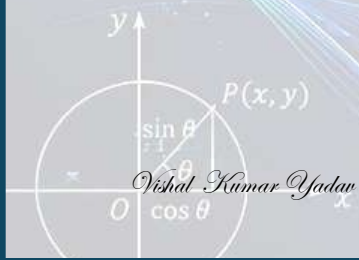
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$x =$



CIMAGE
 Knowledge . Skill . Success



$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$